

OGOR Well Service Practice Test Day 3

OWST3.1 **When planning a wireline operation, what specific wireline equipment must be supported or anchored for safe operations?**

Answer

- B**
- (A) Wireline
 - (B) Lubricator and Wireline unit
 - (C) Wireline unit
 - (D) Lubricator

1

OWST3.2 **What type of buckling occurs when tubing loads cause permanent failure?**

Answer

- C**
- (A) Short column buckling
 - (B) Long column buckling (helical buckling)
 - (C) Intermediate column buckling
 - (D) Transverse buckling

2

OWST3.3 **Before initiating open-hole wireline operations, why is it necessary to circulate bottoms-up?**

Answer

- D**
- (A) Ensure lubricator is clean
 - (B) Prepare for perforating
 - (C) Stimulate the well
 - (D) Clean and condition the hole

3

OWST3.4 **As braided-cable or E-line travels through upper sheave, what prevents imbedded gases from exiting from wire strands?**

Answer

- A** (A) Grease Injection
- (B) Hydraulic Pack Off
- (C) Tool Trap
- (D) Control Head

4

OWST3.5 **Which of the following statements presents a higher risk when a lubricator riser or extension is used?**

Answer

- C** (A) Leak between rams on wireline valve
- (B) Leak on wing valve
- (C) Unseen leak between swab valve and wireline valve
- (D) Leak in the wireline control head

5

OWST3.6 **What is the purpose of the braided line rams within a wireline valve?**

Answer

- C** (A) Provides a positive seal around the wire while running in the hole
- (B) Provides a means to cut the wire
- (C) Makes a seal around a stationary wire
- (D) Provides means to splice the braided cable

6

OWST3.7 **According to the workover program, when testing the integrity of the casing to 3,500 psi, we can expect a compressibility of 3 bbls. After the test is completed, what is the expected volume after bleed off?**

Answer

- B** (A) 1.3 bbl
- (B) 3 bbl
- (C) 4.7 bbls
- (D) None of the above

7

OWST3.8 **For wireline operations, why is a stuffing box used?**

Answer

- B** (A) Keep grease from blowing onto the location
- (B) Dynamic and static seal around wire which contains well pressure
- (C) Contains well pressure and provides only a static seal around wire
- (D) Dynamic and static seal around wire which allows well pressure to be bled off

8

OWST3.9 **What is the purpose of a check valve in a BHA as an internal primary barrier?**

Answer

- B** (A) Allow fluid and pressure flow into the coiled tubing
- (B) Prevent fluids from flowing from the well into the coiled tubing
- (C) Allows for fluid circulation above the BHA
- (D) Allows two-way flow as determined by differential pressure

9

OWST3.10 **What is proper statement which describes the primary reason to perform a function test along with pressure testing a wireline valve upon installation?**

Answer

- A** (A) Verify ram functionality and seal integrity
- (B) Verify ram functionality, seal integrity, and shear value
- (C) Verify seal integrity and accumulator capability
- (D) Verify seal integrity of the stuffing box only

10

OWST3.11 **Exceeding the tubing tensile strength of any tubular due to frictional forces will result in the following**

Answer

- C** (A) Reduction in fluid weight
- (B) Loss of circulation
- (C) Necking down
- (D) Ballooning of tubing

11

OWST3.12 **Which statement best describes what is needed to successfully perform fishing operations?**

Answer

- B** (A) Rotating overshot with internal grapple
- (B) Dimensions of fish, size and length of lubricator, proper fishing tool, proper rams for wire
- (C) Magnetic fishing tool, proper rams for wire, size of fish
- (D) Length from blind ram to rotary table, flange size, quick test sub, magnetic fishing tool

12

OWST3.13 **Which statement best describes how to secure a well without a tree installed and after the wireline is sheared?**

Answer

- A** (A) Closing blind ram
- (B) Closing pipe rams
- (C) Pumping kill weight fluid
- (D) Closing annular ram

13

OWST3.14 **Which statement best describes the most important consideration when determining whether wellhead or Christmas tree bending stress analysis is required?**

Answer

- C** (A) Type of tree connection
- (B) Bottom sheave placement
- (C) Lubricator height
- (D) Depth and angle of the well

14

OWST3.15 **For wireline operations, what is the function of the quick test sub?**

Answer

- C** (A) It keeps the pressure from going too high on the lubricator
- (B) Used to perform initial pressure test
- (C) It allows pressure testing integrity of the connection without testing the entire lubricator assembly
- (D) It ensures no cross contamination between testing fluids and wellbore fluids

15

OWST3.16 **When would we need to close the shear seal ram?**

Answer

- A** (A) Leak at connection between production tree and wireline valve
- (B) Leak at stuffing box
- (C) Leak at lubricator
- (D) No wireline in well

16

OWST3.17 **If the coiled tubing is near surface, what is the proper response for a leak in the coiled tubing between the gooseneck and stripper?**

Answer

- B** (A) Turn the pumps off and close the slip rams
- (B) Turn off the pumps and run the leak below the stripper and pipe rams
- (C) Turn off the pumps and close the blind/shear ram
- (D) Turn off the pumps and immediately come out of the hole

17

OWST3.18 **What is the proper response when the coiled tubing is near the bottom of the well and the stripper assembly fails?**

Answer

- A** (A) Shut in pipe rams
- (B) Pull out of hole
- (C) Speed up pumps
- (D) Shut in the blind/shear or cutter rams

18

OWST3.19 **For coiled tubing operations, which statement best describes when two strippers are installed?**

Answer

- C** (A) When using an internal tapered work string
- (B) Long BHA length
- (C) When dealing with corrosive environments
- (D) Short BHA Length

19

OWST3.20 **If the coiled tubing in near the bottom of the well, what is the proper response for a coiled tubing leak between the gooseneck and stripper?**

Answer

- A** (A) Secure well
- (B) Pump high pressure Nitrogen
- (C) Close blind rams
- (D) Lock injector

20

OWST3.21 **Under what conditions is the installation of a blind/shear ram to be added to the coiled tubing BOP stack?**

Answer

- C** (A) As a part of any normal closure sequence
- (B) During fishing operations
- (C) Additional blind shears are recommended by API when coiled tubing is run without BPVs
- (D) Blind/Shear rams are required on all coiled tubing operations

21

OWST3.22 **For coiled tubing operations with coiled tubing in the hole, which statement describes how to secure the well?**

Answer

- B** (A) Upper master valve
- (B) Emergency Shut-Down (ESD)
- (C) Injector head
- (D) Lower master valve

22

OWST3.23 **Which of the following equipment descriptions provides sealing as a tool joint is passed through it?**

Answer

- C** (A) Stripper Ram
- (B) Safety ram
- (C) Annular
- (D) Master valve

23

OWST3.24 **For coiled tubing operations, which statement best describes the function of strippers?**

Answer

- B** (A) Keep grease from blowing onto the location
- (B) Create a seal around tubing and allow for movement while the well has pressure on it
- (C) Provide shearing and sealing capability
- (D) Used as a frictional brake during operations

24

OWST3.25 **What is the proper response when the coiled tubing is near the bottom of the well and a leak develops between the gooseneck and the reel?**

Answer

- A** (A) Shut off pumps and check for flow
- (B) Go in the hole
- (C) Pump faster
- (D) Pump slower

25

OWST3.26 **For coiled tubing operations, which statement best describes when two strippers are installed?**

Answer

- C** (A) When using a tapered work string
- (B) Long BHA length
- (C) High pressure environments
- (D) Sour service wells

26

OWST3.27 **For coiled tubing operations with coiled tubing in the hole, which of the following equipment is used to secure a well.**

Answer

- B** (A) Upper master valve
- (B) Pipe ram
- (C) Lower master valve
- (D) Blind ram

27

OWST3.28 **For snubbing operations, which statement best describes the most important consideration when determining whether wellhead or Christmas tree bending stress analysis is required?**

Answer

- C**
- (A) High well fluid weight and viscosity
 - (B) Any time tubulars are snubbed into the hole
 - (C) Fishing jobs with significant dynamic loads
 - (D) High pressure, high temperature

28

OWST3.29 **During well control operations while snubbing, what is the proper response for a power or hydraulic failure?**

Answer

- D**
- (A) Pull out of the hole with BOP's open for faster response
 - (B) Switch snubbing controls to BOP accumulator unit
 - (C) Switch controls to secondary HPU (Hydraulic Power Unit)
 - (D) Close safety BOPs that are controlled by closing unit (accumulator unit)

29

OWST3.30 **If tubing integrity is exceeded due to frictional forces during snubbing operations, what impact will this have on tubulars?**

Answer

- A**
- (A) Pipe buckling
 - (B) Incorrect pipe tally
 - (C) Reduced pump rate through tubing
 - (D) Pipe collapse

30

OWST3.31 **During snubbing operations, what is the proper response for a leak in a stripper ram?**

Answer

- C**
- (A) Close annular BOP
 - (B) Close guide ram
 - (C) Close safety ram BOP
 - (D) Close slip rams

31

OWST3.32 **For snubbing operations, which statement best describes an internal snubbing barrier?**

Answer

- D** (A) Shear ram
- (B) Stripper ram
- (C) Annular
- (D) Back Pressure or Non-Return Valve

32

OWST3.33 **For snubbing operations, which statement best describes an internal snubbing barrier?**

Answer

- C** (A) Annular
- (B) Fluid weight
- (C) Stab-in/safety valve
- (D) Shear ram

33

OWST3.34 **What is the maximum well working pressure for an annular preventer used for snubbing or stripping operations for a 10,000 psi BOP stack?**

Answer

- C** (A) 5,000 psi
- (B) 10,000 psi
- (C) 2,000psi
- (D) 15,000 psi

34

OWST3.35 **For snubbing operations, which statement best defines the use of a safety ram?**

Answer

- D** (A) To close and shear pipe with the well under pressure
- (B) To close and seal around a range of pipe sized including tool joints
- (C) To close and move pipe into the well under pressure
- (D) As a backup barrier to allow repairs to stripping rams or annular

35

OWST3.36 **For snubbing operations, which statement best describes the use for a stripping ram?**

Answer

- A** (A) To move pipe into the well when under pressure
- (B) To close and use as a backup when repairing the master rams
- (C) To close and seal around a range of pipe sized including tool joints
- (D) To close and shear pipe with the well under pressure

36

OWST3.37 **For coiled tubing operations, which statement greatly affects the collapse pressure rating of the coiled tubing?**

Answer

- D** (A) Wellbore pressures
- (B) Surface temperatures
- (C) Fluid viscosity
- (D) Corrosion/Erosion

37

OWST3.38 **What measurement technique ensures tubing quality control?**

Answer

- C** (A) Visual inspection only
- (B) Weight measurement exclusively
- (C) Physical measurements with precision calipers and sophisticated laser techniques
- (D) Pressure testing only

38

OWST3.39 **What is NOT a typical remedial operation during snubbing on live wells?**

Answer

- D** (A) Mechanical failure repairs
- (B) Water production isolation
- (C) Tubing and completion equipment replacement
- (D) Continuous automated well monitoring systems

39

OWST3.40 **Which is NOT a standard well control requirement?**

Answer

- D** (A) Multiple barrier implementation
- (B) Personnel communication and training
- (C) Regular equipment testing
- (D) Continuous wellbore pressure increase

40

OWST3.41 **Which apply to barrier verification?**

Answer

- D** (A) Systematic leak detection and identification
- (B) Structural integrity assessment of equipment
- (C) Confirmation of proper valve positioning
- (D) All the above

41

OWST3.42 **Which elements must be considered in design?**

Answer

- D** (A) Wellbore configuration and geometry
- (B) Hydraulically applied loads from pressures
- (C) Only configuration needed
- (D) Both A and B

42

OWST3.43 **Which apply to wireline equipment?**

Answer

- D** (A) Only breaking strength
- (B) Breaking strength of selected wireline
- (C) Proper installation of BOP systems
- (D) Both B and C

43

OWST3.44 **In snubbing operations, when are spherical preventers typically used as strippers?**

Answer

- D** (A) Wells operating above 2,000 PSI
- (B) Wells operating above 5,000 PSI
- (C) All pressure ranges uniformly
- (D) Wells operating under 2,000 PSI

44

OWST3.45 **Which design consideration is critical for floating rig coiled tubing operations?**

Answer

- D** (A) Minimum reel diameter
- (B) Maximum pump capacity
- (C) Highest operating pressure
- (D) Heave compensation equipment to counteract rig movement

45

OWST3.46 **Despite opposing force directions in pipe light and pipe heavy, why is snubbing still required?**

Answer

- D** (A) For monitoring purposes only
- (B) For pressure reduction exclusively
- (C) For increasing circulation rates
- (D) To provide necessary control mechanism managing differential forces safely

46

OWST3.47 **The Drillers Method requires recording which specific pressures at the start of circulation?**

Answer

- A** (A) Shut-in Coiled Tubing Pressure (SICTP) and Shut-in Casing Pressure (SICP)
- (B) Circulation pressure and formation pressure only
- (C) Annular pressure and tubing pressure combined
- (D) Surface pressure and pump pressure exclusively

47

OWST3.48 **What critical step must follow After closing Safety Rams?**

Answer

- C** (A) Increase wellbore pressure
- (B) Activate emergency shutdown immediately
- (C) Bleed off pressures above Upper Safety Ram before stripper maintenance
- (D) Increase pump rates

48

OWST3.49 **In snubbing operations, when are spherical preventers typically used as strippers?**

Answer

- D** (A) Wells operating above 2,000 PSI
- (B) Wells operating above 5,000 PSI
- (C) All pressure ranges uniformly
- (D) Wells operating under 2,000 PSI

49

OWST3.50 **How does hydraulic stuffing box operation differ from manual stuffing box?**

Answer

- B** (A) It requires no sealing elements
- (B) Hydraulic pressure is applied automatically to create and maintain pressure seal
- (C) It cannot withstand high pressures
- (D) It has no operator control mechanism

50

OWST3.51 **At the balance point during snubbing, what condition exists?**

Answer

- D** (A) Pipe weight exceeds wellbore pressure
- (B) Wellbore pressure far exceeds pipe weight
- (C) External forces dominate all conditions
- (D) Gravitational force of pipe string exactly equals upward force from wellbore pressure

51

OWST3.52 **The Drillers Method requires recording which specific pressures at the start of circulation?**

Answer

- A** (A) Shut-in Coiled Tubing Pressure (SICTP) and Shut-in Casing Pressure (SICP)
- (B) Circulation pressure and formation pressure only
- (C) Annular pressure and tubing pressure combined
- (D) Surface pressure and pump pressure exclusively

52

OWST3.53 **What is the purpose of using gradually tapered insertion neck in weld connections?**

Answer

- C** (A) To increase connection strength by 50 percent
- (B) To reduce welding time and cost
- (C) To provide support preventing concentration of bending loads at single point
- (D) To facilitate field assembly operations

53

OWST3.54 **What is the primary advantage of braided wireline in well interventions?**

Answer

- C** (A) Cannot be sheared by rams
- (B) Eliminates pressure control requirements
- (C) Provides enhanced flexibility and reduced diameter for certain well geometries
- (D) Increases circulation rates exclusively

54

OWST3.55 **What is the primary function of the Lubricator in wireline well control operations?**

Answer

- A** (A) Dynamic Pressure Control Equipment creating sealed pathway allowing safe passage of wireline through pressure barrier
- (B) To increase circulation rates exclusively
- (C) To reduce wellbore pressure
- (D) To monitor wire tension

55

OWST3.56 **What is the primary purpose of pre-job communication in workover operations?**

Answer

- A** (A) To ensure all stakeholders are aligned on operational plans, roles, and associated risks
- (B) To reduce personnel numbers
- (C) To simplify operational procedures
- (D) To eliminate equipment requirements

56

OWST3.57 **Which describe Wait and Weight Method?**

Answer

- D** (A) Kill weight fluid calculated before circulation starts
- (B) Only calculation
- (C) Step-down chart followed during displacement
- (D) Both A and C

57

OWST3.58 **What condition occurs with External slip type connections during high stress?**

Answer

- D** (A) They become stronger
- (B) They require additional lubrication
- (C) They automatically adjust
- (D) Sharp spiraled teeth grip outer diameter (OD) of tube body for pressure seal

58

OWST3.59 **What percentage ovality is accepted by industry standards for normal manufacturing?**

Answer

- D** (A) 2% ovality
- (B) 3% ovality
- (C) 8% ovality
- (D) Approximately 5% ovality from normal manufacturing processes

59

OWST3.60 **What condition most accurately describes pipe light situations?**

Answer

- B** (A) When wellbore pressure equals zero
- (B) When weight of buoyant workstring is less than upward force from wellbore pressure
- (C) When circulation is stopped completely
- (D) When pipe weight exceeds upward forces

60

OWST3.61 **Where is the thickest diameter of tapered coiled tubing normally located?**

Answer

- A** (A) At the inner end of the reel at the top of the well where tensile forces are greatest
- (B) At the outer end of the reel where tubing enters the well
- (C) In the middle section for balanced load distribution
- (D) Along the entire length uniformly

61

OWST3.62 **Which factors affect coiled tubing deployment?**

Answer

- D** (A) Only pressures affect coiled tubing displacement
- (B) Operating pressures and wellbore conditions
- (C) Support structures and anti-buckling guides
- (D) Both B and C

62

OWST3.63 **What dangerous scenario occurs during pipe light conditions if unrestrained?**

Answer

- B** (A) The pipe settles slowly to the bottom
- (B) The pipe is dangerously ejected from the well creating significant safety hazard
- (C) The pipe remains suspended indefinitely
- (D) The pipe gently rises at controlled rate

63

OWST3.64 **What is NOT a standard component of coiled tubing equipment systems?**

Answer

- D** (A) Services Power Reel
- (B) Hydraulic Power Pack
- (C) Injector Head
- (D) Automated decompression chamber

64

OWST3.65 **Where is the critical depth for buckling prevention in wells?**

Answer

- B** (A) Only in vertical sections
- (B) Deeper wells where increased coiled tubing length establishes buckling conditions
- (C) Only in shallow wells
- (D) Only in atmospheric pressure wells

65

OWST3.66 **What is the primary distinction between coiled tubing and conventional pipe?**

Answer

- B** (A) Conventional pipe is always stronger
- (B) Coiled tubing maintains continuous circulation while conventional pipe requires connections
- (C) They perform identical functions
- (D) Coiled tubing cannot be used on live wells

66

OWST3.67 **What are the primary components of dynamic wireline pressure control equipment?**

Answer

- A** (A) Control heads (Dynamic PCE) including manual and hydraulic stuffing boxes and grease injection systems
- (B) Ram assemblies exclusively
- (C) Blind rams and shear rams only
- (D) Flow spools and equalizing valves

67

OWST3.68 How do operators evaluate well integrity in workover operations?

Answer

- A** (A) By meticulously reviewing historical well records, diagnostic test results, and real-time data
- (B) Through pressure readings only
- (C) Only from surface pressure monitoring
- (D) By well depth alone

68

OWST3.69 What is the primary purpose of the Bullhead Method?

Answer

- A** (A) Using high-pressure pumps to force heavier kill fluid to displace wellbore fluids
- (B) Gradual pressure reduction through circulation
- (C) Maintaining constant circulation rate
- (D) Reducing tubing weight

69

OWST3.70 Which elements must be verified during shut-in?

Answer

- D** (A) Systematic leak detection across system
- (B) Only leak detection needed
- (C) Comprehensive valve positioning confirmation
- (D) Both A and C

70

OGOR Workover Study Guide Day 1

OWO1.1 **What is the reason to consider a negative pressure test on a packer or liner lap?**

Answer

- A** (A) It tests in the direction of flow
- (B) Negative integrity tests are no longer carried out
- (C) Packers can only be tested in a negative direction
- (D) Packers can only be tested in a positive direction

1

OWO1.2 **What reduces friction as tubing passes through the injector system?**

Answer

- C** (A) Fluid lubrication
- (B) Reduced pump pressure
- (C) Integrated cam rollers in the gripper block assembly
- (D) Manual guidance

2

OWO1.3 **Why are polished bore receptacles (PBR) and seal units used?**

Answer

- D** (A) Allows lower circulation velocities
- (B) Resists H₂S embrittlement
- (C) Allows higher circulation velocities
- (D) Allows for tubing movement with changes in pressure, temperature, and flow

3

OWO1.4 **According to API 16ST, what is the appropriate action if a leak develops between the tree flange and the well control stack?**

Answer

- A** (A) Kill well
- (B) Pump lost circulation material (LCM)
- (C) Continue operations
- (D) Proceed to Muster Point

4

OWO1.5 **What gases are used to energize workover and well service completion fluids?**

Answer

- C** (A) Air
- (B) Helium
- (C) CO₂
- (D) H₂S

5

OWO1.6 **What are typical coiled tubing applications?**

Answer

- D** (A) Structural support only
- (B) Pressure testing only
- (C) Data collection only
- (D) Well cleanouts, stimulation, drilling, and production enhancement

6

OWO1.7 **What well control risk could arise if a well has the potential to form hydrates?**

Answer

- D** (A) Hydrates circulated to surface may catch fire
- (B) Hydrates can cause clay formations to swell
- (C) Blockages can cause well to flow
- (D) Trapped pressure below hydrate giving incorrect surface pressure reading

7

OWO1.8 **When would a reduction in annular regulator closing pressure occur?**

Answer

- C** (A) While making repairs to the production tree
- (B) During wireline operations
- (C) While recovering casing from a well under pressure
- (D) While making repairs to the Annular Preventer

8

OWO1.9 **What provides multiple redundant barrier elements in coiled tubing?**

Answer

- D** (A) Surface pipelines
- (B) Fluid circulation only
- (C) Pressure gauges
- (D) Quad Blowout Preventer System

9

OWO1.10 **What is MAWP in equipment rating?**

Answer

- B** (A) Minimum Allowable Working Pressure
- (B) Maximum Allowable Working Pressure of the lowest rated component
- (C) Measured Absolute Water Pressure
- (D) Maximum Annual Work Period

10

OWO1.11 **How much do excessive loads count toward service life derating?**

Answer

- C** (A) 1 cycle equivalent
- (B) 3 cycle equivalents
- (C) 2 cycle equivalents
- (D) 5 cycle equivalents

11

OWO1.12 **What allows Tandem Stripper replacement while maintaining service?**

Answer

- D** (A) Complete disassembly required
- (B) Pressure venting alone
- (C) Manual removal only
- (D) Split inserts allowing replacement without complete system disassembly

12

OWO1.13 **What typical service life cycle limitation does coiled tubing have?**

Answer

- D** (A) Approximately 50 cycles
- (B) Unlimited cycles
- (C) One-time use only
- (D) Approximately 80 cycles per standard tubing string

13

OWO1.14 **Which statement is NOT a characteristic of an annular preventer?**

Answer

- D** (A) Annular pressures can be adjusted when stripping tooljoints through element
- (B) Close on a range of ODs
- (C) Hydraulically actuated
- (D) Instant actuation

14

OWO1.15 **What special feature do Quad Stacks have for emergency situations?**

Answer

- B** (A) Pressure relief only
- (B) Manual override capability for emergency situations
- (C) Electronic shutdown
- (D) Automatic closure only

15

OWO1.16 **What does a sub-surface safety valve failure require as a secondary barrier?**

Answer

- A** (A) Mechanical or fluid barrier
- (B) Another valve
- (C) Pressure regulator
- (D) Pump system

16

OWO1.17 **What can compromise fluid barrier effectiveness during operations?**

Answer

- B** (A) Reduced pump speed
- (B) Fluid losses in the wellbore during circulation
- (C) Lower density
- (D) Decreased viscosity only

17

OWO1.18 **What is the minimum operating pressure for accumulators?**

Answer

- B** (A) Zero psi
- (B) Typically 1,200 to 1,500 psi preventing complete discharge
- (C) Maximum only
- (D) Variable only

18

OWO1.19 **What occurs when the salt concentration of a brine is increased beyond the saturation level?**

Answer

- A** (A) Salt will drop out of the solution
- (B) Viscosity will begin to decrease
- (C) Freezing point will be affected
- (D) Boiling point will be affected

19

OWO1.20 **What can enhance the carrying capacity of completion fluid during milling processes such as milling a packer?**

Answer

- D**
- (A) Asphlatines
 - (B) Density
 - (C) Crystallization point
 - (D) Annular velocity

20

OWO1.21 **Is _____, maintenance, inspection, and _____ of well barriers are essential components in managing well integrity?**

Answer

- A**
- (A) verification, testing
 - (B) installation, defeating
 - (C) installation, removal
 - (D) testing, installation

21

OWO1.22 **What is critical for emergency response preparedness?**

Answer

- B**
- (A) Theory only
 - (B) Regular practice and drilling until crews are proficient before operations begin
 - (C) Verbal training alone
 - (D) Written procedures only

22

OWO1.23 **What is the core advantage of coiled tubing operations?**

Answer

- A**
- (A) Reduced cost and high portability
 - (B) Increased well depth only
 - (C) Simplified procedures
 - (D) Manual operation only

23

OWO1.24 **What determines the MAWP of a surface system?**

Answer

- B** (A) Highest rated component
- (B) Lowest pressure rated component in the system
- (C) Average of all components
- (D) Operator preference

24

OWO1.25 **What will happen at the bottom of a well if we reduce hydrostatic pressure?**

Answer

- C** (A) Increase permeability
- (B) Formation fracturing
- (C) Undesired flows
- (D) Increase porosity

25

OWO1.26 **What advantage does tapered coiled tubing provide?**

Answer

- D** (A) Uniform weight throughout
- (B) Reduced pressure capacity
- (C) Simplified connections
- (D) Optimized strength with reduced overall weight

26

OWO1.27 **What is the primary function of an accumulator in BOP systems?**

Answer

- B** (A) Fluid storage only
- (B) Provides emergency pressure for BOP operation if main system fails
- (C) Temperature control
- (D) Data logging only

27

OWO1.28 **Which condition does not affect requirements to be a barrier?**

Answer

- C** (A) Sufficient hydrostatic pressure
- (B) Sufficient fluid density
- (C) Sufficient viscosity
- (D) Sufficient porosity

28

OWO1.29 **Referencing a BOP Closing unit (Accumulator), which test provides a physical verification of the unit is performance to ensure the unit has sufficient pressure and hydraulic fluid volumes?**

Answer

- C** (A) BOP function test
- (B) Total System Check
- (C) BOP Accumulator Drawdown Test
- (D) Pneumatic function test

29

OWO1.30 **What does the Injector Head accomplish?**

Answer

- D** (A) Measures tubing strength
- (B) Records pressure data
- (C) Calculates fluid volumes
- (D) Delivers surface driving force to push/pull tubing string

30

OWO1.31 **When is a BOP drawdown test is required, according to API Standard 53?**

Answer

- D** (A) The accumulator drawdown test shall be performed every 14 days
- (B) The accumulator drawdown test shall be performed every 21 days
- (C) The accumulator drawdown test shall be performed every 7 days
- (D) After any repairs that required isolation/partial isolation of the system

31

OWO1.32 **What do Telescopic Legs provide for coiled tubing equipment?**

Answer

- B** (A) Automatic shutdown
- (B) Enhanced operational stability during static and dynamic phases
- (C) Pressure control only
- (D) Data collection only

32

OWO1.33 **What is the function of Slip Rams in the Quad Stack?**

Answer

- D** (A) Cut tubing
- (B) Regulate pressure
- (C) Seal the wellbore
- (D) Hold tubulars in place during shearing operations

33

OWO1.34 **Is Bottom Hole Pressure (BHP) the combined total of fluid hydrostatic pressure, surface-applied pressure, and any friction pressure acting downward during circulation?**

Answer

- A** (A) TRUE
- (B) False
- (C)
- (D)

34

OWO1.35 **What determines if a mechanical barrier passes testing?**

Answer

- B** (A) Visual appearance
- (B) Ability to withstand all anticipated pressure conditions and maintain integrity
- (C) Weight only
- (D) Color of equipment

35

OWO1.36 **What signs suggest that the primary barrier has failed?**

Answer

- D** (A) H2S alarms are not set
- (B) Test fluid density with a scale
- (C) Increase in fluid temperature
- (D) Complete fluid losses to the well

36

OWO1.37 **What is the maximum pumping pressure when coiled tubing is stationary?**

Answer

- C** (A) 3,000 psi
- (B) 4,000 psi
- (C) 5,000 psi
- (D) 7,500 psi

37

OWO1.38 **What is the typical pumping pressure limit while moving coiled tubing?**

Answer

- D** (A) 2,000 psi
- (B) 5,000 psi
- (C) 10,000 psi
- (D) 4,000 psi

38

OWO1.39 **What optional connection capability do Quad Stacks provide?**

Answer

- B** (A) Single connection only
- (B) Ported connections for downhole fluid injection
- (C) Gravity connections only
- (D) Manual connections alone

39

OWO1.40 **What safety feature does the BOP Console incorporate?**

Answer

- B** (A) Single operator control
- (B) Dual control requirement preventing accidental operation
- (C) Automatic activation only
- (D) Manual-only activation

40

OWO1.41 **What is tubular collapse pressure?**

Answer

- B** (A) Pressure gauge reading
- (B) Maximum external pressure before tubular implodes
- (C) Internal pressure only
- (D) Temperature rating

41

OWO1.42 **In the barrier hierarchy, which barrier serves as the first line of defense?**

Answer

- C** (A) Secondary barrier
- (B) Tertiary barrier
- (C) Primary barrier
- (D) Temporary barrier

42

OWO1.43 **Based on the following statements, which statement best describes influx indicators?**

Answer

- C** (A) Increase in pump pressure, changes in fluid density, and change in string weight
- (B) Fluid loss with pump off, excess fill volume when tripping out
- (C) Decrease in pump pressure, change in string weight, and oil and gas shows during circulation
- (D) All the above

43

OWO1.44 **Why is crew proficiency of well control procedures and responsibilities related to detection, well shut-in and control important?**

Answer

- A** (A) Minimizes amount of exposure to the influx
- (B) Manages cost of the completion fluids
- (C) Ensures that drills and training only adress each crew member's specific role
- (D) Ensures proper handover procedures are understood

44

OWO1.45 **What is activated if a secondary barrier fails?**

Answer

- A** (A) Emergency response plan
- (B) Production operations
- (C) Normal circulation
- (D) Crew break

45

OWO1.46 **Which statement best describes a Top Drive Valve installed with a Top Drive?**

Answer

- C** (A) Exempt from stack pressure testing requirements
- (B) Holds pressure from one direction only
- (C) Are full open safety valves
- (D) Are high pressure, high temperature Inside Blowout Preventers (IBOP)

46

OWO1.47 **Which equations determine the potential force below a back pressure valve?**

Answer

- D** (A) Area x Tubing Weight
- (B) Pressure ÷ Area
- (C) Pressure x Tubing Weight
- (D) Pressure × Area

47

OWO1.48 **What component activates the Tandem Stripper sealing mechanism?**

Answer

- D** (A) Hydraulic pump
- (B) Pressure gauge
- (C) Manual lever
- (D) Energizer element

48

OWO1.49 **What redundancy systems does the BOP Console feature?**

Answer

- B** (A) Single system only
- (B) Multiple redundancy maintaining pressure control under all scenarios
- (C) Manual-only backup
- (D) Gravity backup alone

49

OWO1.50 **For a 3,000 psi BOP Closing Unit (Accumulator) with a 1,000 psi precharge pressure and 200 psi of the precharge pressure is lost, what happens to the usable fluid volume (assume minimum operating pressure remains at 1,200 psi)?**

Answer

- C** (A) Volume will increase
- (B) Volume will stay the same
- (C) Volume will decrease
- (D) None of the above

50

OWO1.51 **Referencing a BOP Closing unit (Accumulator), what will occur if a loss of precharge pressure is lost in one accumulator bottle?**

Answer

- A** (A) Rams and annular can still be functioned
- (B) Rams and annular cannot be functioned
- (C) Shear rams cannot be functioned
- (D) Annular preventer cannot be functioned

51

OWO1.52 **What does the BOP Console accumulator bank provide?**

Answer

- B** (A) Pressure measurement
- (B) Emergency operational capability even with power loss
- (C) Temperature control only
- (D) Flow rate indication alone

52

OWO1.53 **What is essential after installing the Stripper Console?**

Answer

- B** (A) Daily calibration only
- (B) Mandatory function testing to verify operational readiness
- (C) Visual inspection only
- (D) Manual operation confirmation alone

53

OWO1.54 **Which is an example of Static Pressure Control Equipment (PCE)?**

Answer

- C** (A) Rotating control device
- (B) Injection equipment
- (C) Blowout preventer or wellhead seal
- (D) Circulation pump

54

OWO1.55 **What is the correct method used to close a BOP from the remote BOP panel?**

Answer

- A** (A) Activate the air source and close the selected BOP components
- (B) Always function from Koomey, do not use remote panel
- (C) Consult the remote panel operations manual before taking any action to shut in the well
- (D) Close function, then activate air source

55

OWO1.56 **Which device is an example of Dynamic Pressure Control Equipment (PCE)?**

Answer

- C** (A) Wellhead seal
- (B) Blowout preventer
- (C) Rotating control device
- (D) Plug device

56

OWO1.57 **What information does the Operators Console provide?**

Answer

- B** (A) Only pressure readings
- (B) Weight indicators and real-time monitoring with data recording
- (C) Temperature only
- (D) Fluid density only

57

OWO1.58 **What is tubular burst pressure?**

Answer

- B** (A) Minimum pressure
- (B) Maximum internal pressure before tubular ruptures
- (C) External pressure only
- (D) Flow rate measurement

58

OWO1.59 **What does the Support Structure accomplish?**

Answer

- B** (A) Measures tubing length
- (B) Transfers loads to ground and provides leveling on uneven surfaces
- (C) Stores fluids only
- (D) Calculates pressures

59

OWO1.60 **What must be done if a primary barrier fails during operations?**

Answer

- A** (A) Immediately reinforce or replace it with another barrier element
- (B) Continue operations normally
- (C) Wait for next shift
- (D) Document and ignore

60

OWO1.61 **What is the typical maximum number of wall thickness sections in tapered tubing?**

Answer

- C** (A) Two sections
- (B) Three sections
- (C) Five to six sections
- (D) Eight sections

61

OWO1.62 **What is the advantage of the Side-Door Stripper design?**

Answer

- D** (A) Permanent closure
- (B) Reduced cost only
- (C) Increased pressure capacity
- (D) Improved accessibility for maintenance without complete disassembly

62

OWO1.63 **What crystalline compound can impede fluid flow and resembles snow or ice?**

Answer

- A** (A) Hydrates
- (B) Brine
- (C) H₂S
- (D) Diesel

63

OWO1.64 **What is used to seal the wellbore without pipe in the well?**

Answer

- D** (A) Annular preventer
- (B) Pipe ram
- (C) Slip ram
- (D) Blind ram

64

OWO1.65 **What should be monitored continuously in fluid barriers?**

Answer

- A** (A) Fluid losses
- (B) Pump efficiency
- (C) Tool weight
- (D) Ambient temperature

65

OWO1.66 **What reduces the effectiveness of fluid barriers during circulation?**

Answer

- C** (A) Lower pump rates
- (B) Increased flow
- (C) Fluid losses in the wellbore
- (D) Higher density

66

OWO1.67 **What material is used for wide temperature range Tandem Stripper applications?**

Answer

- C** (A) Nitrile elastomer
- (B) Viton elastomer
- (C) Urethane elastomer
- (D) Rubber polymer

67

OWO1.68 **Which statement best describes fluid gas separators?**

Answer

- A** (A) Typically, a low pressure vessel
- (B) Used to remove cuttings from fluid
- (C) Typically, a high pressure vessel
- (D) Typically, the tertiary separation component downstream of flare stack

68

OWO1.69 **What are positive kick indicators?**

Answer

- C** (A) Drilling break, change in background gas, flow with the pumps off
- (B) Pit gain, erratic torque and drag, increase in background gas
- (C) Flow with pumps off, increase in flow, pit gain
- (D) Everyone is at the Muster station

69

OWO1.70 **What is accumulator precharge pressure?**

Answer

- B** (A) Operating pressure
- (B) Initial gas pressure establishing energy capacity
- (C) Maximum pressure only
- (D) Flow rate indicator

70

OWO1.71 **Which type of barrier is a physical device like a packer or BOP?**

Answer

- C** (A) Fluid barrier
- (B) Operational barrier
- (C) Mechanical barrier
- (D) Procedural barrier

71

OWO1.72 **Who is the person-in-charge (PIC) during a well control event?**

Answer

- A** (A) Company man
- (B) Tool pusher
- (C) Rig Operator
- (D) Assistant Driller

72

OWO1.73 **Which statement best describes a Top Drive Valve installed with a Top Drive?**

Answer

- A** (A) Can normally hold pressure from both directions
- (B) Holds pressure from one direction only
- (C) Exempt from stack pressure testing requirements
- (D) Are high pressure, high temperature Inside Blowout Preventers (IBOP)

73

OWO1.74 **What pressure ratings do Quad and Dual Stacks provide?**

Answer

- C** (A) 2,000 to 5,000 psi
- (B) 5,000 to 10,000 psi
- (C) 5,000 to 15,000 psi
- (D) 10,000 to 20,000 psi

74

OWO1.75 **What do torsional loading cells monitor in the injector head?**

Answer

- C** (A) Tubing temperature
- (B) Fluid viscosity
- (C) Forces applied to the tubing string for optimal tension
- (D) Ambient conditions

75

OWO1.76 **During completions and workovers, what is the minimum number of independent barriers required?**

Answer

- A** (A) Two independent barriers
- (B) One barrier
- (C) Three independent barriers
- (D) Four independent barriers

76

OWO1.77 **What effect does temperature increase have on formation fluid movement?**

Answer

- B** (A) Decreases flow
- (B) Can increase influx risk requiring density adjustment
- (C) Stops flow
- (D) No effect

77

OWO1.78 **From the following list of annular sizes, which annulus would have the lowest frictional loss while forward circulating?**

Answer

- D** (A) 3-1/2" in OD tubing within 7 in OD casing
- (B) 3-1/2" in OD tubing within 9-5/8 in OD casing
- (C) 3-1/2" in OD tubing within 5 in OD casing
- (D) 3-1/2" in OD tubing within 13-3/8 in OD casing

78

OWO1.79 **What ram is designed to seal on pipe?**

Answer

- D** (A) Annular Preventer
- (B) Slip ram
- (C) Blind ram
- (D) Pipe ram

79

OWO1.80 **In what position is a FOSV stabbed into the well?**

Answer

- B** (A) Closed position
- (B) Open position allowing unrestricted flow initially
- (C) Neutral position only
- (D) Pressurized position

80

OWO1.81 **What creates a fluid barrier in a well?**

Answer

- C** (A) Air
- (B) Gas injection
- (C) Weighted brine or workover fluid
- (D) Cement paste

81

OWO1.82 **What does a positive pressure test on a barrier confirm?**

Answer

- B** (A) Pressure capacity upward
- (B) Pressure capacity downward
- (C) Wellbore stability only
- (D) Fluid viscosity

82

OWO1.83 **Which of the following statements do NOT cause de-rating tubular ratings?**

Answer

- A** (A) Internal plastic coating
- (B) Improper make-up torque and procedures
- (C) Sand erosion
- (D) Rust

83

OWO1.84 **What is the main factor which limits a pumping operation?**

Answer

- A** (A) Maximum Allowable Working Pressure
- (B) Formation Pressure
- (C) Maximum Anticipated Surface Pressure (MASP)
- (D) Maximum Anticipated Annular Surface Pressure (MAASP)

84

OWO1.85 **What is the correct name for the pressure check valve which is installed into a wellhead outlet to allow safe removal of a casing valve?**

Answer

- A** (A) Valve removal (VR) plug
- (B) Wireline plug
- (C) Two-way check valve
- (D) Gas lift mandrel

85

OWO1.86 **What non-sealable or non-shearable tools could potentially lead to shut-in problems?**

Answer

- C** (A) Running wireline through a BOP stack
- (B) Running production tubing across the BOP stack
- (C) Running a screen across the BOP stack
- (D) Running a perforation gun through BOP stack

86

OWO1.87 **What is the appearance of hydrate formations?**

Answer

- B** (A) Liquid
- (B) Crystalline compound resembling snow or ice
- (C) Gas bubble
- (D) Fluid suspension

87

OWO1.88 **Which type of salt will achieve a dramatic increase in density of a completion fluid?**

Answer

- C** (A) Nitrates
- (B) Chlorides
- (C) Bromides
- (D) Sea Salts

88

OWO1.89 **What material is used for low-wear Tandem Stripper applications?**

Answer

- C** (A) Viton elastomer
- (B) Urethane elastomer
- (C) Nitrile elastomer
- (D) Polyurethane

89

OWO1.90 **What does Static Pressure Control Equipment (PCE) maintain?**

Answer

- D** (A) Active circulation
- (B) Fluid density
- (C) Pump efficiency
- (D) Wellbore pressure containment when not circulating

90

OWO1.91 **What is the core component of coiled tubing equipment?**

Answer

- D** (A) Control panel only
- (B) Single hydraulic pump
- (C) Manual winch system
- (D) Services Power Reel and supporting systems

91

OWO1.92 **What is the importance of testing the pressure integrity of a mechanical barrier over predetermined period of time?**

Answer

- B** (A) To ensure that barrier components cannot withstand anticipated wellbore pressure
- (B) To ensure that barrier components can withstand anticipated wellbore pressure
- (C) To verify that certain barriers can hold pressure from one direction only
- (D) To verify pump capacity

92

OWO1.93 **What is the primary purpose of a barrier in well operations?**

Answer

- A** (A) To prevent unwanted fluid movement and maintain wellbore stability
- (B) To increase fluid viscosity
- (C) To reduce pump pressure
- (D) To increase well depth

93

OWO1.94 **What is the limitation of a blind ram?**

Answer

- A** (A) Blind rams cannot seal around pipe
- (B) Blind rams cannot seal an open hole
- (C) Blind rams can be used when testing casing
- (D) Blind ram cannot hold pressure both ways

94

OWO1.95 **What condition confirms secondary barrier has failed?**

Answer

- A** (A) Pressure loss
- (B) Pressure gain
- (C) Loss of hydrostatic pressure
- (D) Gain in hydrostatic pressure

95

OWO1.96 **How many barriers are necessary during well servicing operations?**

Answer

- A** (A) Two barriers
- (B) One barrier
- (C) Three barriers
- (D) Four barriers

96

OWO1.97 **What is considered a non-shearable component when positioned within the BOP stack?**

Answer

- D** (A) Nipple
- (B) Blank pipe
- (C) Tubing
- (D) Packer

97

OWO1.98 **What is the best way to reduce tubing movement during stimulation or testing procedures?**

Answer

- D** (A) Ensure tubing slips are engaged
- (B) Increase pump rate
- (C) Tubing won't move because it is anchored by tubing hangar and production packer
- (D) Applying annular pressure

98

OWO1.99 **For a 10,000 psi rated BOP stack, what is the maximum pressure the crew's should use to test the stack?**

Answer

- C** (A) 7,500 psi
- (B) 5,000 psi
- (C) 10,000 psi
- (D) 12,500 psi

99

OWO1.100 **What advantage does coiled tubing provide in well operations?**

Answer

- D** (A) Increased crew size
- (B) Reduced safety requirements
- (C) Manual power only
- (D) Ability to work in pressurized wellbores without killing the well

100

OWO1.101 **What mechanism enhances the Stripper Packing seal?**

Answer

- D** (A) External weights
- (B) Gravity alone
- (C) Reduced pressure settings
- (D) Well-assisted closure mechanism leveraging wellbore pressure

101

OWO1.102 **What is the function of the Tubing Arch Guide?**

Answer

- B** (A) Measures tubing diameter
- (B) Supports tubing allowing sufficient 90-degree bending radius
- (C) Stores fluids
- (D) Calculates pressure

102

OWO1.103 **Where is the Stripper Packing positioned?**

Answer

- C** (A) At surface
- (B) Downhole in wellbore
- (C) Below the injector head connecting to riser or BOPs
- (D) On the reel assembly

103

OWO1.104 **What allows fluid pumping during coiled tubing deployment?**

Answer

- D** (A) Secondary storage tank
- (B) Flexible hose system
- (C) Top drive system
- (D) Swivel connection on reel

104

OWO1.105 **What is the minimum bending radius for the Tubing Arch Guide?**

Answer

- C** (A) 10 times the outer diameter
- (B) 20 times the outer diameter
- (C) 30 times the outer diameter
- (D) 50 times the outer diameter

105

OWO1.106 **What indicates a well has become secure after shut-in?**

Answer

- B** (A) Increase in pit volume
- (B) No visible leaks at surface and pressure readings stabilize
- (C) Pressure increases
- (D) Temperature increases

106

OWO1.107 **What is occurring below the annular preventer, if a workover crew has been stripping the work string out of the hole and the tool joint reaches the annular preventer?**

Answer

- C** (A) Force has no effect on work string
- (B) Force on work string is decreasing
- (C) Force on work string is increasing
- (D) None of the above

107

OWO1.108 **Emergency response plans should be practiced and drilled until crews are proficient before beginning a well operation?**

Answer

- A** (A) True
- (B) False
- (C)
- (D)

108

OWO1.109 **Which statement describes a water-based fluid which is NOT considered to be a common workover/well service completion fluid?**

Answer

- D** (A) Sodium Chloride
- (B) Zinc Bromide
- (C) Calcium Chloride
- (D) Fresh water with no additives

109

OWO1.110 **What factors affect actual coiled tubing running speeds?**

Answer

- C** (A) Crew experience only
- (B) Equipment manufacturer recommendations only
- (C) Specific wellbore conditions, pressure differentials, and mechanical properties
- (D) Safety regulations only

110

OWO1.111 **What must the PIC ensure during emergency situations?**

Answer

- B** (A) Crew comfort
- (B) Personnel safety and implementation of emergency response procedures
- (C) Minimal documentation
- (D) Quick return to normal operations

111

OWO1.112 **What prevents mistakes in operational barriers?**

Answer

- D** (A) Peer review
- (B) Equipment redundancy
- (C) Automation alone
- (D) Correct procedures and personnel training

112

OWO1.113 **What does the Injector Head roller assembly accomplish?**

Answer

- C** (A) Calculates tubing depth
- (B) Measures fluid pressure
- (C) Guides tubing while minimizing surface damage
- (D) Records operational data

113

OWO1.114 **What can be added to a completion fluid to reduce fluid loss?**

Answer

- C** (A) Diesel
- (B) Antifreeze
- (C) Calcium Carbonate
- (D) Nut plug

114

OWO1.115 **True or False. Is the rated working pressure of surface equipment is lower than the maximum allowable working pressure due to design safety factors?**

Answer

- A** (A) TRUE
- (B) FALSE
- (C)
- (D)

115

OWO1.116 **What is included in pre-job safety measures for operational barriers?**

Answer

- D** (A) Cost cutting
- (B) Equipment upgrades only
- (C) Management approval only
- (D) Pre-job meetings and safety briefings

116

OWO1.117 **What is essential for Static and Dynamic Pressure Control Equipment (PCE)?**

Answer

- D** (A) Daily inspections only
- (B) Manual operation only
- (C) Automatic shutdown only
- (D) Regular maintenance, testing, and monitoring

117

OWO1.118 **Which statement best describes fluid gas separators?**

Answer

- D** (A) Typically, the tertiary separation component downstream of flare stack
- (B) Used to remove cuttings from fluid
- (C) Typically, a high pressure vessel
- (D) Typically, the first separation component downstream of the choke

118

OWO1.119 **An increase in differential pressure across the tubing reduces the string's allowable tensile load.**

Answer

- A** (A) TRUE
- (B) FALSE
- (C)
- (D)

119

OWO1.120 **What is the function of the Stripper Packing assembly?**

Answer

- C** (A) Measures fluid density
- (B) Regulates pump speed
- (C) Critical pressure control component withstanding 5,000 to 15,000 psi
- (D) Stores hydraulic fluid

120

OWO1.121 **What are operational-procedural barriers based on?**

Answer

- D** (A) Equipment only
- (B) Fluid density
- (C) Pressure ratings
- (D) Human factors, written procedures, and engineered safety standards

121

OWO1.122 **What non-sealable or non-shearable tools could potentially lead to shut-in problems?**

Answer

- A** (A) Running a production packer across the BOP stack
- (B) Running production tubing across the BOP stack
- (C) Running wireline through a BOP stack
- (D) Running a perforation gun through BOP stack

122

OWO1.123 **Which statement best describes a Top Drive Valve installed with a Top Drive?**

Answer

- D** (A) Are high pressure, high temperature Inside Blowout Preventers (IBOP)
- (B) Holds pressure from one direction only
- (C) Exempt from stack pressure testing requirements
- (D) Can be used to shut off flow from the work string

123

OWO1.124 **What characterizes a hydrate formation?**

Answer

- B** (A) Gas only
- (B) Water and low boiling point gas combine into crystalline compound
- (C) Oil mixture
- (D) Pressure increase alone

124

OWO1.125 **As the temperature increases downhole, what happens to the fluid barrier?**

Answer

- C** (A) Increase in hydrostatic pressure
- (B) Temperature has no effect on a fluid barrier
- (C) Decrease hydrostatic pressure
- (D) Increased temperatures bake the barrier and increases it's physical sealing properties.

125

OWO1.126 **During circulation operations, which sensor locations is most likely to identify the presence of H2S first?**

Answer

- A** (A) Flow line
- (B) Crown
- (C) Cellar
- (D) Fluid gas separator

126

OWO1.127 **What is the advantage of having multiple barriers?**

Answer

- A** (A) A backup for a failed barrier system
- (B) Reduces need for visual checks
- (C) Multiplies production from the well
- (D) None of the above

127

OWO1.128 **Which statement best describes fluid gas separators?**

Answer

- A** (A) Used to remove large volume of gas from the fluid
- (B) Used to remove cuttings from fluid
- (C) Typically, a high pressure vessel
- (D) Typically, the tertiary separation component downstream of flare stack

128

OWO1.129 **What feature prevents the gripper blocks from damaging the coiled tubing?**

Answer

- C** (A) Rubber coating
- (B) Plastic sleeves
- (C) Serrated slips engineered with precisely calculated pressure
- (D) Lubricant film

129

OWO1.130 **What occurs at the surface, when running sub-surface safety valve into the well and control line pressure is lost?**

Answer

- C** (A) The SCSSV will remain open
- (B) Additional surface fluid gain due to SCSSV fail safe open
- (C) Additional surface fluid gain due to SCSSV fail safe closed
- (D) Reduction in fluid gain due to the SCSSV fail safe closed

130

OWO1.131 **What bore sizes are typical for Quad Stack systems?**

Answer

- D** (A) 1-1/16 to 3-1/16
- (B) 7-1/16 to 11-1/16
- (C) 5-1/16 to 9-1/16
- (D) 3-1/16 to 7-1/16

131

OWO1.132 **True or False. Is the rated working pressure of surface equipment always equal to maximum allowable working pressure.**

Answer

- B** (A) True
- (B) False
- (C)
- (D)

132

OWO1.133 **What advantage does the Stripper Packing design provide?**

Answer

- C** (A) Permanent installation
- (B) Manual operation only
- (C) Top-down servicing design allowing maintenance without complete disassembly
- (D) Emergency shutdown only

133

OWO1.134 **Bowen / Quick-Type connections should always be fastened by?**

Answer

- A** (A) By hand
- (B) Pipe Wrench
- (C) Torqued to 8,500 ft-lbs
- (D) Crescent Wrench

134

OWO1.135 **What is the primary responsibility of the PIC during well control?**

Answer

- B** (A) Cost monitoring
- (B) Coordination and command of all well control operations
- (C) Equipment maintenance only
- (D) Data recording alone

135

OWO1.136 **During workover operations, what fluid properties are considered when pressure testing a deep-set plug?**

Answer

- A** (A) Fluid compressibility
- (B) Surface Volume Limitation
- (C) Fluid viscosity
- (D) Fluid temperatures

136

OWO1.137 **Which of the following statements best describes why the Full Opening Safety Valve (FOSV) is used?**

Answer

- D** (A) Can be used as a choke to apply variable flows from the work string.
- (B) Can be used to shut off flow from the annulus
- (C) Can be used to shut off flow from the flowline
- (D) Can be used to shut off flow from the work string

137

OWO1.138 **What is the Control Cabin in coiled tubing operations?**

Answer

- D** (A) Crew quarters
- (B) Storage facility
- (C) Emergency shelter
- (D) Operational nerve center where technicians direct intervention

138

OWO1.139 **What benefit does the Side-Door Stripper provide during maintenance?**

Answer

- D** (A) Increased complexity
- (B) Equipment replacement only
- (C) Manual operation only
- (D) Significantly simplified procedures with reduced downtime

139

OWO1.140 **What is the Tandem Stripper used for?**

Answer

- D** (A) Tubing measurement
- (B) Pressure relief only
- (C) Fluid storage
- (D) Critical backup system providing redundancy during high-pressure interventions

140

OWO1.141 **Why is it important to monitor fluid volume at the surface?**

Answer

- A** (A) Determines condition of well or operation and helps plan forward operations
- (B) Monitoring fluid volume at surface is not necessary for wireline runs
- (C) Fluid tank volume can provide valuable information about bottomhole temperature
- (D) Offset tidal forces caused by the moon's phases

141

OWO1.142 **Before opening the BOP, which of the following statement best describes how to verify the well is dead?**

Answer

- A** (A) Line up to trip tank and monitor for flow or losses
- (B) Attempt to circulate a reduced fluid weight and monitor pressures
- (C) Monitor for increase in string weight
- (D) All the above

142

OWO1.143 **During which operation should a closing-time test be performed?**

Answer

- C** (A) Every 21 days
- (B) After running casing
- (C) Upon rig up
- (D) Every 14 days

143

OWO1.144 **What is the second component in the Quad Stack configuration from the top?**

Answer

- D** (A) Blind Rams
- (B) Pipe Rams
- (C) Slip Rams
- (D) Shear Rams designed to cut tubulars in emergency

144

OWO1.145 **What is the fourth component in the Quad Stack from top?**

Answer

- C** (A) Blind Rams
- (B) Shear Rams
- (C) Pipe Rams that create seal around tubing during operations
- (D) Slip Rams

145

OWO1.146 **What happens if accumulator precharge pressure is lost?**

Answer

- B** (A) No effect
- (B) Usable fluid volume decreases reducing emergency capability
- (C) Increases volume
- (D) Improves performance

146

OWO1.147 **What must be identified to determine the safe working pressures of a pumping system?**

Answer

- A** (A) Lowest pressure rated component of connections, wellhead pressure gauges, hoses, and fittings that will experience pressure
- (B) Pressure rating of the chickens iron being used
- (C) The wellhead or tree rated working pressure
- (D) Wellhead ring gasket

147

OWO1.148 **What is the main function of the main control panel for the BOP Closing Unit (Accumulator)?**

Answer

- A** (A) Primary BOP operating panel
- (B) Adjusts accumulator pressure
- (C) Maintains pump pressure, accumulator fluid
- (D) Allows full operating pressure to be applied to Annular Preventer

148

OWO1.149 **How does circulating fluid with entrained solids (sand or other fill) affect bottomhole pressure?**

Answer

- C** (A) Bottomhole pressure stays the same
- (B) Bottomhole pressure decreases
- (C) Bottomhole pressure increases
- (D) None of the above

149

OWO1.150 **What does FOSV stand for?**

Answer

- B** (A) Fluid Operating Safety Valve
- (B) Full Opening Safety Valve
- (C) Flow Opening Shut-off Valve
- (D) Fluid Only Safety Valve

150

OWO1.151 **For wellhead and BOP equipment, why are routine visual inspections needed?**

Answer

- C** (A) Visual inspections are more efficient than function tests
- (B) Visual inspections must be performed every 7 days, per API STD 53
- (C) Verification of maintenance
- (D) To ensure rams can be locked

151

OWO1.152 **What hydraulic working pressure does a Quad Stack systems operate at?**

Answer

- C** (A) 500 psi
- (B) 1,000 psi
- (C) 1,500 psi
- (D) 2,000 psi

152

OWO1.153 **What is the function of non-extrusion rings in the Tandem Stripper?**

Answer

- D** (A) Guide tubing
- (B) Control flow rate
- (C) Measure temperature
- (D) Maintain seal integrity under extreme pressures

153

OWO1.154 **According to API RP 5C7, what is the first component in the Quad Stack from top?**

Answer

- C** (A) Shear Rams
- (B) Pipe Rams
- (C) Blind Rams
- (D) Slip Rams

154

OWO1.155 **How many independently functioning elements does a Tandem Stripper have?**

Answer

- D** (A) One element
- (B) Four elements
- (C) Three elements
- (D) Two independent elements

155

OWO1.156 **What does Pressure Control Equipment (PCE) stand for in well operations?**

Answer

- D** (A) Pressure Casing Equipment
- (B) Primary Control Element
- (C) Pipe Compression Equipment
- (D) Pressure Control Equipment

156

OWO1.157 **What is an External-Internal Slip connection?**

Answer

- B** (A) Welded connection
- (B) Uses mechanical gripping mechanisms for connection
- (C) Permanent bonding
- (D) Fluid seal only

157

OWO1.158 **What demonstrates proper personnel communication during drills?**

Answer

- B** (A) Radio silence
- (B) Clear understood instructions and feedback mechanisms
- (C) Written notes only
- (D) Silent hand signals

158

OWO1.159 **What is an issue in which hydrates may cause a well control problem to arise?**

Answer

- A** (A) Trapped pressure giving false surface pressure reading
- (B) Increase in temperature
- (C) Increases fluid weight
- (D) Decrease in temperatures

159

OWO1.160 Which of the following signs suggest the well is secure after being shut-in?

Answer

- C** (A) Reduction in pressures
- (B) Rising tank volume
- (C) No leak at surface
- (D) Reduction in tank volumes

160

OWO1.161 What is the primary function of a FOSV?

Answer

- B** (A) Pressure measurement
- (B) Can shut off flow from the work string in emergency situations
- (C) Temperature control
- (D) Data logging only

161

OWO1.162 What does comprehensive barrier management provide?

Answer

- B** (A) Cost reduction only
- (B) Ensured well integrity and minimized risks throughout all operational phases
- (C) Speed only
- (D) Simplified procedures

162

OWO1.163 How is a fluid barrier effected by crystallation of the fluid?

Answer

- C** (A) No change in hydrostatic pressure
- (B) Increase hydrostatic pressure
- (C) Reduce hydrostatic pressure
- (D) Increased temperatures cause fluid barrier withstand greater downhole pressures

163

OWO1.164 **What determines if a fluid barrier will function effectively?**

Answer

- A** (A) Adequate height and density
- (B) Fluid color
- (C) Container material
- (D) Pump type

164

OWO1.165 **What can be added to fluid barriers to mitigate losses?**

Answer

- A** (A) Bridging agents or viscous pills
- (B) Sand
- (C) Grease
- (D) Cement

165

OWO1.166 **What type of testing validates a mechanical barrier?**

Answer

- A** (A) Pressure tests (positive and negative)
- (B) Visual inspection only
- (C) Temperature checks
- (D) Flow rate testing

166

OWO1.167 **What material provides resistance to high-temperature wear?**

Answer

- C** (A) Nitrile elastomer
- (B) Urethane elastomer
- (C) Viton elastomer
- (D) Standard rubber

167

OWO1.168 **Complete the following definition. Maximum Anticipated Surface Pressure (MASP) in cased hole is?**

Answer

A (A) Highest pressure predicted to be encountered at the surface of the well

(B) Highest anticipated pressure at the shoe

(C) Highest annular pressure minus the fluid column minus surface pressure

(D) Highest pressure predicted to be encountered at the downhole production packer or perforations

168

OWO1.169 **What is the minimum reel core radius to coil diameter ratio?**

Answer

D (A) 10:1

(B) 30:0

(C) 50:1

(D) 20:1

169

OWO1.170 **What is a primary indicator of well control effectiveness?**

Answer

B (A) Speed of operations

(B) Ability to prevent unwanted fluid movement and maintain safe barriers

(C) Low cost operations

(D) Minimal equipment use

170

OWO1.171 **How much do high-pressure applications count toward service life derating?**

Answer

C (A) 1 cycle equivalent

(B) 2 cycle equivalents

(C) 3 cycle equivalents

(D) 5 cycle equivalents

171

OWO1.172 **What are the factors needed to increase the differential pressure across the tubing to lower the allowable tensile load of the string?**

Answer

- C** (A) Pump rate; pressure in the well; depth of the well
- (B) Temperature; depth of well; height of U-Tube
- (C) Height of U-Tube; density of fluid in U-Tube; pressure in the well
- (D) Pressure; measured depth of well, viscosity of U-tube

172

OWO1.173 **Primary fluid barrier failure would be identified by the following?**

Answer

- D** (A) Less than expected displaced fluid while tripping in
- (B) Increase in fluid temperature
- (C) Decrease in fluid temperature
- (D) Greater than expected displaced fluid while tripping in

173

OWO1.174 **To accept fluid as a barrier, what following statements are required to be met?**

Answer

- A** (A) A fluid level height in the wellbore that can be monitored
- (B) The required viscosity and yield point required to maintain hydrostatic pressure equal to or greater than the estimated or measured pore pressure or reservoir pressure plus a safety margin
- (C) The required viscosity required to maintain hydrostatic pressure equal to or greater than the estimated or measured pore pressure or reservoir pressure plus a safety margin
- (D) The required viscosity required to decrease hydrostatic pressure less than than the estimated or measured pore pressure or reservoir pressure plus a safety margin

174

OWO1.175 **What is a primary fluid barrier characteristic?**

Answer

- B** (A) Compressibility
- (B) Adequate density to maintain hydrostatic pressure equal to or exceeding formation pressure with safety margin
- (C) Low viscosity
- (D) High temperature only

175

OWO1.176 **What does the Stripper Console manage?**

Answer

- B** (A) Surface pumps only
- (B) Critical pressure control equipment through dual action Pack/Retract
- (C) Tubing deployment only
- (D) Data logging alone

176

OWO1.177 **How long does a fluid barrier provide protection after circulation stops?**

Answer

- C** (A) Indefinitely
- (B) 12 hours
- (C) Temporarily only
- (D) Until next circulation

177

OWO1.178 **What is the purpose of gripper blocks in coiled tubing?**

Answer

- C** (A) Measure fluid temperature
- (B) Calculate pressure gradients
- (C) Serve as critical contact interface between injector and coiled tubing
- (D) Store operational fluids

178

OWO1.179 **What function does the Hydraulic Power Pack provide?**

Answer

- D** (A) Tubing storage only
- (B) Data collection only
- (C) Communication services
- (D) Directed energy to power various components

179

OWO1.180 **What operational problem can hydrate blockage create?**

Answer

- B** (A) Reduced flow rate
- (B) Prevents pumping into the well creating control problems
- (C) Increases flow only
- (D) Reduces pressure minimally

180

OWO1.181 **What should be recorded immediately after shut-in?**

Answer

- B** (A) Crew names only
- (B) Time, pressure readings, pit gain, and tubing information
- (C) Equipment serial numbers
- (D) Weather conditions

181

OWO1.182 **What is the behavior of water as a fluid barrier component?**

Answer

- B** (A) Highly compressible
- (B) Considered non-compressible having stable properties under pressure
- (C) Sensitive to temperature
- (D) Variable density

182

OWO1.183 **Why must both Static and Dynamic Pressure Control Equipment (PCE) be tested?**

Answer

- D** (A) For compliance filing
- (B) For insurance purposes
- (C) To calculate costs
- (D) To confirm effective function under operational pressures

183

OWO1.184 **What is the maximum coiled tubing diameter typically used?**

Answer

- C** (A) 0.75 inches
- (B) 2 inches
- (C) 4 inches
- (D) 6 inches

184

OWO1.185 **What is Dynamic Pressure Control Equipment (PCE) used for?**

Answer

- D** (A) Static pressure containment
- (B) Emergency procedures
- (C) Data recording only
- (D) Active pressure management during operations

185

OWO1.186 **What statement best describes why stroke rate indicators and pump stroke counters are needed?**

Answer

- D** (A) Notoriously unreliable, must be replaced several times during the circulation
- (B) Stroke or volume counters can be used to accurately measure downhole temperature
- (C) Stroke or volume counters cannot be used to accurately measure volume pumped
- (D) Stroke counters and stroke rate indicators can be useful in ensuring proper rates are pumped

186

OWO1.187 **What factors can lead to the settling of solids?**

Answer

- A** (A) Using low viscosity fluids and circulating at a slow velocity
- (B) Using high viscosity fluids and circulating at rate to cause turbulent flow
- (C) Using fluid of a higher density than the solids
- (D) Temperature inversions as heated fluids are circulated to surface

187

OWO1.188 **What is ideal for the Side-Door Stripper configuration?**

Answer

- D** (A) Operations with unlimited vertical clearance
- (B) Surface operations only
- (C) Deep offshore wells only
- (D) Operations with limited vertical clearance where conventional servicing is challenging

188

OWO1.189 **What do stabilized pressure readings indicate?**

Answer

- B** (A) Pump failure
- (B) Well has reached pressure equilibrium and is under control
- (C) Formation flow only
- (D) Equipment malfunction

189

OWO1.190 **What condition does NOT de-rate tubular pressure ratings?**

Answer

- B** (A) Sand erosion
- (B) Internal plastic coating applied to prevent corrosion
- (C) Improper make-up torque
- (D) Rusted pipe

190

OWO1.191 **What yield strength range is typical for coiled tubing?**

Answer

- D** (A) 20,000 to 40,000 psi
- (B) 150,000 to 200,000 psi
- (C) 250,000 to 300,000 psi
- (D) 55,000 to 120,000 psi

191

OWO1.192 **How does the Stripper Packing create a reliable seal?**

Answer

- D** (A) Mechanical spring force only
- (B) Manual compression
- (C) Fluid pressure alone
- (D) Packer mechanism compresses with upward piston movement

192

OWO1.193 **What does the Pumping Tee serve as in coiled tubing operations?**

Answer

- B** (A) Measurement device only
- (B) Primary interface point for fluid circulation and recovery
- (C) Pressure regulator only
- (D) Storage tank

193

OWO1.194 **What backup security measure is incorporated in Quad Stacks?**

Answer

- D** (A) Electronic locks
- (B) Gravity alone
- (C) Mechanical springs
- (D) Manual locks should hydraulic pressure be lost

194

OWO1.195 **What is the purpose of well control drills?**

Answer

- B** (A) Entertainment
- (B) Increase crew competency and communication proficiency
- (C) Reduce operating costs
- (D) Minimize paperwork

195

OWO1.196 **What do negative pressure tests on barriers simulate?**

Answer

- A** (A) Pressure loss above barrier
- (B) Formation gain
- (C) Pump failure
- (D) Ambient conditions

196

OWO1.197 **What effects the crystallization point in a fluid?**

Answer

- C** (A) Height of fluid column
- (B) Type of annular flow
- (C) Temperature
- (D) Surface pressure

197

OWO1.198 **What is a key characteristic of coiled tubing?**

Answer

- D** (A) Permanent connection required
- (B) Requires daily assembly
- (C) Manual handling only
- (D) Continuous single string deployment

198

OWO1.199 **Negating well bore conditions, what is the expected maximum running speed of coiled tubing?**

Answer

- D** (A) 50 feet per minute
- (B) 100 feet per minute
- (C) 150 feet per minute
- (D) 250 feet per minute

199

OWO1.200 Which of the following statements best describes a Fluid Gas Separator (sometimes called the MGS)?

Answer

- A** (A) It is typically, a low pressure vessel
- (B) It is typically, a high pressure vessel
- (C) It is typically, a moving vessel
- (D) It is typically, a vessel located downstream of flare stack

200

OGOR Workover Practice Test Day 1

OWOT1.1 **What is the most common sign that a fluid barrier has failed?**

Answer

- B** (A) Test fluid density with a scale
- (B) Flow from the well
- (C) Check fluid viscosity
- (D) Leak in BOP stack or wellhead gate valves

1

OWOT1.2 **What could be an added problem during well control if hydrates form?**

Answer

- A** (A) Hydrates can create blockage which would prevent pumping into the well
- (B) Hydrates can create blockage which would cause the well to flow
- (C) Hydrates can cause clay formations to swell
- (D) Hydrates circulated to surface may catch fire

2

OWOT1.3 **Who is the person-in-charge (PIC) during a well control event?**

Answer

- B** (A) Toolpusher
- (B) Company Man
- (C) Rig Operator
- (D) Assistant Driller

3

OWOT1.4 **What is the definition of Well Integrity?**

Answer

- C** (A) The ability of the well to prevent fluid flow at surface
- (B) The ability of the well to prevent flow from the formation
- (C) The ability of the well to contain fluids within the intended barrier envelope
- (D) Ability to separate gases from fluids

4

OWOT1.5 **What changes in the height of fluid in the pits indicate?**

Answer

- D** (A) Height of fluids can be used to estimate viscosity
- (B) Fluid tank volume can provide valuable information about bottomhole temperature
- (C) Monitoring fluid volumes at surface is not necessary during dead-well operations
- (D) Tank volume changes can indicate flow or losses in the well

5

OWOT1.6 **During an accumulator drawdown test, which of the following is not required by API Standard 53?**

Answer

- B** (A) Close Annular and pipe rams (maximum of 4), simulate closing blind rams (by opening a set of rams)
- (B) Allow a full recharge of system between each step
- (C) Check that final pressures are 200 psi above pre-charge pressure
- (D) Ensure final accumulator pressure is equal to operating pressure of the vessel.

6

OWOT1.7 **A solid formed when water and a gas with a low boiling point combine into a crystalline compound is called what?**

Answer

- D** (A) Grease
- (B) Brine
- (C) Completion Fluid
- (D) Hydrate

7

OWOT1.8 **Visual inspections of BOP and wellhead equipment are performed routinely. Why is this a good idea?**

Answer

- A** (A) Visual inspections can help detect fluid or pressure leaks
- (B) Visual inspections are more efficient than function tests
- (C) Visual inspections can take the place of pressure tests
- (D) Visual inspections ensures BOP and valves are closed

8

OWOT1.9 **What is a property of water?**

Answer

- D** (A) Considered to be highly compressible
- (B) Sensitive to pressure changes
- (C) Relatively unstable
- (D) Considered to be non-compressible

9

OWOT1.10 **What should we record after we have taken a kick and shut-in the well?**

Answer

- D** (A) Pit volume and reset stroke counters. Replace SITP and SICP gauges, for more accurate readings
- (B) Time of shut in, initial pit gain, pipe length in hole and initial casing pressure
- (C) Initial casing pressure, initial tubing pressure, total volume in pits and temperature of well fluids
- (D) Time of shut-in, initial pit gain, initial casing pressure and initial tubing pressure

10

OWOT1.11 **Shear rams are used for what purpose?**

Answer

- C** (A) Shears tubing and casing simultaneously
- (B) Shears bottom hole assembly
- (C) Shears single pipe tube
- (D) Close in on open hole

11

OWOT1.12 **Trip drills are performed for what reason?**

Answer

- B** (A) Increase productivity of the well
- (B) Increases crew competency and observe communication
- (C) To make sure the rig operator is testing well integrity
- (D) Remove trip hazards

12

OWOT1.13 **Blind shear rams cannot do which of the following operations?**

Answer

- D** (A) Blind Shear rams cannot be closed on tubulars
- (B) Blind Shears can shear pipe and seal open hole
- (C) Blind Shears can close and seal an open hole
- (D) Blind Shears can close and seal on pipe without damage to the pipe

13

OWOT1.14 **If there is a total loss of hydraulic pressure from the accumulator, will the manual locking pins keep the rams closed?**

Answer

- A** (A) True
- (B) False
- (C)
- (D)

14

OWOT1.15 **Are the ram locks used on ram preventers to ensure that the rams stay closed?**

Answer

- A** (A) True
- (B) False
- (C)
- (D)

15

OWOT1.16 **The purpose of a blind/shear ram is what?**

Answer

- D** (A) Shear on any tooljoint
- (B) Shear BHA
- (C) Seals annulus with pipe in the well
- (D) Shear pipe

16

OWOT1.17 **Tubular pressure and strength ratings are not de-rated for which of the following cases?**

Answer

- A** (A) Internal plastic coating
- (B) Sand erosion
- (C) Improper make-up torque and procedures
- (D) Rusted pipe

17

OWOT1.18 **Pit drills are used to?**

Answer

- B** (A) To see how much fluid it takes to fill the pits
- (B) Increase crew competency with kick detection
- (C) Increase crew competency with tripping tubing
- (D) Remove trip hazards on pits

18

OWOT1.19 **API 53 requires which test to check the performance of the hydraulic operations of the BOP closing system?**

Answer

- D** (A) BOP isolation valve test
- (B) BOP function Test
- (C) BOP pressure test
- (D) Accumulator Drawdown Test

19

OWOT1.20 **What is the time interval to perform a drawdown test according to API Standard 53?**

Answer

- A** (A) The accumulator drawdown test shall be performed no more than 6 months from the previous test.
- (B) The accumulator drawdown test shall be performed every 7 days.
- (C) The accumulator drawdown test shall be performed every 21 days.
- (D) Once, every 100 rig moves

20

OWOT1.21 **What happens to bottom hole pressure when the fluid level in the well drops?**

Answer

- B** (A) No change
- (B) Decreases
- (C) Increases
- (D) Both A and B

21

OWOT1.22 **Shear rams are used for what purpose?**

Answer

- A** (A) To cut pipe
- (B) To seal the well
- (C) To hang off pipe
- (D) To cut tooljoints

22

OWOT1.23 **If using fluid as a barrier which of the following conditions must be met?**

Answer

- B** (A) The viscosity is sufficient to maintain hydrostatic pressure equal to or greater than the estimated or measured pore pressure or reservoir pressure plus a safety margin
- (B) The density is sufficient to maintain hydrostatic pressure equal to or greater than the estimated or measured pore pressure or reservoir pressure plus a safety margin
- (C) The downhole fluid temperature is sufficient to maintain hydrostatic pressure equal to or greater than the estimated or measured pore pressure or reservoir pressure plus a safety margin
- (D) The yield point is sufficient to maintain hydrostatic pressure equal to or greater than the estimated or measured pore pressure or reservoir pressure plus a safety margin

23

OWOT1.24 **To prevent or stop uncontrolled flow is a barrier envelope which consists of a combination of components and/or practices is used?**

Answer

- A** (A) True
- (B) False
- (C)
- (D)

24

OWOT1.25 **If we have flow coming from inside the workstring at the surface, what should we do?**

Answer

- B** (A) Stab in closed safety valve then open valve
- (B) Stab in safety valve in the open position and then close
- (C) Close annular
- (D) Close Blind/Shear ram

25

OWOT1.26 **What is the purpose of well integrity?**

Answer

- C** (A) Desired fluids are prevented from reaching the surface
- (B) Produced water and sand is prevented from corroding or eroding the production tubing
- (C) Unwanted formation fluids are unable to escape at surface
- (D) To provide means to reduce casing scale

26

OWOT1.27 **Which of the following fluids is the most compressible?**

Answer

- B** (A) Cement slurry
- (B) Oil-based fluids
- (C) Water-based fluids
- (D) None of the above

27

OWOT1.28 **How is a barrier test done?**

Answer

- D** (A) Placing 2,000 lbs of weight directly on top the barrier
- (B) Decrease accumulator pressure until leakage occurs
- (C) Decrease amount of fluid below the barrier and look for leaks
- (D) Increase differential pressure across a barrier

28

OWOT1.29 **Which components dictates what the Maximum Allowable Working Pressure (MAWP) is using the following equipment? BOP – 10,000 psi; Choke Manifold – 12,000 psi; Pumps – 7,500 psi; Surface Pump Line – 13,000 psi; Wellhead – 5,000 psi Tubular Burst Pressure**

Answer

- A** (A) 5,000 psi
- (B) 7,500 psi
- (C) 12,000 psi
- (D) 24,000 psi

29

OWOT1.30 **A liquid seal in a fluid gas separator is dependent on which of the following?**

Answer

- B** (A) Depth of well; height of u-tube; density of fluid in u-tube
- (B) Density of fluid in u-tube, pressure in the vessel, height of u-tube
- (C) Temperature; depth of well; height of u-tube
- (D) The viscosity of fluid in u-tube only

30

OWOT1.31 **What effect do fluids coming to the surface have on bottomhole pressure while reverse circulating versus forward circulation?**

Answer

- C** (A) Increased clearance, higher friction loss, higher bottomhole pressure
- (B) Reduced clearance, lower friction loss, lower bottomhole pressure
- (C) Reduced clearance, higher friction loss, higher bottomhole pressure
- (D) Increased clearance, lower friction loss, higher bottomhole pressure

31

OWOT1.32 **Is there an increase or decrease of the allowable tensile load of the tubing when there is a differential pressure across the tubing?**

Answer

- B** (A) Increase
- (B) Decrease
- (C)
- (D)

32

OWOT1.33 **Which statement describes the best influx indicators?**

Answer

- A** (A) Flow with pump off, decrease in pump pressure, change in string weight, formation fluids show up during circulation
- (B) Flow with the pump off, volume displacement changes during tubular movement, and density remains the same
- (C) Increase in pump pressure, changes in fluid density, and change in string weight
- (D) Decrease in pump pressure, fluid density remains the same and string weight decreases then increases

33

OWOT1.34 **What is used as the operating limit for BOP equipment?**

Answer

- B** (A) Mud weight
- (B) Test pressure limit of equipment
- (C) Maximum Anticipated Surface Pressure (MASP)
- (D) Maximum Anticipated Annular Surface Pressure (MAASP)

34

OWOT1.35 **A common environmental factor that would require lowering the maximum working pressure of equipment is _____?**

Answer

- C** (A) Leaking seal
- (B) Incorrect elastomers
- (C) Temperature
- (D) Leaking ring gasket

35

OWOT1.36 In a cased hole what does MASP refer to?

Answer

- D** (A) The highest annular pressure minus the fluid column
- (B) The highest annular pressure minus the fluid column minus surface pressure of the well before injection
- (C) The highest anticipated total pressure at the perforations before injection
- (D) The highest amount of pressure predicted at the surface of the well upon shut in and stabilization

36

OWOT1.37 Which of the following operations would be a problem if the well flowed and had to be shut-in?

Answer

- B** (A) Running large tubing size
- (B) Running SSSV control lines
- (C) Running very small tubing size
- (D) Tubing within BOP stack

37

OWOT1.38 An upward force in a well is caused by _____?

Answer

- C** (A) Tubing weight
- (B) Bottom hole temperature
- (C) Well pressure
- (D) Bottom hole porosity

38

OWOT1.39 In what position is a FOSV used?

Answer

- D** (A) Inside Blowout Preventer is first installed followed by a closed FOSV
- (B) Are always stabbed in the closed position
- (C) Are always stabbed in the neutral position
- (D) Are always stabbed in the open position

39

OWOT1.40 Which of the following barriers can be physically tested?

Answer

- B** (A) Fluid Barrier
- (B) Mechanical Barrier
- (C) Procedural Barrier
- (D) None of the above

40

OWOT1.41 Keeping the hole full affects what?

Answer

- D** (A) Fluid viscosity
- (B) Fluid weight
- (C) Formation pressure
- (D) Hydrostatic pressure

41

OWOT1.42 A fluid gas separator is located where?

Answer

- A** (A) Typically, the first separation component downstream of the choke
- (B) Typically, the first separation component upstream of the choke
- (C) Typically, the second separation component downstream of the choke
- (D) Typically, the tertiary separation component downstream of flare stack

42

OWOT1.43 How does viscosity affect frictional pressure loss?

Answer

- C** (A) Viscosity increases, frictional pressure loss decreases
- (B) Viscosity decreases, no effect on frictional pressure loss
- (C) Viscosity decreases, frictional pressure loss decreases
- (D) Viscosity increases, no effect on frictional pressure loss

43

OWOT1.44 **If one accumulator bottle loses its precharge, what would be the effect on the accumulator system?**

Answer

- D** (A) With multiple redundant accumulator bottles, nothing would happen
- (B) Usable amount of fluid in bottles will increase
- (C) The bypass valve will have to be functioned
- (D) Usable amount of fluid in bottles will decrease

44

OWOT1.45 **A hydrostatic column in a static condition that exerts an overbalance to the formation pressure is known as what?**

Answer

- A** (A) Fluid Barrier
- (B) Procedural Barrier
- (C) Barrier Envelope
- (D) Both B & A

45

OWOT1.46 **A typical workover fluid would exhibit which of the following properties?**

Answer

- B** (A) Hydrate sensitive
- (B) Low solids
- (C) High solids
- (D) High viscosity

46

OWOT1.47 **Opening the bypass on an accumulator sends higher pressure to which components?**

Answer

- C** (A) Air and hydraulic system and manifold
- (B) Pipe rams, blind rams, and annular manifold
- (C) HCR, pipe rams, blind rams, manifold
- (D) Annular preventer

47

OWOT1.48 **In a production well what should be done prior to pulling tubing ensure the well is balanced?**

Answer

- D** (A) Set a plug
- (B) Pump a slug
- (C) Maintain the clean well
- (D) Replace fluids with kill fluid

48

OWOT1.49 **If the working pressure of the BOP stack is 5,000 psi, to what pressure should the pipe rams be tested?**

Answer

- A** (A) 5,000 psi
- (B) 2,500 psi
- (C) 7,500 psi
- (D) 10,000 psi

49

OWOT1.50 **Which of the following is the primary factor for determining if a bending stress analysis should be performed prior to beginning operations.**

Answer

- B** (A) Milling/drilling cement
- (B) Overall weight of stack and equipment
- (C) Pump pressure
- (D) Land subsidence

50

OWOT1.51 **Which of the following is a fluid loss additive for completion fluids?**

Answer

- D** (A) Cement
- (B) Zinc
- (C) Oil based mud
- (D) Calcium carbonate

51

OWOT1.52 **Pulling tubing at a fast rate may case _____.**

Answer

- A** (A) Swabbing
- (B) Surging
- (C) Bullheading
- (D) Lube and Bleed

52

OWOT1.53 **Which of the following statements best describes the most accurate way to measure completion fluid density?**

Answer

- B** (A) Pressurized mud balance
- (B) Mud balance
- (C) Hydrometer
- (D) Fann VG meter

53

OWOT1.54 **What effects the crystallization point of any brine fluid?**

Answer

- C** (A) Type of annular flow
- (B) Height of fluid column
- (C) Density
- (D) Ambient temperature

54

OWOT1.55 **When checking if a fluid can work as a barrier, which fluid property would NOT matter?**

Answer

- D** (A) The density of the fluid
- (B) The viscosity of the fluid
- (C) The true vertical depth of the fluid in the well
- (D) None of the above

55

OWOT1.56 **On a BOP Closing Unit, which statement best describes what the annular pressure regulator does?**

Answer

- C**
- (A) Regulate opening pressure of the HCR
 - (B) Regulate closing pressure of rams and annular preventer
 - (C) Regulate the closing pressure of annular preventer
 - (D) Regulates flow of fluids from reservoir to accumulator bottle

56

OWOT1.57 **Why are barriers tested?**

Answer

- D**
- (A) Verifies wellbore pressure cycles
 - (B) Lowers maintenance costs
 - (C) Increases life of the well
 - (D) Verify barrier integrity

57

OWOT1.58 **What is the purpose of a sliding sleeve?**

Answer

- A**
- (A) Allows circulation without unseating a packer
 - (B) Inject chemicals into the well
 - (C) Measure downhole pressure
 - (D) Allows circulation by unseating the packer

58

OWOT1.59 **Which statement best describes the process for reducing tubing collapse due to excessive tensile loads and differential pressure?**

Answer

- B**
- (A) Decrease tubing pressure
 - (B) Increase tubing pressure
 - (C) Increase annulus pressure
 - (D) Decrease annulus pressure

59

OWOT1.60 **Which device manages back pressure from a remote panel instead of at the choke manifold?**

Answer

- B** (A) Manual adjustable choke
- (B) Remote adjustable choke
- (C) Fixed choke
- (D) None of the above

60

OWOT1.61 **What should be checked first before starting operations to enter a well?**

Answer

- D** (A) H2S on surface and in cellar, spark arrestors on all internal combustion equipment
- (B) Ambient temperature, drill pipe size, and pit volume
- (C) Pit volume, maximum anticipated surface pressure (MASP), and formation pressure
- (D) Formation pressure, fluid type, WH/tree integrity and maximum anticipated surface pressure (MASP)

61

OWOT1.62 **When should adding an extra float or downhole check valve be considered?**

Answer

- A** (A) When working in high Pressure environments
- (B) When pumping high viscosity fluid
- (C) When pumping low viscosity fluid
- (D) When well is flowing

62

OWOT1.63 **What could lead to tubing failure?**

Answer

- B** (A) Improper joint length
- (B) Incorrect make-up torque
- (C) Low pump rate
- (D) Freezing temperatures

63

OWOT1.64 **What is required if there is a high level of H2S present?**

Answer

- C** (A) Only carbon steel can be used
- (B) Only stainless steel can be used
- (C) Steel components must include nickel and chromium
- (D) Must use proper PPE

64

OWOT1.65 **What is an operating limit of BOP equipment?**

Answer

- D** (A) Maximum Anticipated Annular Surface Pressure (MAASP)
- (B) Mud weight
- (C) Measured depth
- (D) Rated working pressure

65

OWOT1.66 **What considerations may be made for pump stroke counters and stroke rate indicators?**

Answer

- A** (A) If pump efficiency is known, Stroke or volume counters can be used to accurately measure volume pumped
- (B) Stroke or volume counters cannot be used to accurately measure volume pumped
- (C) Stroke or volume counters can be used to accurately measure downhole temperature
- (D) Notoriously unreliable, must be replaced several times during the circulation

66

OWOT1.67 **Which of the following items are most likely to first identify the presence of H2S?**

Answer

- C** (A) Pipe deck
- (B) Cellar
- (C) Flow line
- (D) Dead wildlife

67

OWOT1.68 **Which of the following statements would not be included within an accumulator drawdown test?**

Answer

- B** (A) Prior to test, allow accumulator system to charge to max pressure
- (B) Allow system pressure to recharge in-between each step
- (C) Check that final pressures are 200 psi above pre-charge pressure
- (D) None of the above

68

OWOT1.69 **What can improve the completion fluid's ability to carry cuttings while milling out a packer?**

Answer

- A** (A) Density, viscosity, and annular fluid velocity
- (B) Drilling mud, formation fluids, and annular pressure loss
- (C) pH, chlorine content, and temperature of fluids
- (D) Adding cement to completion fluids

69

OWOT1.70 **Who is the person-in-charge (PIC) during a well control event?**

Answer

- B** (A) Rig Operator
- (B) Company Man
- (C) Toolpusher
- (D) Assistant Driller

70

OGOR Workover Study Guide Day 2

OWO2.1 **What is the primary difference between MAWP and MAOP?**

Answer

- A** (A) MAWP is maximum safety limit while MAOP is operational limit
- (B) MAWP is always higher than MAOP
- (C) They are identical
- (D) MAOP is equipment rating

1

OWO2.2 **Shear rams are designed to perform which function?**

Answer

- A** (A) Cut through tubulars
- (B) Seal open hole
- (C) Direct mud circulation
- (D) Measure pressure

2

OWO2.3 **What is a limitation of shear rams in well control?**

Answer

- A** (A) Difficulty with thick-walled or high-strength materials
- (B) Cannot cut at any depth
- (C) Can only work with light tubing
- (D) Cannot function under pressure

3

OWO2.4 **Blind rams are designed to close on which scenario?**

Answer

- A** (A) Open hole
- (B) Inside tubing
- (C) Around closed pipe
- (D) Annular space only

4

OWO2.5 **A safety margin of 20% applied to a 10,000 psi MAWP reduces working pressure to?**

Answer

- C** (A) 9,000 psi
- (B) 8,500 psi
- (C) 8,000 psi
- (D) 7,500 psi

5

OWO2.6 **Which of the following is a mechanical barrier?**

Answer

- A** (A) Packer or bridge plug
- (B) Hydrostatic pressure
- (C) Circulation pressure
- (D) Surface pressure

6

OWO2.7 **Fluid barriers can be validated through which method?**

Answer

- D** (A) Pressure testing only
- (B) Volume monitoring only
- (C) Surface pressure only
- (D) Continuous monitoring of fluid volume and surface pressure

7

OWO2.8 **Differential pressure plays a role in which function?**

Answer

- A** (A) Energizing packers and elastomers
- (B) Reducing formations pressure
- (C) Decreasing hydrostatic pressure
- (D) Eliminating surface pressure

8

OWO2.9 **Prior to operations, pressure across devices should be?**

Answer

- A** (A) Equalized to prevent sudden pressure changes
- (B) Unbalanced for accuracy
- (C) Kept at surface values
- (D) Not monitored

9

OWO2.10 **Sustained casing pressure exceeding 500 psi could indicate?**

Answer

- A** (A) Breach in casing or cement integrity
- (B) Normal operation
- (C) Proper well control
- (D) Low formation pressure

10

OWO2.11 **Boyle's Law demonstrates which relationship?**

Answer

- A** (A) Inverse relationship between pressure and volume
- (B) Direct relationship between pressure and volume
- (C) No relationship between pressure and volume
- (D) Linear relationship only

11

OWO2.12 **$P_1V_1 = P_2V_2$ represents which gas law principle?**

Answer

- A** (A) Boyle's Law
- (B) Charles' Law
- (C) Gay-Lussac's Law
- (D) Combined Gas Law

12

OWO2.13 **In Boyle's Law application, P_1 represents?**

Answer

- A** (A) Formation pressure (initial kick pressure)
- (B) Surface pressure
- (C) Applied pressure
- (D) Maximum casing pressure

13

OWO2.14 **Total Pit Gain is calculated using which formula?**

Answer

- A** (A) $(\text{FormP} \times \text{Kick Volume}) \div \text{Max CP}$
- (B) Formation Pressure $\div$ Kick Volume
- (C) Max CP $\div$ Formation Pressure
- (D) Kick Volume $\times$ Max CP

14

OWO2.15 **Before well shut-in during a kick, what occurs with gas?**

Answer

- A** (A) Gas expands freely as it rises
- (B) Gas remains stationary
- (C) Gas compresses
- (D) Gas dissolves in fluids

15

OWO2.16 **As gas rises and expands during circulation, what occurs to hydrostatic pressure?**

Answer

- A** (A) Decreases
- (B) Increases continuously
- (C) Remains unchanged
- (D) Becomes zero

16

OWO2.17 **Even after shut-in, gas continues to migrate due to which force?**

Answer

- D** (A) Friction
- (B) Adhesion
- (C) Compression
- (D) Buoyancy

17

OWO2.18 **Maximum Casing Pressure during a kick can be estimated using?**

Answer

- D** (A) Formation Pressure only
- (B) Kick Volume only
- (C) Surface pressure only
- (D) Boyle's Law calculations

18

OWO2.19 **What is the critical risk if BHP falls below formation pressure during gas circulation?**

Answer

- A** (A) Additional influx can occur
- (B) Equipment failure only
- (C) Formation fracture
- (D) Mud degradation

19

OWO2.20 **Additional Pit Gain is calculated as?**

Answer

- D** (A) Kick Volume only
- (B) Total Pit Gain only
- (C) Total Pit Gain plus Kick Volume
- (D) Total Pit Gain minus Kick Volume

20

OWO2.21 **Buoyancy in well operations refers to which principle?**

Answer

- A** (A) Upward force exerted by fluid on immersed object
- (B) Weight of equipment in air
- (C) Downward force from pressure
- (D) Frictional resistance

21

OWO2.22 **The density of low carbon steel (AISI 1018) is approximately?**

Answer

- D** (A) 60.4 ppg
- (B) 62.4 ppg
- (C) 68.4 ppg
- (D) 65.4 ppg

22

OWO2.23 **Snub force is calculated to overcome which factors?**

Answer

- D** (A) Buoyancy only
- (B) Wellhead pressure only
- (C) Formation pressure only
- (D) Buoyancy effect and wellhead pressure

23

OWO2.24 **The balance point marks the transition where?**

Answer

- D** (A) Pipe becomes heavier
- (B) Pipe becomes lighter
- (C) Pressure equalizes
- (D) Downward forces equal upward forces

24

OWO2.25 **Pipe is considered 'pipe heavy' when?**

Answer

- A** (A) Buoyed weight exceeds opposing forces
- (B) Buoyed weight is less than balance point
- (C) Wellhead pressure is high
- (D) Formation pressure increases

25

OWO2.26 **'Pipe light' conditions require which operational approach?**

Answer

- A** (A) Mechanical snubbing force
- (B) Free fall of pipe
- (C) No force application
- (D) Reduced surface pressure

26

OWO2.27 **Total volume of tubular is calculated using which formula?**

Answer

- D** (A) Depth only
- (B) Area only
- (C) Depth ÷ Area
- (D) Length × Cross-sectional area

27

OWO2.28 **Open-ended displacement refers to volume displaced by?**

Answer

- A** (A) Pipe metal only
- (B) Internal fluid only
- (C) Entire tubular section
- (D) Annular space

28

OWO2.29 **Annular capacity is expressed in which units?**

Answer

- A** (A) Barrels per foot
- (B) Barrels only
- (C) Gallons
- (D) Cubic feet

29

OWO2.30 **The conversion factor used in volume calculations is?**

Answer

- D** (A) 1000.4
- (B) 1015.4
- (C) 1050.4
- (D) 1029.4

30

OWO2.31 **What is the first response when a kick is suspected?**

Answer

- A** (A) Immediately shut-in the well
- (B) Continue circulation at high rate
- (C) Increase pump pressure
- (D) Pull pipe from well

31

OWO2.32 Which pressure indicators help determine kick size and nature?

Answer

- D** (A) Only surface pressure
- (B) Only formation pressure
- (C) Only hydrostatic pressure
- (D) SITP and SICP

32

OWO2.33 When using forward circulation to kill a well, the ECD is typically?

Answer

- A** (A) Calculated with annular pressure loss
- (B) Unable to be determined with reverse circulation
- (C) Not calculated
- (D) Same as hydrostatic pressure

33

OWO2.34 In reverse circulation, where are pressures typically higher?

Answer

- A** (A) In the annulus
- (B) In the tubing
- (C) At surface only
- (D) In the formation

34

OWO2.35 The Volumetric Method is used when which scenario exists?

Answer

- A** (A) Circulation is not possible
- (B) Circulation is optimal
- (C) Well is flowing
- (D) Gas reaches surface

35

OWO2.36 **Bullheading should be selected when which condition is present?**

Answer

- A** (A) Unable to handle influx at surface
- (B) Quick kill is desired
- (C) Normal circulation is preferred
- (D) Gas at surface is manageable

36

OWO2.37 **During stripping operations, pressure regulation is accomplished through?**

Answer

- D** (A) Pump pressure alone
- (B) Formation pressure only
- (C) Surface valve opening
- (D) Annular regulator adjustment

37

OWO2.38 **When running pipe into a pressurized well, mud must be bled to?**

Answer

- A** (A) Keep bottomhole pressure nearly constant
- (B) Increase surface pressure
- (C) Reduce fluid density
- (D) Increase kick potential

38

OWO2.39 **Temperature correction of kill fluid is necessary because?**

Answer

- A** (A) Fluid density changes with temperature
- (B) Surface and downhole temps are identical
- (C) Temperature does not affect operations
- (D) Only for measurement purposes

39

OWO2.40 **The main cause of influx during workover operations is?**

Answer

- D** (A) Equipment failure only
- (B) Surface equipment malfunction
- (C) Excessive mud weight
- (D) Wellbore pressure less than formation pressure

40

OWO2.41 **The four principles of risk management are?**

Answer

- A** (A) Identify, quantify/qualify, mitigate, control
- (B) Assess, test, analyze, execute
- (C) Plan, implement, verify, complete
- (D) Design, build, test, operate

41

OWO2.42 **Pre-job meetings should address?**

Answer

- D** (A) Operational sequence only
- (B) Pressure limitations only
- (C) Equipment inventory only
- (D) Operational sequence, pressure limits, and emergency protocols

42

OWO2.43 **MAWP of equipment must exceed which pressure during wireline operations?**

Answer

- A** (A) Anticipated surface pressure
- (B) Formation pressure
- (C) Hydrostatic pressure
- (D) Pump pressure

43

OWO2.44 **Well integrity maintenance is essential to prevent?**

Answer

- D** (A) Pressure fluctuations only
- (B) Temperature changes only
- (C) Fluid level changes
- (D) Well control incidents

44

OWO2.45 **Night shift well control operations require special attention to?**

Answer

- A** (A) Adequate lighting and visibility of hazards
- (B) Pressure readings only
- (C) Fluid temperature only
- (D) Pump function only

45

OWO2.46 **Spill response protocols are critical for which reason?**

Answer

- A** (A) Environmental protection and regulatory compliance
- (B) Cost containment only
- (C) Equipment protection only
- (D) Fluid recovery only

46

OWO2.47 **A pressure control device differs from a barrier because it?**

Answer

- A** (A) Manages and regulates wellbore pressures dynamically
- (B) Isolates wellbore fluids
- (C) Provides static isolation only
- (D) Never requires monitoring

47

OWO2.48 **Wellhead and tree bending stress analysis is necessary when?**

Answer

- D** (A) Pressure is above formation
- (B) Depth exceeds 5,000 ft
- (C) Circulation rate increases
- (D) Applied loads may exceed design limits

48

OWO2.49 **In the Lube and Bleed method, the primary advantage is?**

Answer

- D** (A) Faster than all methods
- (B) No wellhead equipment needed
- (C) Lower cost only
- (D) Usable when unable to circulate

49

OWO2.50 **After forward circulation kill, if SITP and SICP are not equal, the safest action is?**

Answer

- A** (A) Circulate another bottoms-up
- (B) Pull out of hole immediately
- (C) Continue with other operations
- (D) Bleed off pressure quickly

50

OWO2.51 **U-tube principle applies to which scenario?**

Answer

- A** (A) Wellbore with different fluid densities in tubing and annulus
- (B) Surface operations only
- (C) Single density systems
- (D) Open hole wells only

51

OWO2.52 **In U-tube principle, equilibrium is reached when?**

Answer

- A** (A) Pressure forces balance
- (B) Pressures are unequal
- (C) Volumes are equal
- (D) Densities are identical

52

OWO2.53 **BHP in a wellbore with multiple fluid densities requires?**

Answer

- A** (A) Summing pressure from each fluid column
- (B) Single density calculation
- (C) Average density only
- (D) Heaviest fluid density

53

OWO2.54 **As wellbore temperature increases, fluid weight typically?**

Answer

- A** (A) Decreases
- (B) Increases
- (C) Remains constant
- (D) Doubles

54

OWO2.55 **Brine saturation refers to?**

Answer

- D** (A) Temperature of brine
- (B) Depth of brine
- (C) Pressure of brine
- (D) Amount of dissolved salt before crystallization

55

OWO2.56 **Hydrate formation in wellbores is prevented by?**

Answer

- D** (A) Increasing pressure only
- (B) Rapid circulation
- (C) Using high density mud
- (D) Managing temperature and pressure conditions

56

OWO2.57 **Gas compressibility is significant because?**

Answer

- D** (A) It is constant
- (B) Only matters at surface
- (C) It has no effect on pressure
- (D) Volume changes dramatically with pressure

57

OWO2.58 **Frictional pressure losses increase with which factor?**

Answer

- D** (A) Decreased flow rate
- (B) Larger pipe diameter
- (C) Lower fluid viscosity
- (D) Increased flow rate

58

OWO2.59 **Laminar flow is characterized by?**

Answer

- A** (A) Smooth, predictable fluid movement
- (B) Chaotic turbulent motion
- (C) No flow resistance
- (D) High pressure losses

59

OWO2.60 **Turbulent flow results in?**

Answer

- D** (A) Lower pressure losses
- (B) Reduced friction
- (C) Smoother operations
- (D) Higher and more variable pressure losses

60

OWO2.61 **Workover and completion fluids serve which primary function?**

Answer

- D** (A) Lubrication only
- (B) Cool equipment
- (C) Reduce friction only
- (D) Maintain well control and barriers

61

OWO2.62 **Critical fluid properties include all EXCEPT?**

Answer

- D** (A) Density
- (B) Viscosity
- (C) Compressibility
- (D) Color only

62

OWO2.63 **Fluid barrier validation involves?**

Answer

- D** (A) One-time testing
- (B) No ongoing monitoring
- (C) Initial pressure test only
- (D) Continuous monitoring of critical parameters

63

OWO2.64 **Fluid barrier effectiveness is compromised by?**

Answer

- D** (A) Higher density only
- (B) Lower temperature only
- (C) Increased depth
- (D) Contamination or degradation

64

OWO2.65 **Mud Gas Separators are essential during?**

Answer

- D** (A) Circulating kill fluid only
- (B) Only at surface
- (C) Static operations only
- (D) Kick circulation to separate gas from brine

65

OWO2.66 **Packers function as which type of barrier?**

Answer

- D** (A) Temporary barrier only
- (B) Pressure relief device
- (C) Fluid barrier only
- (D) Mechanical wellbore sealing device

66

OWO2.67 **Bridge plugs are used to?**

Answer

- D** (A) Circulate fluid
- (B) Measure depth
- (C) Increase pressure
- (D) Isolate and contain wellbore sections

67

OWO2.68 **Pre-recorded well information should be verified for?**

Answer

- D** (A) Cost purposes only
- (B) Regulatory filing only
- (C) Record keeping only
- (D) Accuracy to ensure safe operations

68

OWO2.69 **Downhole completion equipment is tested to?**

Answer

- D** (A) Establish pressure ratings only
- (B) Compare with other wells
- (C) Record historical data only
- (D) Verify integrity and functionality under operational pressures

69

OWO2.70 **Barrier management requires?**

Answer

- D** (A) Single verification
- (B) No ongoing attention
- (C) Periodic testing only
- (D) Continuous monitoring and maintenance

70

OWO2.71 **Coiled tubing operations are advantageous for?**

Answer

- D** (A) Deep wells only
- (B) Shallow wells only
- (C) High pressure wells only
- (D) Confined space operations and through-tubing work

71

OWO2.72 **Wireline operations require pressure control due to?**

Answer

- D** (A) Shallow depth only
- (B) Equipment requirements only
- (C) Low pressures only
- (D) Exposed wellbore to surface atmosphere

72

OWO2.73 **Snubbing operations are employed when?**

Answer

- D** (A) Normal tripping is safer
- (B) Pressure is low
- (C) Pipe is heavy only
- (D) Wellbore is pressurized and pipe is light

73

OWO2.74 **Slip rams in snubbing are used to?**

Answer

- D** (A) Only cut pipe
- (B) Measure pressure
- (C) Block flow only
- (D) Gradually feed pipe into well under controlled pressure

74

OWO2.75 **Influx detection methods include all EXCEPT?**

Answer

- D** (A) Pit gain monitoring
- (B) Shut-in pressure change
- (C) Flow rate changes
- (D) Formation fracture only

75

OWO2.76 **Trapped pressure between barriers can cause?**

Answer

- D** (A) No effect on well control
- (B) No operational concerns
- (C) Only surface pressure changes
- (D) Dangerous pressure buildup exceeding wellbore strength

76

OWO2.77 **Wellbore restrictions affect pressure by?**

Answer

- D** (A) Reducing flow rate only
- (B) No pressure effect
- (C) Reducing pressure only
- (D) Creating flow restrictions that increase pressure losses

77

OWO2.78 **Formation pressure estimates from well records are used for?**

Answer

- D** (A) Verification only
- (B) Regulatory filing only
- (C) Cost estimation only
- (D) Well control planning and operational decisions

78

OWO2.79 **During fracturing operations, exceeding formation pressure causes?**

Answer

- D** (A) Well improvement
- (B) Better well control
- (C) Increased productivity
- (D) Formation damage and potential fluid losses

79

OWO2.80 **Offset well data is reviewed to?**

Answer

- B** (A) Only compare production
- (B) Estimate formation pressures and well behavior
- (C) Determine equipment only
- (D) Assess drilling costs only

80

OWO2.81 **BOP Accumulators store?**

Answer

- B** (A) Drilling mud
- (B) Pressurized hydraulic fluid for closing rams
- (C) Gas only
- (D) Water only

81

OWO2.82 **BOP control systems require?**

Answer

- B** (A) Manual operation only
- (B) Redundant systems for reliability
- (C) Single control line
- (D) No backup systems

82

OWO2.83 **Accumulator pre-charge must be verified?**

Answer

- B** (A) Once per year only
- (B) Prior to each well intervention
- (C) Never required
- (D) Only after failures

83

OWO2.84 **Production tree valves should be used infrequently to ensure?**

Answer

- B** (A) Cost savings only
- (B) Valves remain functional for emergencies
- (C) Pressure maintenance only
- (D) Wear prevention only

84

OWO2.85 **Equipment selection for well control must account for?**

Answer

- D** (A) Cost only
- (B) Speed only
- (C) Availability only
- (D) Anticipated maximum pressures and operational conditions

85

OWO2.86 **Lubricators are required for which operations?**

Answer

- B** (A) Surface work only
- (B) Through-tubing wireline operations
- (C) Drilling only
- (D) Monitoring only

86

OWO2.87 **Choke manifold MAWP limits which operation?**

Answer

- B** (A) Circulation rate only
- (B) Maximum applicable backpressure during well control
- (C) Pump selection only
- (D) Fluid type only

87

OWO2.88 **Using non-OEM replacement parts may result in?**

Answer

- B** (A) Cost savings only
- (B) Premature wear or system failure
- (C) Improved performance
- (D) No operational impact

88

OWO2.89 **Pressure gauges must be calibrated?**

Answer

- D** (A) Once per year
- (B) Only after failures
- (C) Never required
- (D) Before each operation

89

OWO2.90 **Post-closure verification after well shut-in includes?**

Answer

- B** (A) Pressure reading only
- (B) Pressure reading and check for leaks
- (C) Visual inspection only
- (D) No verification

90

OWO2.91 **Management of Change (MOC) during well operations is important for?**

Answer

- B** (A) Documentation only
- (B) Identifying impacts of procedure changes on well control
- (C) Cost tracking only
- (D) Schedule management

91

OWO2.92 **When H2S is present, which kill method is preferred?**

Answer

- B** (A) Reverse circulation
- (B) Bullheading
- (C) Forward circulation
- (D) Lube and Bleed

92

OWO2.93 **H2S considerations require which operational adjustment?**

Answer

- D** (A) No change from standard procedures
- (B) Only increased monitoring
- (C) No special precautions needed
- (D) Modified well control procedures and protective equipment

93

OWO2.94 **Hydrate formation risk increases at?**

Answer

- D** (A) High temperatures
- (B) Shallow depths only
- (C) Warm depths
- (D) Low temperatures and high pressures

94

OWO2.95 **Well intervention preparation involves?**

Answer

- B** (A) Jumping directly to intervention
- (B) Securing well with appropriate barriers and documenting conditions
- (C) Only reviewing surface equipment
- (D) No advance preparation

95

OWO2.96 **Contingency procedures should address which scenarios?**

Answer

- D** (A) Only primary methods
- (B) No alternative methods
- (C) Routine operations only
- (D) Potential well control failures and equipment problems

96

OWO2.97 **Well control planning must include?**

Answer

- D** (A) Pressure data only
- (B) Equipment inventory only
- (C) No advance planning
- (D) Comprehensive pressure analysis and procedure development

97

OWO2.98 **Well control drills are conducted to ensure?**

Answer

- C** (A) Meeting regulations only
- (B) Filling shift time only
- (C) Personnel readiness and procedure familiarity
- (D) Equipment testing only

98

OWO2.99 **Operational documentation should include?**

Answer

- C** (A) Only pressure readings
- (B) Only fluid volumes
- (C) All pressures, volumes, procedures, and observations
- (D) No documentation needed

99

OWO2.100 **Post-operation review is critical for?**

Answer

- C** (A) Cost assessment only
- (B) Meeting reporting requirements only
- (C) Identifying issues and improving future well control procedures
- (D) Archive purposes only

100

OWO2.101 **What is a primary advantage of coiled tubing in well control operations?**

Answer

- C** (A) Continuous string reduces connection points
- (B) Allows single-pass deployment capability
- (C) Both advantages
- (D) Only for drilling operations

101

OWO2.102 **Which of the following is NOT a component of coiled tubing pressure control equipment (PCE)?**

Answer

- D** (A) Shear blind ram
- (B) Lubricator
- (C) Annular preventer
- (D) Power drill bit

102

OWO2.103 **What is the primary function of the injector head in coiled tubing operations?**

Answer

- A** (A) Provides gripping and feeding of the tubing
- (B) Controls pressure at wellhead
- (C) Prevents tubing collapse
- (D) Measures fluid volume

103

OWO2.104 **In coiled tubing operations, what does the stripper element primarily do?**

Answer

- B** (A) Strips mud from the tubing
- (B) Provides pressure control around the coiled tubing
- (C) Removes debris from fluid
- (D) Cools the tubing

104

OWO2.105 **The quad blowout preventer (QUAD BOP) in coiled tubing operations typically includes?**

Answer

- C** (A) Shear ram only
- (B) Pipe ram only
- (C) Shear rams, pipe rams, and blind rams
- (D) Check valves only

105

OWO2.106 **What is the purpose of the tubing arch guide in coiled tubing equipment?**

Answer

- C** (A) Guides tubing through the injector head
- (B) Prevents tubing kinking
- (C) Both A and B
- (D) Measures pressure

106

OWO2.107 **Which statement about coiled tubing disadvantages is MOST accurate?**

Answer

- C** (A) High pumping pressures required
- (B) Reduced strength compared to conventional tubing
- (C) Both limitations apply
- (D) Superior to all conventional methods

107

OWO2.108 **The flow check assembly in dynamic coiled tubing PCE serves to?**

Answer

- B** (A) Check fluid flow rate
- (B) Prevent backflow and control fluid direction
- (C) Measure fluid density
- (D) Adjust system pressure

108

OWO2.109 **In coiled tubing operations, what happens during a kick with pipe to bottom?**

Answer

- D** (A) Well must be shut-in immediately
- (B) Pressure builds at surface
- (C) Formation pressure exceeds hydrostatic pressure
- (D) All of the above

109

OWO2.110 **What is the primary advantage of using nitrogen with coiled tubing?**

Answer

- B** (A) Reduces wellbore pressure
- (B) Lightens the fluid column for underbalanced operations
- (C) Prevents formation damage
- (D) Improves tubing strength

110

OWO2.111 **In snubbing operations, a 'balance point' represents?**

Answer

- B** (A) Maximum safe pressure
- (B) Depth where downward and upward forces equalize
- (C) Surface pressure reading
- (D) Formation fracture point

111

OWO2.112 **Which factor is MOST critical in snubbing force calculations?**

Answer

- B** (A) Well temperature
- (B) Wellhead pressure and tubing cross-sectional area
- (C) Ambient surface conditions
- (D) Number of tools in wellbore

112

OWO2.113 **What does 'pipe heavy' mean in snubbing operations context?**

Answer

- B** (A) Pipe weight exceeds equipment capacity
- (B) Buoyed weight exceeds upward forces and pipe descends naturally
- (C) Pipe is damaged or heavy
- (D) Pipe weight in air

113

OWO2.114 **The slip interlock system in snubbing equipment prevents?**

Answer

- B** (A) Pressure from escaping
- (B) Uncontrolled pipe movement during slipping operations
- (C) Fluid from reaching surface
- (D) Formation collapse

114

OWO2.115 **In snubbing operations, buckling complications occur when?**

Answer

- B** (A) Pipe is too long
- (B) Compressive forces exceed critical buckling load
- (C) Wellhead pressure is too high
- (D) Fluid density is too high

115

OWO2.116 **What is the primary sealing element in a snubbing stripper assembly?**

Answer

- B** (A) Metal rings
- (B) Elastomer packings
- (C) Mechanical rams
- (D) Spring-loaded plungers

116

OWO2.117 **Which is the CORRECT formula for calculating balance point?**

Answer

- A** (A) $BP = \text{Snub Force} / \text{Buoyed Wt of Tubulars}$
- (B) $BP = \text{Wellhead Pressure} \times \text{Tubing Area}$
- (C) $BP = \text{Tubing Weight} \times \text{Formation Pressure}$
- (D) $BP = \text{TVD} \times \text{Fluid Density}$

117

OWO2.118 **In snubbing operations, annular stripping allows?**

Answer

- B** (A) Removal of annulus fluids only
- (B) Running pipe into pressurized well with annular preventer closed
- (C) Separation of tubing and casing
- (D) Pressure relief through the annulus

118

OWO2.119 **The equalizing loop and bleed-off line in snubbing PCE are used to?**

Answer

- A** (A) Equalize pressure and allow controlled fluid bleed-off
- (B) Increase system pressure
- (C) Separate different fluid types
- (D) Measure fluid density

119

OWO2.120 **What does collapse resistance in snubbing refer to?**

Answer

- C** (A) Ability to resist external pressure from formation
- (B) Resistance to tubing collapse under compression
- (C) Both A and B
- (D) Surface pressure limitations

120

OWO2.121 **The manual stuffing box in wireline operations provides?**

Answer

- B** (A) Automatic pressure control
- (B) Manual pressure control around wireline cable
- (C) Thermal protection only
- (D) Electrical isolation

121

OWO2.122 **What is the primary function of the lubricator in wireline operations?**

Answer

- B** (A) Reduces friction on wireline cable
- (B) Creates low-pressure working space for tool deployment
- (C) Prevents pressure spikes
- (D) Measures pressure

122

OWO2.123 **In wireline operations, a shear BOP is used to?**

Answer

- A** (A) Sever wireline cable and close well in emergency
- (B) Cut tools at depth
- (C) Prevent pressure rise
- (D) Measure well pressure

123

OWO2.124 Which wireline line type provides the smallest breaking strength?

Answer

- B** (A) Electric line
- (B) Solid wireline (slickline)
- (C) Braided line
- (D) Coated wireline

124

OWO2.125 The grease injection system in wireline operations serves to?

Answer

- A** (A) Provide lubrication and pressure control around cable
- (B) Lubricate winch bearings
- (C) Prevent corrosion only
- (D) Cool the wireline

125

OWO2.126 In wireline operations, what is the tool trap (or hydraulic tool catcher) designed to do?

Answer

- B** (A) Trap tools at surface
- (B) Catch and hold tools if cable breaks
- (C) Measure tool weight
- (D) Prevent tool rotation

126

OWO2.127 The quick unions in wireline pressure control equipment allow?

Answer

- A** (A) Rapid connection and disconnection of components
- (B) Quick pressure adjustment
- (C) Easy tool replacement
- (D) Rapid well control

127

OWO2.128 **What is the primary purpose of line wipers in wireline operations?**

Answer

- C** (A) Wipe debris from wireline cable
- (B) Clean fluid from surface
- (C) Both A and B
- (D) Measure cable thickness

128

OWO2.129 **Emergency Shut-Down Devices (ESDS) on wireline equipment are designed to?**

Answer

- C** (A) Automatically shut down operations if power is lost
- (B) Stop winch rotation during emergencies
- (C) Both A and B
- (D) Measure system pressure

129

OWO2.130 **The equalizing valve in wireline PCE is used to?**

Answer

- C** (A) Equalize pressure between lubricator and wellbore
- (B) Release pressure safely
- (C) Both A and B
- (D) Measure formation pressure

130

OWO2.131 **What is the primary hazard associated with hydrogen sulfide (H₂S)?**

Answer

- B** (A) Causes equipment corrosion only
- (B) Toxic gas that can be fatal at low concentrations
- (C) Increases formation pressure only
- (D) Reduces fluid viscosity

131

OWO2.132 **In H2S environments, which protective equipment is ESSENTIAL?**

Answer

- B** (A) Standard hard hats only
- (B) Self-contained breathing apparatus (SCBA)
- (C) Rubber gloves only
- (D) Safety glasses

132

OWO2.133 **What is the purpose of H2S drills in well control training?**

Answer

- B** (A) Test equipment strength
- (B) Practice emergency procedures in H2S conditions
- (C) Measure gas concentrations
- (D) Check equipment calibration

133

OWO2.134 **During H2S well control events, evacuation plans should prioritize?**

Answer

- A** (A) Wind direction and upwind evacuation routes
- (B) Nearest exit regardless of wind direction
- (C) Equipment recovery first
- (D) Personnel communication only

134

OWO2.135 **Blockages in wellbore or tubing during workover operations can result from?**

Answer

- B** (A) Sand accumulation only
- (B) Cement plugs, debris, or formation collapse
- (C) Pressure increase only
- (D) Temperature changes alone

135

OWO2.136 **When trapped pressure is identified below a barrier, the PRIMARY concern is?**

Answer

- B** (A) Loss of circulation
- (B) Potential overpressure leading to barrier failure
- (C) Equipment shutdown
- (D) Fluid viscosity change

136

OWO2.137 **Hydrates in wellbore operations typically form when?**

Answer

- B** (A) Temperature increases
- (B) Water and hydrocarbon gases combine under high pressure and low temperature
- (C) Pressure decreases
- (D) Only at surface

137

OWO2.138 **Management of Change (MOC) during a well kill is important because?**

Answer

- C** (A) It documents plan deviations
- (B) Ensures all personnel are informed of operational changes
- (C) Both A and B
- (D) It is optional

138

OWO2.139 **When performing wireline shear seal operations, the primary objective is?**

Answer

- B** (A) Measure formation pressure
- (B) Sever wireline and seal well if wireline operations fail
- (C) Test formation strength
- (D) Adjust wellhead pressure

139

OWO2.140 **Fracturing operations during workover pose risk because?**

Answer

- C** (A) Fracturing pressure can exceed well control equipment capacity
- (B) Can damage nearby wellbores
- (C) Both A and B
- (D) Reduces fluid viscosity

140

OWO2.141 **The Maximum Allowable Working Pressure (MAWP) represents?**

Answer

- B** (A) Operating limit of equipment
- (B) Maximum pressure equipment is designed to safely withstand
- (C) Fracture pressure of formation
- (D) Surface pressure reading

141

OWO2.142 **What is the difference between MAWP and MAOP?**

Answer

- A** (A) MAWP is design limit; MAOP is operational limit and may be lower
- (B) They are identical
- (C) MAOP is formation pressure related
- (D) MAWP is temporary

142

OWO2.143 **In a BOP system stack, the component limiting system MAOP is?**

Answer

- B** (A) The strongest component
- (B) The weakest component in the flow path
- (C) The lubricator
- (D) The accumulator

143

OWO2.144 **BOP closing units (accumulators) serve to?**

Answer

- A** (A) Store hydraulic energy for closing BOPs
- (B) Measure system pressure
- (C) Only cool equipment
- (D) Store fluid only

144

OWO2.145 **What is the primary function of a check valve in pressure control equipment?**

Answer

- D** (A) Prevents backflow of fluid
- (B) Measures pressure
- (C) Allows only one-way flow
- (D) Both A and C

145

OWO2.146 **Workstrings in workover operations are designed primarily to?**

Answer

- B** (A) Carry maximum load only
- (B) Deliver control and allow manipulation of tools while maintaining pressure control
- (C) Prevent well flow only
- (D) Measure depth

146

OWO2.147 **In pressure control equipment, redundancy means?**

Answer

- B** (A) Having extra equipment available
- (B) Multiple barriers or safety systems to prevent single-point failure
- (C) Unnecessary duplication
- (D) Only one backup system

147

OWO2.148 **What does 'electrical classification' of workover equipment address?**

Answer

- B** (A) Equipment power rating only
- (B) Equipment suitability for potentially explosive atmospheres (e.g., H2S zones)
- (C) Equipment color coding
- (D) Operating temperature range

148

OWO2.149 **Ram-type preventers in pressure control equipment are characterized by?**

Answer

- B** (A) Sliding elements that cut through pipe
- (B) Ram elements that close through center of stack
- (C) Spring-loaded mechanisms only
- (D) Always blind rams

149

OWO2.150 **Annular preventers differ from ram preventers because?**

Answer

- C** (A) They close around tubing elastically
- (B) They seal annular space around tool
- (C) Both A and B
- (D) They cannot handle backpressure

150

OWO2.151 **Kill Fluid Weight (KFW) calculation requires knowing?**

Answer

- B** (A) Formation pressure and TVD only
- (B) Formation pressure, TVD, hydrostatic pressure, and shut-in tubing pressure
- (C) Surface temperature only
- (D) Wellbore diameter

151

OWO2.152 **When correcting KFW for temperature, denser brines lose weight at HIGHER temperatures because?**

Answer

- A** (A) Fluid expands, reducing mass per unit volume
- (B) Pressure decreases with depth
- (C) Surface conditions change
- (D) Gravity increases

152

OWO2.153 **The Bullheading method of well control involves?**

Answer

- A** (A) Pumping kill fluid down tubing to displace influx into formation
- (B) Circulating kill fluid to bottom through annulus
- (C) Rotating pipe while pumping
- (D) Negative pressure technique

153

OWO2.154 **In the Volumetric Method of well control?**

Answer

- C** (A) Gas is allowed to migrate to the surface in a controlled manner
- (B) Well is shut-in and pressure monitored
- (C) Both A and B
- (D) Pipe rotation is required

154

OWO2.155 **The Lube and Bleed method allows?**

Answer

- C** (A) Lubricating annulus while bleeding off gas pressure slowly
- (B) Removal of gas from well without circulation
- (C) Both A and B
- (D) Rotating pipe only

155

OWO2.156 **Forward circulation (Driller's Method) kills well by?**

Answer

- C** (A) Circulating kill fluid from surface down through tubing to annulus
- (B) Removing influx and replacing wellbore fluid in one circulation
- (C) Both A and B
- (D) Pulling tubing from well

156

OWO2.157 **Reverse circulation differs from forward circulation because?**

Answer

- C** (A) Fluid flow direction is reversed
- (B) Kill fluid returns up through tubing instead of annulus
- (C) Both A and B
- (D) Pipe rotation is required

157

OWO2.158 **SITP (Shut-in Tubing Pressure) readings increase during a kick because?**

Answer

- C** (A) Formation influx enters tubing side
- (B) Pressure builds in tubing from gas migration
- (C) Both A and B
- (D) Surface pressure increased

158

OWO2.159 **The correct action when well control is lost during operations is?**

Answer

- A** (A) Shut-in well immediately at wellhead and subsurface isolating devices
- (B) Continue pumping
- (C) Open choke and reduce pressure
- (D) Increase fluid circulation rate

159

OWO2.160 **When transitioning from Volumetric to circulating kill method, priority is?**

Answer

- C** (A) Maintaining pressure control
- (B) Establishing safe circulation rates
- (C) Both A and B
- (D) Reducing system pressure

160

OWO2.161 **Well Integrity refers to the ability of well to?**

Answer

- B** (A) Produce hydrocarbon only
- (B) Contain and control pressure, fluids, and gases safely
- (C) Withstand maximum pressure only
- (D) Allow fluid circulation

161

OWO2.162 **Mechanical barriers in well control include?**

Answer

- B** (A) Packers and plugs only
- (B) Casing, tubing, packers, and bridge plugs
- (C) Only weighted fluids
- (D) Pressure gauges

162

OWO2.163 **Operational barriers in workover operations primarily consist of?**

Answer

- B** (A) Equipment components only
- (B) Hydrostatic pressure from weighted fluids
- (C) Casing connections only
- (D) BOP systems only

163

OWO2.164 **Sustained casing pressure (CP) readings exceeding 500 psi may indicate?**

Answer

- B** (A) Normal well condition
- (B) Breach in casing or cement integrity
- (C) Formation pressure decrease
- (D) Increased wellhead pressure

164

OWO2.165 **Barrier Management in well control requires?**

Answer

- B** (A) Identifying single barriers only
- (B) Establishing multiple barriers and validating each
- (C) Only surface barriers
- (D) Pressure monitoring only

165

OWO2.166 **Testing downhole completion equipment barriers involves?**

Answer

- C** (A) Pressure testing to verify seal integrity
- (B) Confirming equipment can handle anticipated pressures
- (C) Both A and B
- (D) Visual inspection only

166

OWO2.167 **A 'barrier' differs from a 'pressure control device' in that?**

Answer

- C** (A) Barriers contain fluids; PCE manages pressure
- (B) Barriers isolate zones; PCE regulates flow
- (C) Both A and B
- (D) They are identical

167

OWO2.168 **The wellbore fluid serves as a barrier when?**

Answer

- C** (A) Its hydrostatic pressure exceeds formation pressure
- (B) It maintains overbalanced condition
- (C) Both A and B
- (D) Only at surface

168

OWO2.169 **Bridge plugs and packers as barriers function to?**

Answer

- C** (A) Isolate and seal specific wellbore sections
- (B) Prevent fluid migration between zones
- (C) Both A and B
- (D) Only block flow

169

OWO2.170 **In barrier management, pressure differential across barriers must be monitored because?**

Answer

- C** (A) High differential may damage equipment
- (B) Low differential may compromise seal
- (C) Both pressure extremes create risks
- (D) It is not necessary

170

OWO2.171 **When a kick is detected, the first action is?**

Answer

- B** (A) Call for assistance
- (B) Shut-in well at surface and close downhole barriers
- (C) Increase circulation rate
- (D) Open choke

171

OWO2.172 Monitoring during shut-in requires recording of?

Answer

- C** (A) SITP and SICP readings at regular intervals
- (B) Pit gain and wellbore fluid levels
- (C) Both A and B
- (D) Only mud gas detector

172

OWO2.173 Stripping operations are procedures used to?

Answer

- A** (A) Run or pull pipe through pressurized wellbore
- (B) Maintain pressure control with annular element closed
- (C) Both A and B
- (D) Only pulling operations

173

OWO2.174 Preparing for well entry requires?

Answer

- C** (A) Verifying well is stable and secure
- (B) Confirming barriers are in place and tested
- (C) Both A and B
- (D) Pressurizing wellbore

174

OWO2.175 Pre-job communication meetings must address?

Answer

- C** (A) Operational procedures and pressure limits
- (B) Emergency protocols and personnel roles
- (C) Both A and B
- (D) Equipment only

175

OWO2.176 **Verification of shut-in means confirming that?**

Answer

- C** (A) All surface and downhole isolating devices are closed
- (B) Pressure is contained and no flow occurs
- (C) Both A and B
- (D) Pit levels are stable

176

OWO2.177 **Coiled tubing failures during operations may result from?**

Answer

- C** (A) Fatigue from cyclic loading
- (B) Buckling or kinking in deviated wells
- (C) Both A and B
- (D) Pressure only

177

OWO2.178 **When handling kill problems during well control, first step is?**

Answer

- B** (A) Increase pump rate
- (B) Identify root cause of problem
- (C) Change kill method
- (D) Reduce wellhead pressure

178

OWO2.179 **Well control risks in wireline operations include?**

Answer

- C** (A) Cable break and loss of tool
- (B) Pressure surge if tool becomes stuck
- (C) Both A and B
- (D) Equipment noise only

179

OWO2.180 **Contingency procedures provide?**

Answer

- C** (A) Predetermined response to anticipated equipment failures
- (B) Backup methods for pressure control
- (C) Both A and B
- (D) Only advisory guidance

180

OWO2.181 **Boyle's Law states that for a gas at constant temperature?**

Answer

- B** (A) Pressure and volume are directly proportional
- (B) Pressure and volume are inversely proportional
- (C) Temperature increases with volume
- (D) Pressure remains constant

181

OWO2.182 **In well control, gas expansion becomes critical when?**

Answer

- C** (A) Gas moves from high pressure to low pressure region
- (B) Total Pit Gain (TPG) indicates influx volume
- (C) Both A and B
- (D) Temperature increases

182

OWO2.183 **Total Pit Gain (TPG) calculation helps determine?**

Answer

- C** (A) Whether gas kick has been taken
- (B) Approximate volume of influx
- (C) Both A and B
- (D) Pressure at surface

183

OWO2.184 **Gas migration can lead to?**

Answer

- C** (A) Reduced hydrostatic pressure in wellbore
- (B) Formation fluid influx into wellbore
- (C) Both A and B
- (D) Increased casing pressure

184

OWO2.185 **Free gas in wellbore behaves differently from liquid because?**

Answer

- C** (A) Gas is compressible while liquid is essentially incompressible
- (B) Gas expands under pressure reduction
- (C) Both A and B
- (D) Gas has higher density

185

OWO2.186 **During influx detection, increase in mud pit level indicates?**

Answer

- C** (A) Possible gas kick
- (B) Fluid influx from formation
- (C) Both A and B
- (D) Normal circulation loss

186

OWO2.187 **Mud gas detector readings provide indication of?**

Answer

- B** (A) Total gas volume only
- (B) Hydrocarbon gas concentration in return flow
- (C) Formation pressure
- (D) Wellbore temperature

187

OWO2.188 **When gas kick is detected while circulating, procedure should?**

Answer

- D** (A) Stop circulation immediately
- (B) Continue controlled circulation and monitor readings
- (C) Open choke fully
- (D) Close well at surface

188

OWO2.189 **Characteristics of wellbore fluids include?**

Answer

- C** (A) Density, viscosity, and gel strength only
- (B) Thermal properties and corrosivity
- (C) All of the above
- (D) Pressure only

189

OWO2.190 **In well control situations, fluid type affects?**

Answer

- C** (A) Hydrostatic pressure calculations
- (B) Formation damage potential
- (C) Both A and B
- (D) Surface pressure only

190

OWO2.191 **Primary functions of workover fluids include?**

Answer

- C** (A) Hydrostatic pressure support only
- (B) Carrying cuttings, heat transfer, and system cooling
- (C) Both A and B
- (D) Pressure reduction only

191

OWO2.192 **Completion fluids are selected based on?**

Answer

- C** (A) Pressure requirements only
- (B) Formation characteristics and expected wellbore conditions
- (C) Both A and B
- (D) Availability only

192

OWO2.193 **Fluid flow behavior in annulus is characterized by?**

Answer

- B** (A) Laminar flow only
- (B) Laminar or turbulent depending on flow rate and wellbore geometry
- (C) Always turbulent
- (D) No flow patterns

193

OWO2.194 **Validating workover fluid as barrier requires confirming?**

Answer

- C** (A) Sufficient density for overbalanced condition
- (B) Stability at anticipated downhole temperatures
- (C) Both A and B
- (D) Only surface pressure

194

OWO2.195 **Fluid properties that impact well control include?**

Answer

- C** (A) Density (ppg) and viscosity (cp)
- (B) Thermal stability and gel strength
- (C) Both A and B
- (D) Only density

195

OWO2.196 **Brine selection for workover operations considers?**

Answer

- C**
- (A) Chemical compatibility with formation
 - (B) Corrosion protection requirements
 - (C) Both A and B
 - (D) Cost only

196

OWO2.197 **Equivalent Circulating Density (ECD) accounts for?**

Answer

- D**
- (A) Hydrostatic pressure of fluid column only
 - (B) Additional pressure from annular friction losses
 - (C) Only casing pressure
 - (D) Both A and B

197

OWO2.198 **Safety margin in fluid selection means?**

Answer

- A**
- (A) Additional hydrostatic pressure above formation pressure
 - (B) Preventing formation fracture and wellbore collapse
 - (C) Both A and B
 - (D) Using maximum rated equipment

198

OWO2.199 **Temperature effects on workover fluids include?**

Answer

- C**
- (A) Density reduction at higher temperatures
 - (B) Viscosity changes affecting circulation
 - (C) Both A and B
 - (D) Pressure reduction only

199

OWO2.200 **Fluid cleanliness in workover operations is important because?**

Answer

- C**
- (A) Contaminants can damage equipment seals
 - (B) Particles may plug formation and restrict flow
 - (C) Both A and B
 - (D) It affects wellhead pressure

200

OGOR Workover Practice Test Day 2

OWOT2.1 **Using Boyle's Law, if a 10 bbl gas influx enters an Open Well at 10,000 ft, what is the expected change in volume when the gas reaches 2,500 ft? (Disregard possible secondary influxes)**

Answer

- D** (A) 80 bbls
(B) 10 bbls
(C) 20 bbls
(D) 40 bbls

1

OWOT2.2 **What is the main cause of well control problems during workover and completion operations?**

Answer

- B** (A) The fluid density is too heavy
(B) Unrecognized warning signs of influxes
(C) Not enough fluid available at the well site
(D) Abnormal pressure

2

OWOT2.3 **Prior to POOH with tubing, what operation should be performed?**

Answer

- C** (A) Pump diesel for a complete circulation
(B) Pump a high viscosity fluid for a complete circulation
(C) Pump kill fluid for complete circulation
(D) Pump lost circulation material to plug off perforations

3

OWOT2.4 **If a kick migrates half-way up an annulus which is not shut-in, what will its volume do?**

Answer

- B** (A) Volume stays the same
- (B) Volume will double
- (C) Volume will be half
- (D) Volume will triple

4

OWOT2.5 **How is a flow tee/cross used in a BOP stack?**

Answer

- C** (A) Aids in gas separation
- (B) Supports coiled tubing after shearing
- (C) Primary path of flow to circulate without damaging the BOP
- (D) Allows emergency access to BOPs

5

OWOT2.6 **What would be a valid reason to shut down during a forward gas kick circulation before gas is at the surface?**

Answer

- A** (A) Casing pressure and/or pit level is not increasing
- (B) Casing pressure is increasing as gas reaches surface
- (C) Casing pressure and/or pit level is not decreasing
- (D) To switch over to reverse circulating

6

OWOT2.7 **During bullheading if an excess safety margin is added to the pump pressure, what may occur?**

Answer

- D** (A) May cause total losses if pumped slowly
- (B) Increase the bottom hole pressure during the kill operation and damage the formation
- (C) Cause less bottomhole pressure that can fracture the formation
- (D) Not pump at a fast enough rate to push the well fluids back in formation

7

OWOT2.8 **A bullhead kill should take which factors into consideration?**

Answer

- B** (A) Pump at a rate to overcome gas migration, how much fluid is being pumped and the distance from other wells
- (B) Pump at a rate that overcomes gas migration, density of fluids being pumped and maximum pressures
- (C) The density of the fluids being pumped, tailpipe length and rathole depth
- (D) Temperature corrections of surface fluids to downhole temperature environments must be monitored

8

OWOT2.9 **A kill operation should have which of the following sets of information available at the wellsite?**

Answer

- A** (A) Tubing/casing diameters/capacities, depths of tubulars/perforations, kill fluid density, volumes of fluids needed
- (B) Tubing/casing diameters, depths of tubulars/perforations, capacities of tubulars, distance from other locations and kill fluid density
- (C) Tubing/casing diameters/capacities, depths of tubulars/perforations, date of last service and volumes of fluids needed
- (D) Tubing/casing diameters/capacities, depths of only casing, date of casing installation and volumes of fluids needed

9

OWOT2.10 **In which part of a well with no packer is the frictional pressure lowest while forward circulating?**

Answer

- C** (A) Across a flow restriction
- (B) Inside the tubing
- (C) In the annulus
- (D) Across surface flow lines

10

OWOT2.11 **Which method is used to remove gas from a well if it is already at the surface and circulation or bullheading is not possible?**

Answer

- D** (A) Forward Circulation
- (B) Volumetric
- (C) Reverse Circulation
- (D) Lube and bleed

11

OWOT2.12 **Use of the Driller's Method (forward circulation) is used in which case?**

Answer

- C** (A) When Hydrogen Sulfide needs to be prevented from migrating to surface
- (B) When the BHA is packed off and the crew cannot circulate
- (C) When an influx was swabbed in while tripping out of the well
- (D) When casing has ruptured

12

OWOT2.13 **While pumping in reverse between the intermediate and production casing, what would result with increased pressure acting on intermediate casing?**

Answer

- A** (A) The intermediate casing could rupture
- (B) The intermediate casing could collapse
- (C) The intermediate casing could cause swab the well and induce a kick
- (D) The intermediate casing would separate due to high tensile loading

13

OWOT2.14 **Which of the following is true concerning the Lube and Bleed Method?**

Answer

- C** (A) The Lubricate and Bleed method can be used during total loss of fluid in wellbore
- (B) The Lubricate and Bleed method is faster than circulating
- (C) No tubing is needed
- (D) The Lubricate and Bleed method can be used immediately after an influx enters the well

14

OWOT2.15 **What is one reason to use forward circulation?**

Answer

- A** (A) Less likely to collapse tubing
- (B) Higher equivalent circulating density (ECD) than reverse circulating
- (C) Forces influx back into the formation
- (D) Unable to perform reverse circulation

15

OWOT2.16 How is reverse circulation done?

Answer

- A** (A) Pumping down the annulus and taking returns through the tubing
- (B) Pumping down tubing, taking returns from annulus
- (C) Pumping down tubing, displacing well fluid into formation
- (D) Pumping through flow cross from one side and use choke for backpressure

16

OWOT2.17 What can be done if there is a chance that collapsed tubing will occur with a high tensile load?

Answer

- C** (A) Increase annulus pressure
- (B) Decrease tubular pressure
- (C) Bleed off pressure on the annulus
- (D) Pull on tubing to increase high tensile load

17

OWOT2.18 In which well kill method do we use forward circulation?

Answer

- D** (A) Volumetric
- (B) Reverse Circulation
- (C) Lube and Bleed
- (D) Forward Circulation (Driller's Method)

18

OWOT2.19 Prior to bullheading, what does increasing pressure on the annulus above a packer accomplish?

Answer

- A** (A) Determine if the tubing or casing begins to leak
- (B) To determine the annulus volume
- (C) To make sure the annulus equals the formation pressure
- (D) To verify pressure seal of casing and casing cement

19

OWOT2.20 **What limits the maximum pump pressure during a bullhead operation?**

Answer

- A** (A) Lowest pressure rated component
- (B) Allowable pump pressure is limited only by formation permeability
- (C) Highest pressure rated component
- (D) The heaviest fluid to be pumped

20

OWOT2.21 **Why is the forward circulating method used?**

Answer

- A** (A) To avoid the effects on bottomhole pressure of excessive tubing friction pressures while reverse circulating
- (B) When undesirable gas needs to be prevented from migrating to surface
- (C) When a fast removal of gas from the well is desired
- (D) Forward circulation is used to establish low friction rates before switching over to reverse circulating rates

21

OWOT2.22 **Which of the following conditions may cause shut-in complications?**

Answer

- A** (A) Running a production packer across the BOP stack
- (B) Running large tubing size
- (C) Running very small tubing size
- (D) Pulling a production packer through a stuck point in casing

22

OWOT2.23 **Prior to bullheading into a well with a packer set with tubing installed, why is it necessary to add additional annular pressure?**

Answer

- A** (A) Confirm annulus is full and help resist tubing burst
- (B) Make sure the packer is set
- (C) Make sure the annulus equals the formation pressure
- (D) It is not necessary to add additional pressure, as internal and external pressures will always be the same

23

OWOT2.24 **Annular regulator pressure may be needed during this type of operation?**

Answer

- C** (A) Recovering a packer
- (B) Tripping Pipe
- (C) Snubbing
- (D) Reverse circulating

24

OWOT2.25 **Under what conditions would annular regulator pressure adjustment be needed?**

Answer

- D** (A) Reverse circulating
- (B) Running fishing tools across the stack
- (C) Recovering a packer
- (D) Stripping

25

OWOT2.26 **A redundant float or downhole check valve is required during the following operation?**

Answer

- A** (A) Pumping at a high rate for extended periods for time
- (B) Highly deviated wellbore
- (C) Pumping at a low rate for extended periods of time
- (D) Any operations with H₂S

26

OWOT2.27 **Which circulation method has the largest effect on bottomhole pressure?**

Answer

- A** (A) Reverse Circulation
- (B) Forward Circulation
- (C) Lube and Bleed
- (D) Volumetric

27

OWOT2.28 **After completion of the forward circulation method, if the Shut-in Tubing Pressure (SITP) and Shut-in Casing Pressure (SICP) are not equal, what is the safest next step?**

Answer

- D** (A) Pull out of the hole
- (B) Reverse circulate one tubing volume at low rate
- (C) Evacuate to muster point
- (D) Circulate another bottoms-up to ensure influx is removed

28

OWOT2.29 **In the event of a hydrogen sulfide (H₂S) kick, which well control method should be used?**

Answer

- C** (A) Forward Circulate
- (B) Lube and Bleed
- (C) Bullhead
- (D) Reverse Circulate

29

OWOT2.30 **What is the major advantage of using the forward circulating method?**

Answer

- D** (A) Enviromentally safe
- (B) Forces influxes back into the formation
- (C) Typically, faster than a reverse circulation
- (D) Lower equivalent circulating density than reverse circulating

30

OWOT2.31 **For a bullhead kill sheet, which following depth is used to determine Kill Fluid Weight (KFW)?**

Answer

- A** (A) Top perforation TVD
- (B) Bottom perforation TVD
- (C) Bottom perforation MD
- (D) Bottom perforation TVD

31

OWOT2.32 **Based on the following statements, which statement is NOT true?**

Answer

- D**
- (A) When bullheading fluid into the well, it is best to always remain below the fracture pressure in order to minimize damage to the formation(s)
 - (B) Exceeding the minimum stress in the formation is required for both fracturing pressure and performing a leak off test
 - (C) The formation fracture may be permanent or may close once pressure is released
 - (D) When bullheading fluid into the well, it is best to always exceed the fracture pressure to push the fluid into the formation

32

OWOT2.33 **What changes occur when a well is shut in with a 10 bbl gas influx at 5,000 ft as the influx migrates to 2,500 ft?**

Answer

- A**
- (A) Volume stays the same, surface pressure increases
 - (B) Volume stays the same, surface pressure stays the same
 - (C) Volume doubles, surface pressure increases
 - (D) Volume triple, surface pressures increases

33

OWOT2.34 **In which situation is the Driller's Method (forward circulation with current brine weight) most effective?**

Answer

- C**
- (A) A quick well kill is desired with gas at surface in tubing
 - (B) Killing a well with a production packer installed prior to a workover
 - (C) An influx was swabbed in while tripping out of the well
 - (D) Well with influx and total losses

34

OWOT2.35 **In which situation is the Driller's Method (forward circulation with current brine weight) most effective?**

Answer

- D**
- (A) Well with influx and total losses
 - (B) Killing a well with a production packer installed prior to a workover
 - (C) A well was shut in after flowing with no workstring in hole
 - (D) To avoid the effects on bottomhole pressure of excessive tubing friction pressure while reverse circulating

35

OWOT2.36 **During forward circulation, what is the annular effect caused by downhole tools with larger outer diameters (ODs)?**

Answer

- A** (A) Higher annular friction loss
- (B) Lower annular friction loss
- (C) Lower tubing friction pressure
- (D) Higher tubing friction pressure

36

OWOT2.37 **During forward circulation, where is frictional pressure lowest?**

Answer

- B** (A) Across BHA restriction(s)
- (B) In the annulus
- (C) Inside the tubing
- (D) At pump

37

OWOT2.38 **What is the primary function of a sliding sleeve?**

Answer

- C** (A) Real time measurement of downhole pressure
- (B) Allows for artificial lift production
- (C) Allows for communication or isolation of specific interval
- (D) None of the above

38

OWOT2.39 **Which statement best describes the action needed for balancing pressure prior to pulling tubing from a producing well?**

Answer

- D** (A) Displace formation fluid with Nitrogen (NO₂)
- (B) Run a gauge ring
- (C) Fill trip tank with KMW
- (D) None of the above

39

OWOT2.40 **For producing well, when preparing to remove the tubing, why is bullheading chosen over circulation?**

Answer

- A** (A) H₂S gas in well
- (B) Hole in the tubing
- (C) Wireline fish in the hole
- (D) Close proximity to production gathering station

40

OWOT2.41 **Which of the following statements best describes when bullheading should be selected over any circulating method?**

Answer

- D** (A) Inability to pump into well
- (B) Hole in the tubing
- (C) Plugged perforations
- (D) Unable to handle influx at surface

41

OWOT2.42 **For production tree valves, what valve should be used infrequently to ensure the valve will hold pressure during an emergency?**

Answer

- B** (A) Swab Valve
- (B) Lower Master Valve
- (C) Wing Valve
- (D) Upper Master Valve

42

OWOT2.43 **Using Boyle's Law, if a 5 bbl gas influx enters an Open Well at 10,000 ft, what is the expected change in volume when the gas reaches 2,500 ft? (Disregard possible secondary influxes)**

Answer

- D** (A) Volume stays the same
- (B) Volume triples
- (C) Volume gets smaller
- (D) Volume is 4 times larger

43

OWOT2.44 **The Volumetric Method is employed for which of the following situations?**

Answer

- C** (A) When it is not desirable to bring dangerous gases to surface
- (B) When circulation is possible and the well needs to be killed quickly
- (C) Plugged string
- (D) When circulating in possible, but will require two circulations to kill the well (i.e. Driller's Method)

44

OWOT2.45 **During what operation is adjusting the annular pressure regulator done?**

Answer

- C** (A) During wireline operations
- (B) While making repairs to the production tree
- (C) During stripping operations
- (D) During Lube and Bleed operations

45

OWOT2.46 **During snubbing operations, what occurs when the tubing integrity is exceeded due to frictional forces?**

Answer

- D** (A) Collapse of pipe
- (B) Loss of circulation
- (C) Reduction in fluid weight
- (D) Parting pipe

46

OWOT2.47 **The Volumetric Method is employed for which of the following situations?**

Answer

- C** (A) When it is not desirable to bring dangerous gases to surface
- (B) When circulation is possible and the well needs to be killed quickly
- (C) Pipe is out of the hole
- (D) When choke has partially plugged

47

OWOT2.48 **What are the benefits of using the Lube and Bleed Method?**

Answer

- C** (A) The Lubricate and Bleed method is faster than the Driller's Method
- (B) The Lubricate and Bleed method can be used immediately after an influx enters the well
- (C) Lubricate and Bleed method can still be used when unable to circulate
- (D) Simplest well control method

48

OWOT2.49 **What method can be used to remove gas from the well if bullheading is not an option and gas has reached the surface?**

Answer

- B** (A) Bleed off gas head and shut back in
- (B) Lube and bleed
- (C) Volumetric
- (D) None of the above

49

OWOT2.50 **What are the drawbacks when using the Lube and Bleed method?**

Answer

- C** (A) Only works for a oil based mud
- (B) Does not remove gas from the wellbore
- (C) Potential for high surface pressures that exceed maximum allowable annular surface pressure
- (D) Lube and Bleed only works on Steam Wells

50

OWOT2.51 **What is the conversion factor used in the hydrostatic pressure formula?**

Answer

- C** (A) 0.042
- (B) 0.062
- (C) 0.052
- (D) 0.072

51

OWOT2.52 In a 10,000 ft TVD well filled with 10 ppg fluid, what is the hydrostatic pressure?

Answer

- A** (A) 5,200 psi
- (B) 4,200 psi
- (C) 6,200 psi
- (D) 3,200 psi

52

OWOT2.53 Which type of pressure results from external forces acting on the wellbore system?

Answer

- C** (A) Hydrostatic Pressure
- (B) Formation Pressure
- (C) Applied Pressure
- (D) Pore Pressure

53

OWOT2.54 What does differential pressure indicate when it is negative?

Answer

- B** (A) Well is in equilibrium
- (B) Risk of reservoir fluids flowing into wellbore
- (C) Well is overbalanced
- (D) Formation is fractured

54

OWOT2.55 What does SITP stand for?

Answer

- B** (A) Shut-in Total Pressure
- (B) Shut-in Tubing Pressure
- (C) Surface Injection Test Pressure
- (D) Stabilized Inner Tubing Pressure

55

OWOT2.56 **Formation Pressure equals which of the following?**

Answer

- D** (A) HP minus SITP
- (B) HP times SITP
- (C) SITP divided by HP
- (D) HP plus SITP

56

OWOT2.57 **What is the significance of Formation Fracture Pressure in well operations?**

Answer

- D** (A) Indicates pore pressure
- (B) Surface pressure limit
- (C) Maximum hydrostatic pressure
- (D) The pressure threshold where formation begins to fracture

57

OWOT2.58 **Bottomhole Pressure is equal to which formula?**

Answer

- D** (A) Formation Pressure minus SITP
- (B) SICP divided by HP
- (C) Surface Pressure plus HP
- (D) HP plus SITP

58

OWOT2.59 **When BHP equals formation pressure, the well is considered:**

Answer

- C** (A) Underbalanced
- (B) Overbalanced
- (C) Balanced
- (D) Over-pressured

59

OWOT2.60 **An underbalanced condition exists when which scenario occurs?**

Answer

- B** (A) BHP exceeds formation pressure
- (B) BHP is less than formation pressure
- (C) BHP equals formation pressure
- (D) Formation pressure is stable

60

OWOT2.61 **Trapped pressure most commonly occurs when which situation exists?**

Answer

- B** (A) Well is flowing
- (B) Blockage or restriction impedes fluid flow
- (C) Formation is fractured
- (D) Tubing is open-ended

61

OWOT2.62 **What is the primary effect of swabbing during workover operations?**

Answer

- A** (A) Decrease in wellbore pressure
- (B) Increase in wellbore pressure
- (C) Stabilization of wellbore pressure
- (D) No change in wellbore pressure

62

OWOT2.63 **Surging is typically caused by which action?**

Answer

- B** (A) Pulling equipment out of the well
- (B) Pushing equipment into the well
- (C) Maintaining static conditions
- (D) Circulating at low rates

63

OWOT2.64 Which pressure concern is most critical when swabbing?

Answer

- B** (A) Formation fracture
- (B) Influx of formation fluids
- (C) Equipment rupture
- (D) Pressure stabilization

64

OWOT2.65 In reverse circulation, what type of friction pressure result occurs?

Answer

- B** (A) Lower friction losses
- (B) Higher backpressure on formation
- (C) No frictional effects
- (D) Reduced annular pressure

65

OWOT2.66 Equivalent Circulating Density must be monitored against which of the following?

Answer

- B** (A) Formation pressure only
- (B) Equivalent Fluid Weight to prevent fracturing
- (C) Pore pressure only
- (D) Surface pressure only

66

OWOT2.67 How is pressure converted to fluid density and vice versa?

Answer

- C** (A) Using 0.042 conversion factor
- (B) Using 0.062 conversion factor
- (C) Using 0.052 conversion factor
- (D) Using 0.072 conversion factor

67

OWOT2.68 **A gradient represents which measurement?**

Answer

- B** (A) Total pressure at depth
- (B) Rate of change of pressure with respect to depth
- (C) Fluid density only
- (D) Formation pressure change

68

OWOT2.69 **Maximum Anticipated Surface Pressure is calculated as:**

Answer

- B** (A) Formation Pressure plus HP
- (B) Formation Pressure minus HP
- (C) HP plus SITP
- (D) Formation Pressure divided by HP

69

OWOT2.70 **Kill Fluid Weight must be rounded which direction for safety?**

Answer

- B** (A) Down
- (B) Up
- (C) To nearest whole number
- (D) To nearest 0.1 ppg

70

OGOR Well Service Study Guide Day 3

OWS3.1 **What is the Rig Assist Space Saver unit designed for?**

Answer

- C** (A) Arctic conditions exclusively
- (B) Deepwater applications only
- (C) Lower-pressure work typically under 3,500 PSI with minimal tool access
- (D) High-pressure operations above 5,000 PSI

1

OWS3.2 **What indicator might suggest compromised well integrity?**

Answer

- C** (A) Low surface pressure only
- (B) Reduced circulation rates
- (C) Identification of sustained casing pressure (CP) exceeding 500 psi
- (D) Increased wellhead temperature

2

OWS3.3 **Which elements ensure barrier reliability?**

Answer

- D** (A) Only maintenance
- (B) Regular maintenance and inspection schedules
- (C) Documented testing and verification results
- (D) Both B and C

3

OWS3.4 **Which statements are correct about coiled tubing advantages?**

Answer

- C**
- (A) Ability to maintain continuous circulation throughout interventions
 - (B) Capability to work on pressurized live wells without killing production
 - (C) Both A and B
 - (D) Both A and B are incorrect

4

OWS3.5 **At what wellhead pressure are Hydraulic Workover Units (Snubbing Units) typically mobilized?**

Answer

- A**
- (A) When wellhead pressure exceeds 2,000 PSI
 - (B) When pressure exceeds 1,000 PSI
 - (C) When pressure exceeds 3,500 PSI
 - (D) At any measurable pressure above zero

5

OWS3.6 **Which is NOT a standard well control requirement?**

Answer

- A**
- (A) Continuous wellbore pressure increase
 - (B) Personnel communication and training
 - (C) Regular equipment testing
 - (D) Multiple barrier implementation

6

OWS3.7 **What condition most accurately describes pipe light situations?**

Answer

- B**
- (A) When wellbore pressure equals zero
 - (B) When weight of buoyant workstring is less than upward force from wellbore pressure
 - (C) When circulation is stopped completely
 - (D) When pipe weight exceeds upward forces

7

OWS3.8 **Which type of snubbing unit has the largest operational footprint capability?**

Answer

- D** (A) Rig Assist Space Saver units
- (B) Portable snubbing units
- (C) Rig Assist Snubbing Units only
- (D) Stand Alone Snubbing Units (Hydraulic Workover Units)

8

OWS3.9 **How many significant bending events occur during each operational cycle of coiled tubing?**

Answer

- D** (A) Two bending events per cycle
- (B) Four bending events per cycle
- (C) Five bending events per cycle
- (D) Six significant bending events per operational cycle

9

OWS3.10 **Which factors affect coiled tubing operational performance?**

Answer

- C** (A) Maintaining detailed records of deployment cycles
- (B) Implementing maximum cycle limits for each string
- (C) Both A and B
- (D) Both A and B are incorrect

10

OWS3.11 **In pipe heavy conditions, what would happen to unrestrained pipe?**

Answer

- B** (A) The pipe would float at fixed depth
- (B) The pipe would fall into well potentially damaging downhole equipment
- (C) The pipe would remain stationary
- (D) The pipe would rise to the surface

11

OWS3.12 **What is the primary purpose of pressure control equipment in well interventions?**

Answer

- A** (A) To create and maintain effective barriers preventing uncontrolled wellbore fluid flow
- (B) To reduce wellbore temperature
- (C) To eliminate all wellbore pressure
- (D) To increase circulation rates significantly

12

OWS3.13 **What is the minimum recommended core radius diameter ratio for coiled tubing services power reels?**

Answer

- A** (A) At least 20 times the specified coiled tubing diameter
- (B) At least 8 times the coiled tubing diameter
- (C) At least 15 times the tubing wall thickness
- (D) At least 3 times the specified yield strength

13

OWS3.14 **At what wellhead pressure are Hydraulic Workover Units (Snubbing Units) typically mobilized?**

Answer

- A** (A) When wellhead pressure exceeds 2,000 PSI
- (B) When pressure exceeds 1,000 PSI
- (C) When pressure exceeds 3,500 PSI
- (D) At any measurable pressure above zero

14

OWS3.15 **What measurement threshold indicates possible casing integrity compromise?**

Answer

- D** (A) 200 psi sustained casing pressure
- (B) 300 psi sustained casing pressure
- (C) 400 psi sustained casing pressure
- (D) 500 psi or greater sustained casing pressure (CP)

15

OWS3.16 **How does tubing arc guide design address buckling prevention?**

Answer

- C** (A) Increases wall thickness significantly
- (B) Reduces fluid pressure
- (C) Provides support achieving bending radius of 30 times tubing outer diameter
- (D) Eliminates all bending forces

16

OWS3.17 **Which apply to Bullhead operations?**

Answer

- B** (A) High-pressure pumps force kill fluid through wellbore
- (B) Both A and C
- (C) Hydraulics simulator models fluid behavior
- (D) Only pumps

17

OWS3.18 **What critical function does the Equalizing Loop perform in snubbing?**

Answer

- B** (A) Reduces pump pressures only
- (B) Balances pressure between rams allowing controlled transition as workstring moves
- (C) Shears the workstring in emergency
- (D) Isolates the annulus

18

OWS3.19 **Tapered coiled tubing can be manufactured with how many sections of varying wall thickness?**

Answer

- D** (A) Three sections
- (B) Four sections
- (C) Five sections
- (D) Six sections (forming a 5-step taper)

19

OWS3.20 Which requirements govern connection design?

Answer

- B** (A) Gradual tapered insertion necks for force distribution
- (B) Both A and C
- (C) Removal of internal manufacturing welds
- (D) Only taper needed

20

OWS3.21 Where can Bullhead Method fluid be pumped in coiled tubing operations?

Answer

- C** (A) Only through surface delivery
- (B) Only down the coiled tubing conduit
- (C) Down CT itself, through CT annulus, or via kill line when CT unavailable
- (D) Only through the coiled tubing annulus

21

OWS3.22 What is NOT a recognized failure point in coiled tubing operations?

Answer

- D** (A) Ovality-induced collapse
- (B) Bending fatigue failures
- (C) Connection failures
- (D) Unexplained atmospheric pressure changes

22

OWS3.23 Which factors impact coiled tubing service life?

Answer

- C** (A) Accumulated cycle counts and operational history
- (B) Exposure to corrosive fluids during procedures
- (C) Both A and B
- (D) Only cycles matter

23

OWS3.24 **Which is NOT a function of equipment in snubbing operations?**

Answer

- A** (A) Continuous automated decision-making
- (B) Pressure monitoring and control
- (C) Barrier maintenance
- (D) Pipe movement regulation

24

OWS3.25 **Which design consideration is critical for floating rig coiled tubing operations?**

Answer

- A** (A) Heave compensation equipment to counteract rig movement
- (B) Maximum pump capacity
- (C) Highest operating pressure
- (D) Minimum reel diameter

25

OWS3.26 **Stand Alone Snubbing Units differ from other systems in what primary capability?**

Answer

- C** (A) They can operate at lower pressures only
- (B) They require less support equipment
- (C) They have the added capability to rotate tubulars
- (D) They are limited to vertical wells exclusively

26

OWS3.27 **Snubbing is most accurately defined as which operation?**

Answer

- A** (A) The process of forcing pipe or tubulars into a wellbore while overcoming wellbore pressure
- (B) The circulation of kill fluid throughout the wellbore
- (C) The temporary isolation of well zones
- (D) The rapid removal of tubing from a well under pressure

27

OWS3.28 **How does barrier system design ensure well control during operations?**

Answer

- B** (A) Through single barrier implementation only
- (B) Through multiple redundant barriers providing layers of protection
- (C) By eliminating all pressure monitoring
- (D) By increasing wellbore pressure continuously

28

OWS3.29 **What condition occurs with External slip type connections during high stress?**

Answer

- D** (A) They become stronger
- (B) They require additional lubrication
- (C) They automatically adjust
- (D) Sharp spiraled teeth grip outer diameter (OD) of tube body for pressure seal

29

OWS3.30 **Where is the critical depth for buckling prevention in wells?**

Answer

- B** (A) Only in shallow wells
- (B) Deeper wells where increased coiled tubing length establishes buckling conditions
- (C) Only in vertical sections
- (D) Only in atmospheric pressure wells

30

OWS3.31 **What must be avoided when applying lubrication to stripper elements?**

Answer

- D** (A) Excessive lubrication volume
- (B) Regular maintenance schedules
- (C) Pressure testing procedures
- (D) Pipe dope application (prevents damage to elastomeric elements)

31

OWS3.32 **What prevents stripper element damage in coiled tubing operations?**

Answer

- B**
- (A) High pressure application
 - (B) Proper lubrication while avoiding pipe dope application
 - (C) Increased circulation rates
 - (D) Pressure reduction procedures

32

OWS3.33 **What must pressure inside the coiled tubing remain above during Bullhead Method?**

Answer

- C**
- (A) Hydrostatic pressure of the wellbore
 - (B) The annular pressure
 - (C) The coiled tubing collapse rating
 - (D) Formation pressure

33

OWS3.34 **Where can Bullhead Method fluid be pumped in coiled tubing operations?**

Answer

- C**
- (A) Only down the coiled tubing conduit
 - (B) Only through the coiled tubing annulus
 - (C) Down CT itself, through CT annulus, or via kill line when CT unavailable
 - (D) Only through surface delivery

34

OWS3.35 **Which requirements apply to shut-in procedures?**

Answer

- D**
- (A) Activating BOP slip rams to secure workstring
 - (B) Engaging pipe rams to create reliable seal,
 - (C) Verifying valve positioning throughout system
 - (D) All the above

35

OWS3.36 **How many significant bending events occur during each operational cycle of coiled tubing?**

Answer

- A** (A) Six significant bending events per operational cycle
- (B) Five bending events per cycle
- (C) Four bending events per cycle
- (D) Two bending events per cycle

36

OWS3.37 **What does the Annular Preventer (Stripper) accomplish in snubbing operations?**

Answer

- A** (A) Provides primary pressure barrier creating continuous seal around workstring for wells under 2,000 PSI
- (B) Isolates open hole sections
- (C) Shears pipe in emergencies only
- (D) Regulates flow rates

37

OWS3.38 **Where can Bullhead Method fluid be pumped in coiled tubing operations?**

Answer

- C** (A) Only through the coiled tubing annulus
- (B) Only down the coiled tubing conduit
- (C) Down CT itself, through CT annulus, or via kill line when CT unavailable
- (D) Only through surface delivery

38

OWS3.39 **What critical function does the Equalizing Loop perform in snubbing?**

Answer

- B** (A) Isolates the annulus
- (B) Balances pressure between rams allowing controlled transition as workstring moves
- (C) Shears the workstring in emergency
- (D) Reduces pump pressures only

39

OWS3.40 **Which is NOT a recognized failure mechanism in coiled tubing?**

Answer

- D** (A) Helical buckling from friction
- (B) Bending fatigue from operational cycles
- (C) Ovality-induced collapse under pressure
- (D) None of the above

40

OWS3.41 **What effect does ovality have on coiled tubing collapse rating?**

Answer

- C** (A) No measurable impact on ratings
- (B) Minimal effect only
- (C) Tubing with 12% ovality experiences 50% decrease in collapse rating
- (D) Improves collapse resistance

41

OWS3.42 **Which statements are correct about well barriers?**

Answer

- B** (A) Internal barriers such as Back Pressure Valves prevent flow inside
- (B) Both A and C
- (C) External barriers contain wellbore fluids outside
- (D) Only internal barriers are needed

42

OWS3.43 **How does hydraulic stuffing box operation differ from manual stuffing box?**

Answer

- B** (A) It requires no sealing elements
- (B) Hydraulic pressure is applied automatically to create and maintain pressure seal
- (C) It cannot withstand high pressures
- (D) It has no operator control mechanism

43

OWS3.44 **How does barrier system design ensure well control during operations?**

Answer

- B** (A) Through single barrier implementation only
- (B) Through multiple redundant barriers providing layers of protection
- (C) By increasing wellbore pressure continuously
- (D) By eliminating all pressure monitoring

44

OWS3.45 **What does grease injection accomplish in wireline operations?**

Answer

- A** (A) Applies lubricating grease along wireline maintaining seal integrity and reducing friction
- (B) Prevents wireline corrosion only
- (C) Reduces wellbore pressure significantly
- (D) Increases wireline tension exclusively

45

OWS3.46 **What effect does ovality have on coiled tubing collapse rating?**

Answer

- C** (A) Minimal effect only
- (B) Minimal effect only
- (C) Tubing with 12% ovality experiences 50% decrease in collapse rating
- (D) Improves collapse resistance

46

OWS3.47 **Which elements characterize the Drillers Method?**

Answer

- C** (A) Two distinct circulation phases for well control
- (B) Precise recording of shut-in pressures
- (C) Both A and B
- (D) Only phases matter

47

OWS3.48 **How do well control risk management principles apply in operations?**

Answer

- B** (A) Eliminate all pressure
- (B) Identify, quantify, mitigate, and control risks systematically
- (C) Monitor pressure changes only
- (D) Identify hazards only

48

OWS3.49 **Under what circumstances would a substructure be added to snubbing unit BOP systems?**

Answer

- A** (A) If casing is compromised and additional structural support is necessary
- (B) For aesthetic purposes
- (C) To increase operational cost
- (D) To reduce equipment height

49

OWS3.50 **Which must be addressed in contingency planning?**

Answer

- D** (A) Equipment failure scenarios and responses
- (B) Personnel safety and evacuation procedures,
- (C) Communication protocols and notifications
- (D) All the above

50

OWS3.51 **Which apply to Ram-to-Ram snubbing systems?**

Answer

- B** (A) Upper Strip Rams (Master Rams) required
- (B) Both A and C
- (C) Guide Ram with Equalizing Loop needed
- (D) Only Master Rams

51

OWS3.52 **How do well control risk management principles apply in operations?**

Answer

- B** (A) Monitor pressure changes only
- (B) Identify, quantify, mitigate, and control risks systematically
- (C) Identify hazards only
- (D) Eliminate all pressure

52

OWS3.53 **What is the primary function of the Lubricator in wireline well control operations?**

Answer

- A** (A) Dynamic Pressure Control Equipment creating sealed pathway allowing safe passage of wireline through pressure barrier
- (B) To increase circulation rates exclusively
- (C) To reduce wellbore pressure
- (D) To monitor wire tension

53

OWS3.54 **Why are Safety Rams essential in snubbing operations?**

Answer

- B** (A) To regulate wellbore pressure
- (B) To provide two-barrier wellbore isolation essential when performing maintenance on strippers
- (C) To shear stuck pipe exclusively
- (D) To increase circulation capacity

54

OWS3.55 **What specific capability does electric line provide beyond conventional solid wireline?**

Answer

- C** (A) Allows pipe rotation capability
- (B) Prevention of pressure issues
- (C) Enables data transmission for downhole tools including logging cable for operations
- (D) Increased tensile strength only

55

OWS3.56 **Why is an anti-buckling guide used during coiled tubing deployment into pressurized wells?**

Answer

- A** (A) To provide additional lateral support and prevent buckling from compressive loads
- (B) To reduce pump friction pressures
- (C) To increase tubing tensile strength
- (D) To facilitate faster deployment rates

56

OWS3.57 **Despite opposing force directions in pipe light and pipe heavy, why is snubbing still required?**

Answer

- D** (A) For monitoring purposes only
- (B) For pressure reduction exclusively
- (C) For increasing circulation rates
- (D) To provide necessary control mechanism managing differential forces safely

57

OWS3.58 **What indicator might suggest compromised casing integrity?**

Answer

- A** (A) Identification of sustained casing pressure (CP) exceeding 500 psi
- (B) Low surface pressure only
- (C) Reduced circulation rate
- (D) Increased wellhead temperature

58

OWS3.59 **What critical action must be taken when shut-in procedures are completed?**

Answer

- B** (A) Reduce all barrier pressures
- (B) Verify shut-in through systematic checks of all barriers and equipment
- (C) Immediately resume normal operations
- (D) Release wellbore pressure manually

59

OWS3.60 **What is the primary purpose of pressure control equipment in well interventions?**

Answer

- A** (A) To create and maintain effective barriers preventing uncontrolled wellbore fluid flow
- (B) To eliminate all wellbore pressure
- (C) To increase circulation rates significantly
- (D) To reduce wellbore temperature

60

OWS3.61 **What dangerous scenario occurs during pipe light conditions if unrestrained?**

Answer

- B** (A) The pipe gently rises at controlled rate
- (B) The pipe is dangerously ejected from the well creating significant safety hazard
- (C) The pipe remains suspended indefinitely
- (D) The pipe settles slowly to the bottom

61

OWS3.62 **How does hydraulic stuffing box operation differ from manual stuffing box?**

Answer

- B** (A) It cannot withstand high pressures
- (B) Hydraulic pressure is applied automatically to create and maintain pressure seal
- (C) It requires no sealing elements
- (D) It has no operator control mechanism

62

OWS3.63 **What must be avoided when applying lubrication to stripper elements?**

Answer

- D** (A) Regular maintenance schedules
- (B) Pressure testing procedures
- (C) Excessive lubrication volume
- (D) Pipe dope application (prevents damage to elastomeric elements)

63

OWS3.64 **Why is adequate lighting crucial for 24-hour workover operations?**

Answer

- A** (A) To minimize shadows and ensure clear visibility for identifying potential hazards
- (B) For equipment appearance
- (C) For documentation purposes
- (D) For thermal control purposes

64

OWS3.65 **What is the primary purpose of pre-job communication in workover operations?**

Answer

- A** (A) To ensure all stakeholders are aligned on operational plans, roles, and associated risks
- (B) To eliminate equipment requirements
- (C) To simplify operational procedures
- (D) To reduce personnel numbers

65

OWS3.66 **Where is the critical depth for buckling prevention in wells?**

Answer

- B** (A) Only in vertical sections
- (B) Deeper wells where increased coiled tubing length establishes buckling conditions
- (C) Only in shallow wells
- (D) Only in atmospheric pressure wells

66

OWS3.67 **What must pressure inside the coiled tubing remain above during Bullhead Method?**

Answer

- C** (A) Hydrostatic pressure of the wellbore
- (B) Formation pressure
- (C) The coiled tubing collapse rating
- (D) The annular pressure

67

OWS3.68 **What dangerous scenario occurs during pipe light conditions if unrestrained?**

Answer

- B** (A) The pipe settles slowly to the bottom
- (B) The pipe is dangerously ejected from the well creating significant safety hazard
- (C) The pipe remains suspended indefinitely
- (D) The pipe gently rises at controlled rate

68

OWS3.69 **Which statements apply to stripper maintenance?**

Answer

- D** (A) Elastomers and seals require regular inspection
- (B) Routine pressure testing verifies system integrity
- (C) Only inspection matters
- (D) Both A and B

69

OWS3.70 **What effect does ovality have on coiled tubing collapse rating?**

Answer

- C** (A) No measurable impact on ratings
- (B) No measurable impact on ratings
- (C) Tubing with 12% ovality experiences 50% decrease in collapse rating
- (D) Improves collapse resistance

70

OWS3.71 **What critical step must follow After closing Safety Rams?**

Answer

- C** (A) Increase pump rates
- (B) Activate emergency shutdown immediately
- (C) Bleed off pressures above Upper Safety Ram before stripper maintenance
- (D) Increase wellbore pressure

71

OWS3.72 **What does the Guide Ram accomplish in Ram-to-Ram snubbing operations?**

Answer

- B** (A) Isolates open hole sections
- (B) Supports tubulars and prevents buckling during snubbing operations
- (C) Shears the pipe in emergencies
- (D) Regulates fluid flow rates

72

OWS3.73 **What is NOT a standard component of coiled tubing equipment systems?**

Answer

- C** (A) Services Power Reel
- (B) Hydraulic Power Pack
- (C) Automated decompression chamber
- (D) Injector Head

73

OWS3.74 **When performing maintenance on stripper elements, which action occurs first?**

Answer

- B** (A) Activate emergency shutdown
- (B) Close both Upper and Lower Safety Rams to provide two-barrier isolation
- (C) Increase wellbore pressure
- (D) Increase pump rates significantly

74

OWS3.75 **What measurement threshold indicates possible casing integrity compromise?**

Answer

- A** (A) 500 psi or greater sustained casing pressure (CP)
- (B) 400 psi sustained casing pressure
- (C) 300 psi sustained casing pressure
- (D) 200 psi sustained casing pressure

75

OWS3.76 **Which apply to contingency equipment procedures?**

Answer

- B** (A) Pull tubing to safe position when leaks detected,
- (B) Both A and C
- (C) Close all relevant pressure control equipment
- (D) Only repositioning

76

OWS3.77 **Which design consideration is critical for floating rig coiled tubing operations?**

Answer

- A** (A) Heave compensation equipment to counteract rig movement
- (B) Highest operating pressure
- (C) Maximum pump capacity
- (D) Minimum reel diameter

77

OWS3.78 **In the Wait and Weight method, when should kill weight fluid be calculated?**

Answer

- B** (A) Only after influx removal is complete
- (B) After recording shut-in coiled tubing pressure (SICTP) and shut-in casing pressure (SICP)
- (C) Before positioning the coiled tubing at proper kill depth
- (D) During the circulation process

78

OWS3.79 **In pipe heavy conditions, what would happen to unrestrained pipe?**

Answer

- B** (A) The pipe would rise to the surface
- (B) The pipe would fall into well potentially damaging downhole equipment
- (C) The pipe would remain stationary
- (D) The pipe would float at fixed depth

79

OWS3.80 **Which requirements govern operations?**

Answer

- D** (A) Only training
- (B) Personnel training and competency maintenance
- (C) Equipment testing and certification protocols
- (D) Both B and C

80

OWS3.81 **What is NOT a typical remedial operation during snubbing on live wells?**

Answer

- A** (A) Continuous automated well monitoring systems
- (B) Water production isolation
- (C) Tubing and completion equipment replacement
- (D) Mechanical failure repairs

81

OWS3.82 **When performing maintenance on stripper elements, which action occurs first?**

Answer

- B** (A) Increase pump rates significantly
- (B) Close both Upper and Lower Safety Rams to provide two-barrier isolation
- (C) Activate emergency shutdown
- (D) Increase wellbore pressure

82

OWS3.83 **Why are Safety Rams essential in snubbing operations?**

Answer

- B** (A) To increase circulation capacity
- (B) To provide two-barrier wellbore isolation essential when performing maintenance on strippers
- (C) To regulate wellbore pressure
- (D) To shear stuck pipe exclusively

83

OWS3.84 **What critical action must be taken when shut-in procedures are completed?**

Answer

- B**
- (A) Release wellbore pressure manually
 - (B) Verify shut-in through systematic checks of all barriers and equipment
 - (C) Reduce all barrier pressures
 - (D) Immediately resume normal operations

84

OWS3.85 **Which factors must be present for complete well control?**

Answer

- D**
- (A) Pressure control equipment and barriers
 - (B) Trained personnel and procedures
 - (C) Contingency plans and backup systems
 - (D) All the above

85

OWS3.86 **Why is an anti-buckling guide used during coiled tubing deployment?**

Answer

- B**
- (A) To reduce pump friction pressures
 - (B) To provide additional lateral support and prevent buckling from compressive loads
 - (C) To increase tubing tensile strength
 - (D) To facilitate faster deployment rates

86

OWS3.87 **What must be avoided when applying lubrication to stripper elements?**

Answer

- D**
- (A) Excessive lubrication volume
 - (B) Pressure testing procedures
 - (C) Regular maintenance schedules
 - (D) Pipe dope application (prevents damage to elastomeric elements)

87

OWS3.88 **How does tubing arc guide design address buckling prevention?**

Answer

- C** (A) Reduces fluid pressure
- (B) Increases wall thickness significantly
- (C) Provides support achieving bending radius of 30 times tubing outer diameter
- (D) Eliminates all bending forces

88

OWS3.89 **What is the minimum recommended core radius diameter ratio for coiled tubing services power reels?**

Answer

- A** (A) At least 20 times the specified coiled tubing diameter
- (B) At least 9 times the coiled tubing diameter
- (C) At least 14 times the tubing wall thickness
- (D) At least 4 times the specified yield strength

89

OWS3.90 **Which apply to coiled tubing Well control?**

Answer

- D** (A) Removal of influx from wellbore
- (B) Displacement of wellbore with kill fluid
- (C) Verification of successful well control
- (D) All the above

90

OWS3.91 **Which must be addressed in contingency planning?**

Answer

- B** (A) Immediate assessment of failure cause
- (B) Both A and C
- (C) Systematic evaluation of impact scope
- (D) Only assessment

91

OWS3.92 **What well condition requires careful pressure monitoring?**

Answer

- B** (A) Vertical wells only
- (B) All well types to prevent uncontrolled fluid migration during interventions
- (C) Low-pressure wells only
- (D) Shallow wells exclusively

92

OWS3.93 **What indicator might suggest compromised well integrity?**

Answer

- C** (A) Reduced circulation rates
- (B) Low surface pressure only
- (C) Identification of sustained casing pressure (CP) exceeding 500 psi
- (D) Increased wellhead temperature

93

OWS3.94 **What design feature prevents tubing collapse from bending forces?**

Answer

- C** (A) By reducing fluid pressure continuously
- (B) By increasing wall thickness
- (C) By providing bending radius of at least 30 times the specified tubing outer diameter
- (D) By eliminating all bending

94

OWS3.95 **What result occurs once helical buckling locks tubing in the wellbore?**

Answer

- B** (A) The tubing maintains structural integrity
- (B) The tubing is permanently deformed and will not allow retrieval
- (C) The tubing slowly relaxes automatically
- (D) The tubing shrinks and needs resizing

95

OWS3.96 **What is the primary advantage of braided wireline in well interventions?**

Answer

- C** (A) Cannot be sheared by rams
- (B) Eliminates pressure control requirements
- (C) Provides enhanced flexibility and reduced diameter for certain well geometries
- (D) Increases circulation rates exclusively

96

OWS3.97 **What are the primary components of dynamic wireline pressure control equipment?**

Answer

- A** (A) Control heads (Dynamic PCE) including manual and hydraulic stuffing boxes and grease injection systems
- (B) Ram assemblies exclusively
- (C) Blind rams and shear rams only
- (D) Flow spools and equalizing valves

97

OWS3.98 **Which must be managed in operations?**

Answer

- D** (A) Only personnel fatigue
- (B) Personnel fatigue during extended operations
- (C) Environmental conditions affecting equipment
- (D) Both B and C

98

OWS3.99 **What is the minimum recommended core radius diameter ratio for coiled tubing services power reels?**

Answer

- A** (A) At least 20 times the specified coiled tubing diameter
- (B) At least 10 times the coiled tubing diameter
- (C) At least 13 times the tubing wall thickness
- (D) At least 5 times the specified yield strength

99

OWS3.100 How does barrier system design ensure well control during operations?

Answer

- B** (A) By increasing wellbore pressure continuously
- (B) Through multiple redundant barriers providing layers of protection
- (C) By eliminating all pressure monitoring
- (D) Through single barrier implementation only

100

OWS3.101 How do well control risk management principles apply in operations?

Answer

- B** (A) Identify hazards only
- (B) Identify, quantify, mitigate, and control risks systematically
- (C) Monitor pressure changes only
- (D) Eliminate all pressure

101

OWS3.102 Which elements characterize Ram-to-Ram operations?

Answer

- D** (A) Only sequencing
- (B) Proper sequencing of ram activation,
- (C) Maintained equalizing loop functionality
- (D) Both B and C

102

OWS3.103 What dangerous scenario occurs in pipe light conditions if unrestrained?

Answer

- C** (A) Pipe remains suspended indefinitely
- (B) Pipe gently rises at controlled rate
- (C) Pipe is dangerously ejected from well creating significant safety hazard
- (D) Pipe settles slowly to the bottom

103

OWS3.104 **In what wellbore conditions is helical buckling most likely to occur?**

Answer

- C** (A) Only in vertical wells
- (B) Only in shallow wells
- (C) In deeper wells where increased coiled tubing length establishes buckling conditions
- (D) Only in low-pressure wells

104

OWS3.105 **Why is adequate lighting crucial for 24-hour workover operations?**

Answer

- A** (A) To minimize shadows and ensure clear visibility for identifying potential hazards
- (B) For documentation purposes
- (C) For equipment appearance
- (D) For thermal control purposes

105

OWS3.106 **Which factors characterize complete well control systems?**

Answer

- D** (A) Multiple redundant barriers for safety
- (B) Comprehensive monitoring and measurement,
- (C) Regular testing and documentation
- (D) All the above

106

OWS3.107 **Why is an anti-buckling guide used during coiled tubing deployment into pressurized wells?**

Answer

- A** (A) To provide additional lateral support and prevent buckling from compressive loads
- (B) To increase tubing tensile strength
- (C) To facilitate faster deployment rates
- (D) To reduce pump friction pressures

107

OWS3.108 **Which elements characterize effective well control?**

Answer

- B** (A) Multiple redundant barrier layers
- (B) Both A and C
- (C) Comprehensive equipment verification protocols
- (D) Only barriers

108

OWS3.109 **What is the primary purpose of hydraulic systems in snubbing operations?**

Answer

- A** (A) To regulate pressure and force while pushing pipe into the well under controlled conditions
- (B) To monitor wellbore temperature exclusively
- (C) To reduce equipment friction
- (D) To provide emergency power backup only

109

OWS3.110 **Which elements are required for effective well control?**

Answer

- D** (A) Recognition of potential hazards during operations
- (B) Clear communication of risks among all personnel
- (C) Only recognition of potential hazards during operations is important
- (D) Both A and B

110

OWS3.111 **What percentage ovality is accepted by industry standards for normal manufacturing?**

Answer

- C** (A) 2% ovality
- (B) 3% ovality
- (C) Approximately 5% ovality from normal manufacturing processes
- (D) 8% ovality

111

OWS3.112 **Which elements are critical in well interventions?**

Answer

- D** (A) Pressure barrier systems and equipment
- (B) Personnel safety and training
- (C) Contingency procedures and readiness
- (D) All the above

112

OWS3.113 **What must be used to accurately model fluid behavior in Bullhead coiled tubing?**

Answer

- C** (A) Manual calculations only
- (B) Surface pressure readings
- (C) An accurate hydraulics simulator accounting for CT friction losses
- (D) Standard pipe design software

113

OWS3.114 **What critical consideration affects Bullhead pump rates and pressures?**

Answer

- C** (A) Personnel comfort during operations
- (B) Casing pressure exclusively
- (C) Formation damage minimization while accounting for higher frictional pressure losses
- (D) Wellhead design only

114

OWS3.115 **Which apply to pressure control during snubbing?**

Answer

- D** (A) Ram system components must seal properly
- (B) Equalizing loops must balance ram pressures
- (C) Only ram system components are needed to seal properly
- (D) Both A and B

115

OWS3.116 Which considerations affect coiled tubing deployment?

Answer

- D** (A) Preventing tubing collapse through internal pressure
- (B) Preventing buckling from compressive loads
- (C) Accounting for frictional force effects
- (D) All the above

116

OWS3.117 How do operators evaluate well integrity in workover operations?

Answer

- A** (A) By meticulously reviewing historical well records, diagnostic test results, and real-time data
- (B) Through pressure readings only
- (C) Only from surface pressure monitoring
- (D) By well depth alone

117

OWS3.118 What is NOT a factor affecting coiled tubing service life?

Answer

- D** (A) Cycle counts and operational history
- (B) Bending fatigue and ovality
- (C) Exposure to corrosive fluids
- (D) None of the above

118

OWS3.119 Why do snubbing units remain essential control at balance point?

Answer

- B** (A) For monitoring purposes only
- (B) Equilibrium is unstable and can quickly shift with minor changes in pressure or string weight
- (C) For pressure reduction exclusively
- (D) For increasing circulation rates

119

OWS3.120 **Which conditions are found in pipe light scenarios?**

Answer

- D**
- (A) Weight of buoyant workstring is less than upward pressure force
 - (B) Pipe would be ejected if unrestrained
 - (C) Pipe light depends on only the buoyant weight of workstring is less than the upoward pressure.
 - (D) Both A and B

120

OWS3.121 **Where is the thickest diameter of tapered coiled tubing normally located?**

Answer

- A**
- (A) At the inner end of the reel at the top of the well where tensile forces are greatest
 - (B) At the outer end of the reel where tubing enters the well
 - (C) In the middle section for balanced load distribution
 - (D) Along the entire length uniformly

121

OWS3.122 **What is the Rig Assist Space Saver unit designed for?**

Answer

- C**
- (A) Arctic conditions exclusively
 - (B) Deepwater applications only
 - (C) Lower-pressure work typically under 3,500 PSI with minimal tool access
 - (D) High-pressure operations above 5,000 PSI

122

OWS3.123 **Stand Alone Snubbing Units differ from other systems in what primary capability?**

Answer

- C**
- (A) They are limited to vertical wells exclusively
 - (B) They require less support equipment
 - (C) They have the added capability to rotate tubulars
 - (D) They can operate at lower pressures only

123

OWS3.124 **When blockages or trapped pressure occur in wellbore systems, what approach applies?**

Answer

- C** (A) Ignore pressure reading anomalies
- (B) Increase circulation rates immediately
- (C) Use specialized remediation techniques tailored to specific blockage type
- (D) Remove all barriers

124

OWS3.125 **Which ram assembly is specifically designed for solid wireline (slickline)?**

Answer

- C** (A) Annular preventers only
- (B) Flow control rams only
- (C) Wireline Ram Assembly (Slick Line) optimized for solid wireline shearing
- (D) Hydraulic rams exclusively

125

OWS3.126 **What result occurs once helical buckling locks tubing in the wellbore?**

Answer

- B** (A) The tubing maintains structural integrity
- (B) The tubing is permanently deformed and will not allow retrieval
- (C) The tubing slowly relaxes automatically
- (D) The tubing shrinks and needs resizing

126

OWS3.127 **What is the primary purpose of hydraulic systems in snubbing operations?**

Answer

- A** (A) To regulate pressure and force while pushing pipe into the well under controlled conditions
- (B) To provide emergency power backup only
- (C) To monitor wellbore temperature exclusively
- (D) To reduce equipment friction

127

OWS3.128 **Which elements comprise barrier management?**

Answer

- D** (A) Testing of all barrier components
- (B) Regular inspection and maintenance,
- (C) Documented verification protocols
- (D) All the above

128

OWS3.129 **What condition most accurately describes pipe light situations?**

Answer

- B** (A) When pipe weight exceeds upward forces
- (B) When weight of buoyant workstring is less than upward force from wellbore pressure
- (C) When circulation is stopped completely
- (D) When wellbore pressure equals zero

129

OWS3.130 **What is NOT a typical remedial operation during snubbing on live wells?**

Answer

- A** (A) Continuous automated well monitoring systems
- (B) Mechanical failure repairs
- (C) Tubing and completion equipment replacement
- (D) Water production isolation

130

OWS3.131 **Which equipment is NOT part of a complete coiled tubing unit?**

Answer

- D** (A) Services Power Reel
- (B) Injector Head
- (C) Quad BOP System
- (D) None of the above

131

OWS3.132 **What is the purpose of using gradually tapered insertion neck in weld connections?**

Answer

- C** (A) To facilitate field assembly operations
- (B) To increase connection strength by 50 percent
- (C) To provide support preventing concentration of bending loads at single point
- (D) To reduce welding time and cost

132

OWS3.133 **What does grease injection accomplish in wireline operations?**

Answer

- A** (A) Applies lubricating grease along wireline maintaining seal integrity and reducing friction
- (B) Reduces wellbore pressure significantly
- (C) Increases wireline tension exclusively
- (D) Prevents wireline corrosion only

133

OWS3.134 **In the Wait and Weight method, when should kill weight fluid be calculated?**

Answer

- B** (A) During the circulation process
- (B) After recording shut-in coiled tubing pressure (SICTP) and shut-in casing pressure (SICP)
- (C) Before positioning the coiled tubing at proper kill depth
- (D) Only after influx removal is complete

134

OWS3.135 **Which is NOT a component of standard pressure control equipment in well interventions?**

Answer

- D** (A) Barrier systems
- (B) Monitoring devices
- (C) Emergency shutoff systems
- (D) None of the above

135

OWS3.136 Which element is NOT part of coiled tubing pressure control equipment?

Answer

- C** (A) Stripper assembly
- (B) Quad BOP system
- (C) Automated thermal regulation system
- (D) Flow check assembly

136

OWS3.137 Which elements comprise effective pressure management?

Answer

- D** (A) Barrier systems preventing uncontrolled flow
- (B) Equipment to apply and regulate pressures
- (C) Procedures for monitoring and control
- (D) All the above

137

OWS3.138 Tapered coiled tubing can be manufactured with how many sections of varying wall thickness?

Answer

- A** (A) Six sections (forming a 5-step taper)
- (B) Five sections
- (C) Four sections
- (D) Three sections

138

OWS3.139 What does the Annular Preventer (Stripper) accomplish in snubbing operations?

Answer

- A** (A) Provides primary pressure barrier creating continuous seal around workstring for wells under 2,000 PSI
- (B) Isolates open hole sections
- (C) Regulates flow rates
- (D) Shears pipe in emergencies only

139

OWS3.140 **Which elements are part of well integrity assessment?**

Answer

- B** (A) Identification of casing communication through sustained pressure
- (B) Both A and C
- (C) Evaluation of nearby fracturing operations
- (D) Only SITP needed

140

OWS3.141 **Why are spherical preventers typically used as strippers in snubbing?**

Answer

- A** (A) They provide longer contact area which enhances durability and reduces wear during operations
- (B) They operate at lower pressures
- (C) They are less expensive
- (D) They require less maintenance

141

OWS3.142 **Which is NOT a function of equipment in snubbing operations?**

Answer

- A** (A) Continuous automated decision-making
- (B) Barrier maintenance
- (C) Pipe movement regulation
- (D) Pressure monitoring and control

142

OWS3.143 **What is NOT a function of pressure control equipment in operations?**

Answer

- D** (A) Create effective barriers against uncontrolled flow
- (B) Allow controlled equipment movement through barriers
- (C) Maintain well integrity during interventions
- (D) None of the above

143

OWS3.144 **Which type of snubbing unit was developed in the 1920s using cable systems?**

Answer

- C** (A) Hydraulic Workover Units exclusively
- (B) Portable snubbing units
- (C) Rig Assist Snubbing Units utilizing cables connected to traveling blocks
- (D) Stand Alone Snubbing Units

144

OWS3.145 **What prevents stripper element damage in coiled tubing operations?**

Answer

- B** (A) Pressure reduction procedures
- (B) Proper lubrication while avoiding pipe dope application
- (C) Increased circulation rates
- (D) High pressure application

145

OWS3.146 **Which factors impact stripper wear?**

Answer

- B** (A) Operating pressures during circulation
- (B) Both A and C
- (C) Lubrication levels and maintenance conditions
- (D) Only pressure

146

OWS3.147 **What indicator might suggest compromised casing integrity?**

Answer

- A** (A) Identification of sustained casing pressure (CP) exceeding 500 psi
- (B) Increased wellhead temperature
- (C) Reduced circulation rate
- (D) Low surface pressure only

147

OWS3.148 **Which type of snubbing unit has the largest operational footprint capability?**

Answer

- D** (A) Rig Assist Space Saver units
- (B) Rig Assist Snubbing Units only
- (C) Portable snubbing units
- (D) Stand Alone Snubbing Units (Hydraulic Workover Units)

148

OWS3.149 **What is the primary method by which Snubbing Units control pipe movement into pressurized wells?**

Answer

- A** (A) A hydraulic ram system operating through a Ram-to-Ram system for controlled movement
- (B) Gravity assist exclusively
- (C) Cable and block systems only
- (D) Rotational torque application

149

OWS3.150 **In what wellbore conditions is helical buckling most likely to occur?**

Answer

- C** (A) Only in low-pressure wells
- (B) Only in vertical wells
- (C) In deeper wells where increased coiled tubing length establishes buckling conditions
- (D) Only in shallow wells

150

OWS3.151 **What critical consideration affects Bullhead pump rates and pressures?**

Answer

- C** (A) Wellhead design only
- (B) Casing pressure exclusively
- (C) Formation damage minimization while accounting for higher frictional pressure losses
- (D) Personnel comfort during operations

151

OWS3.152 **Despite opposing force directions in pipe light and pipe heavy, why is snubbing still required?**

Answer

- D** (A) For pressure reduction exclusively
- (B) For increasing circulation rates
- (C) For monitoring purposes only
- (D) To provide necessary control mechanism managing differential forces safely

152

OWS3.153 **Which ram assembly is specifically designed for solid wireline (slickline)?**

Answer

- C** (A) Hydraulic rams exclusively
- (B) Flow control rams only
- (C) Wireline Ram Assembly (Slick Line) optimized for solid wireline shearing
- (D) Annular preventers only

153

OWS3.154 **At the balance point during snubbing, what condition exists?**

Answer

- A** (A) Gravitational force of pipe string exactly equals upward force from wellbore pressure
- (B) Wellbore pressure far exceeds pipe weight
- (C) External forces dominate all conditions
- (D) Pipe weight exceeds wellbore pressure

154

OWS3.155 **What is the purpose of using gradually tapered insertion neck in weld connections?**

Answer

- C** (A) To increase connection strength by 50 percent
- (B) To reduce welding time and cost
- (C) To provide support preventing concentration of bending loads at single point
- (D) To facilitate field assembly operations

155

OWS3.156 **Tapered coiled tubing can be manufactured with how many sections of varying wall thickness?**

Answer

D (A) Five sections

(B) Four sections

(C) Three sections

(D) Six sections (forming a 5-step taper)

156

OWS3.157 **What condition most accurately describes pipe light situations?**

Answer

B (A) When circulation is stopped completely

(B) When weight of buoyant workstring is less than upward force from wellbore pressure

(C) When wellbore pressure equals zero

(D) When pipe weight exceeds upward forces

157

OWS3.158 **Which apply to contingency responses?**

Answer

D (A) Only shutdown procedures needed

(B) Rapid system shutdown capabilities,

(C) Equipment repair and replacement readiness

(D) Both B and C

158

OWS3.159 **How does tubing arc guide design address buckling prevention?**

Answer

C (A) Eliminates all bending forces

(B) Increases wall thickness significantly

(C) Provides support achieving bending radius of 30 times tubing outer diameter

(D) Reduces fluid pressure

159

OWS3.160 **Why is an anti-buckling guide used during coiled tubing deployment?**

Answer

- B** (A) To facilitate faster deployment rates
- (B) To provide additional lateral support and prevent buckling from compressive loads
- (C) To increase tubing tensile strength
- (D) To reduce pump friction pressures

160

OWS3.161 **Snubbing is most accurately defined as which operation?**

Answer

- A** (A) The process of forcing pipe or tubulars into a wellbore while overcoming wellbore pressure
- (B) The temporary isolation of well zones
- (C) The circulation of kill fluid throughout the wellbore
- (D) The rapid removal of tubing from a well under pressure

161

OWS3.162 **Snubbing is most accurately defined as which operation?**

Answer

- A** (A) The process of forcing pipe or tubulars into a wellbore while overcoming wellbore pressure
- (B) The rapid removal of tubing from a well under pressure
- (C) The circulation of kill fluid throughout the wellbore
- (D) The temporary isolation of well zones

162

OWS3.163 **Why are dual strippers used in coiled tubing operations?**

Answer

- B** (A) To reduce equipment operating costs
- (B) To ensure operational safety through redundancy if one stripper element fails
- (C) To increase circulation rates
- (D) To eliminate pressure requirements

163

OWS3.164 **What critical requirement applies to internal slip type coiled tubing connections?**

Answer

- A** (A) Internal manufacturing weld seam must be removed including grinding and smoothing
- (B) Maximum weld seam strength required
- (C) Tubing must have zero ovality
- (D) External surface must be polished

164

OWS3.165 **Which is NOT a function of equipment in snubbing operations?**

Answer

- A** (A) Continuous automated decision-making
- (B) Pressure monitoring and control
- (C) Barrier maintenance
- (D) Pipe movement regulation

165

OWS3.166 **What is NOT a recognized failure point in coiled tubing operations?**

Answer

- D** (A) Connection failures
- (B) Bending fatigue failures
- (C) Ovality-induced collapse
- (D) Unexplained atmospheric pressure changes

166

OWS3.167 **Which factors must be managed during operations?**

Answer

- D** (A) Wellbore pressure monitoring and control
- (B) Equipment performance and integrity,
- (C) Personnel positioning and communication
- (D) All the above

167

OWS3.168 **What prevents stripper element damage in coiled tubing operations?**

Answer

- B** (A) High pressure application
- (B) Proper lubrication while avoiding pipe dope application
- (C) Pressure reduction procedures
- (D) Increased circulation rates

168

OWS3.169 **What critical action must be taken when shut-in procedures are completed?**

Answer

- B** (A) Release wellbore pressure manually
- (B) Verify shut-in through systematic checks of all barriers and equipment
- (C) Immediately resume normal operations
- (D) Reduce all barrier pressures

169

OWS3.170 **At what wellhead pressure are Hydraulic Workover Units (Snubbing Units) typically mobilized?**

Answer

- A** (A) When wellhead pressure exceeds 2,000 PSI
- (B) When pressure exceeds 1,000 PSI
- (C) When pressure exceeds 3,500 PSI
- (D) At any measurable pressure above zero

170

OWS3.171 **What dangerous scenario occurs in pipe light conditions if unrestrained?**

Answer

- C** (A) Pipe settles slowly to the bottom
- (B) Pipe remains suspended indefinitely
- (C) Pipe is dangerously ejected from well creating significant safety hazard
- (D) Pipe gently rises at controlled rate

171

OWS3.172 **What percentage ovality is accepted by industry standards for normal manufacturing?**

Answer

- C** (A) 8% ovality
- (B) 2% ovality
- (C) Approximately 5% ovality from normal manufacturing processes
- (D) 3% ovality

172

OWS3.173 **How does coiled tubing differ fundamentally from conventional pipe in operations?**

Answer

- D** (A) Conventional pipe is more flexible
- (B) Coiled tubing requires constant rotation
- (C) Both have identical operational characteristics
- (D) Coiled tubing enables continuous circulation on pressurized live wells

173

OWS3.174 **Why is adequate lighting crucial for 24-hour workover operations?**

Answer

- A** (A) To minimize shadows and ensure clear visibility for identifying potential hazards
- (B) For thermal control purposes
- (C) For equipment appearance
- (D) For documentation purposes

174

OWS3.175 **What design feature prevents tubing collapse from bending forces?**

Answer

- C** (A) By increasing wall thickness
- (B) By eliminating all bending
- (C) By providing bending radius of at least 30 times the specified tubing outer diameter
- (D) By reducing fluid pressure continuously

175

OWS3.176 **What critical requirement applies to internal slip type coiled tubing connections?**

Answer

- A** (A) Internal manufacturing weld seam must be removed including grinding and smoothing
- (B) Tubing must have zero ovality
- (C) External surface must be polished
- (D) Maximum weld seam strength required

176

OWS3.177 **Which is NOT a typical snubbing operation on live wells?**

Answer

- D** (A) Mechanical failure repairs
- (B) Water production isolation
- (C) Tubing replacement procedures
- (D) None of the above

177

OWS3.178 **Which factors define snubbing operations?**

Answer

- D** (A) Forcing pipe against wellbore pressure
- (B) Maintaining safety and pressure control,
- (C) Using specialized hydraulic equipment
- (D) All the above

178

OWS3.179 **What is the primary distinction between coiled tubing and conventional pipe?**

Answer

- B** (A) Conventional pipe is always stronger
- (B) *Coiled tubing maintains continuous circulation while conventional pipe requires connections
- (C) They perform identical functions
- (D) Coiled tubing cannot be used on live wells

179

OWS3.180 **Which is NOT a recognized advantage of coiled tubing operations?**

Answer

- D** (A) Time savings compared to conventional rig operations
- (B) Reduced formation damage through underbalanced working
- (C) Simplified well control with fewer moving parts
- (D) None of the above

180

OWS3.181 **Which is NOT a standard well control requirement?**

Answer

- A** (A) Continuous wellbore pressure increase
- (B) Regular equipment testing
- (C) Multiple barrier implementation
- (D) Personnel communication and training

181

OWS3.182 **Which element is NOT part of coiled tubing pressure control equipment?**

Answer

- C** (A) Quad BOP system
- (B) Stripper assembly
- (C) Automated thermal regulation system
- (D) Flow check assembly

182

OWS3.183 **Which ram assembly is specifically designed for solid wireline (slickline)?**

Answer

- C** (A) Flow control rams only
- (B) Hydraulic rams exclusively
- (C) Wireline Ram Assembly (Slick Line) optimized for solid wireline shearing
- (D) Annular preventers only

183

OWS3.184 **What is the main goal of remedial operations during workover?**

Answer

- A** (A) To safely return well to full production in cost effective manner
- (B) To reduce production permanently
- (C) To eliminate all wellbore pressure
- (D) To abandon the well

184

OWS3.185 **What does grease injection accomplish in wireline operations?**

Answer

- A** (A) Applies lubricating grease along wireline maintaining seal integrity and reducing friction
- (B) Increases wireline tension exclusively
- (C) Reduces wellbore pressure significantly
- (D) Prevents wireline corrosion only

185

OWS3.186 **Which type of snubbing unit was developed in the 1920s using cable systems?**

Answer

- C** (A) Stand Alone Snubbing Units
- (B) Portable snubbing units
- (C) Rig Assist Snubbing Units utilizing cables connected to traveling blocks
- (D) Hydraulic Workover Units exclusively

186

OWS3.187 **What must be used to accurately model fluid behavior in Bullhead coiled tubing?**

Answer

- C** (A) Surface pressure readings
- (B) Manual calculations only
- (C) An accurate hydraulics simulator accounting for CT friction losses
- (D) Standard pipe design software

187

OWS3.188 **What specific capability does electric line provide beyond conventional solid wireline?**

Answer

- C** (A) Increased tensile strength only
- (B) Allows pipe rotation capability
- (C) Enables data transmission for downhole tools including logging cable for operations
- (D) Prevention of pressure issues

188

OWS3.189 **How many significant bending events occur during each operational cycle of coiled tubing?**

Answer

- D** (A) Two bending events per cycle
- (B) Five bending events per cycle
- (C) Four bending events per cycle
- (D) Six significant bending events per operational cycle

189

OWS3.190 **What are the primary components of dynamic wireline pressure control equipment?**

Answer

- A** (A) Control heads (Dynamic PCE) including manual and hydraulic stuffing boxes and grease injection systems
- (B) Ram assemblies exclusively
- (C) Flow spools and equalizing valves
- (D) Blind rams and shear rams only

190

OWS3.191 **Why are Safety Rams essential in snubbing operations?**

Answer

- B** (A) To shear stuck pipe exclusively
- (B) To provide two-barrier wellbore isolation essential when performing maintenance on strippers
- (C) To regulate wellbore pressure
- (D) To increase circulation capacity

191

OWS3.192 **What critical requirement applies to internal slip type coiled tubing connections?**

Answer

- A** (A) Internal manufacturing weld seam must be removed including grinding and smoothing
- (B) Maximum weld seam strength required
- (C) Tubing must have zero ovality
- (D) External surface must be polished

192

OWS3.193 **Under what circumstances would a substructure be added to snubbing unit BOP systems?**

Answer

- A** (A) If casing is compromised and additional structural support is necessary
- (B) To reduce equipment height
- (C) To increase operational cost
- (D) For aesthetic purposes

193

OWS3.194 **Where is the thickest diameter of tapered coiled tubing normally located?**

Answer

- A** (A) At the inner end of the reel at the top of the well where tensile forces are greatest
- (B) Along the entire length uniformly
- (C) In the middle section for balanced load distribution
- (D) At the outer end of the reel where tubing enters the well

194

OWS3.195 **Which type of snubbing unit was developed in the 1920s using cable systems?**

Answer

- C** (A) Hydraulic Workover Units exclusively
- (B) Stand Alone Snubbing Units
- (C) Rig Assist Snubbing Units utilizing cables connected to traveling blocks
- (D) Portable snubbing units

195

OWS3.196 **What is the main goal of remedial operations during workover?**

Answer

- A** (A) To safely return well to full production in cost effective manner
- (B) To reduce production permanently
- (C) To abandon the well
- (D) To eliminate all wellbore pressure

196

OWS3.197 **Which elements are tested in wireline operations?**

Answer

- D** (A) Only stuffing box
- (B) Stuffing box pressure application capability
- (C) Ram assembly sealing effectiveness
- (D) Both B and C

197

OWS3.198 **What is the primary purpose of the Bullhead Method?**

Answer

- A** (A) Using high-pressure pumps to force heavier kill fluid to displace wellbore fluids
- (B) Gradual pressure reduction through circulation
- (C) Maintaining constant circulation rate
- (D) Reducing tubing weight

198

OWS3.199 **What design feature prevents tubing collapse from bending forces?**

Answer

- C** (A) By eliminating all bending
- (B) By increasing wall thickness
- (C) By providing bending radius of at least 30 times the specified tubing outer diameter
- (D) By reducing fluid pressure continuously

199

OWS3.200 **What is NOT a recognized barrier type in well control systems?**

Answer

- D** (A) Internal barriers preventing flow inside
- (B) External barriers containing wellbore fluids
- (C) Mechanical barriers like casings
- (D) None of the above

200

OWS3.201 **How does coiled tubing differ fundamentally from conventional pipe in operations?**

Answer

- D** (A) Coiled tubing requires constant rotation
- (B) Coiled tubing requires constant rotation
- (C) Both have identical operational characteristics
- (D) Coiled tubing enables continuous circulation on pressurized live wells

201

OWS3.202 **Which factors impact equipment selection?**

Answer

- D** (A) Only pressure requirements
- (B) Operating pressure requirements and limits
- (C) Wellbore geometry and accessibility constraints
- (D) Both B and C

202

OWS3.203 **What specific capability does electric line provide beyond conventional solid wireline?**

Answer

- C** (A) Increased tensile strength only
- (B) Prevention of pressure issues
- (C) Enables data transmission for downhole tools including logging cable for operations
- (D) Allows pipe rotation capability

203

OWS3.204 **What is the main goal of remedial operations during workover?**

Answer

- A** (A) To safely return well to full production in cost effective manner
- (B) To reduce production permanently
- (C) To eliminate all wellbore pressure
- (D) To abandon the well

204

OWS3.205 **Which apply to pressure monitoring?**

Answer

- D** (A) Only continuous measurement
- (B) Continuous measurement during operations
- (C) Recording all pressure changes and trends
- (D) Both B and C

205

OWS3.206 **What is the primary function of the Lubricator in wireline well control operations?**

Answer

- A** (A) Dynamic Pressure Control Equipment creating sealed pathway allowing safe passage of wireline through pressure barrier
- (B) To increase circulation rates exclusively
- (C) To reduce wellbore pressure
- (D) To monitor wire tension

206

OWS3.207 **What is the primary method by which Snubbing Units control pipe movement into pressurized wells?**

Answer

- A** (A) A hydraulic ram system operating through a Ram-to-Ram system for controlled movement
- (B) Cable and block systems only
- (C) Gravity assist exclusively
- (D) Rotational torque application

207

OWS3.208 **Which factors characterize balance point conditions?**

Answer

- B** (A) Gravitational force equals upward pressure force
- (B) Both A and C
- (C) Pipe achieves neutral buoyancy state
- (D) Only force equality

208

OWS3.209 **What well condition requires careful pressure monitoring?**

Answer

- B** (A) Shallow wells exclusively
- (B) All well types to prevent uncontrolled fluid migration during interventions
- (C) Vertical wells only
- (D) Low-pressure wells only

209

OWS3.210 **Why is an anti-buckling guide used during coiled tubing deployment into pressurized wells?**

Answer

- A** (A) To provide additional lateral support and prevent buckling from compressive loads
- (B) To increase tubing tensile strength
- (C) To facilitate faster deployment rates
- (D) To reduce pump friction pressures

210

OWS3.211 **What measurement technique ensures tubing quality control?**

Answer

- C** (A) Pressure testing only
- (B) Weight measurement exclusively
- (C) Physical measurements with precision calipers and sophisticated laser techniques
- (D) Visual inspection only

211

OWS3.212 **What measurement technique ensures tubing quality control?**

Answer

- C** (A) Weight measurement exclusively
- (B) Visual inspection only
- (C) Physical measurements with precision calipers and sophisticated laser techniques
- (D) Pressure testing only

212

OWS3.213 **What measurement threshold indicates possible casing integrity compromise?**

Answer

- D** (A) 400 psi sustained casing pressure
- (B) 300 psi sustained casing pressure
- (C) 200 psi sustained casing pressure
- (D) 500 psi or greater sustained casing pressure (CP)

213

OWS3.214 **Which requirements apply to contingency procedures?**

Answer

- D** (A) Immediate assessment of failure mechanism
- (B) Efficient repair or replacement of equipment
- (C) Only assessment matters
- (D) Both A and B

214

OWS3.215 **How does coiled tubing differ fundamentally from conventional pipe in operations?**

Answer

- D** (A) Conventional pipe is more flexible
- (B) Conventional pipe is more flexible
- (C) Both have identical operational characteristics
- (D) Coiled tubing enables continuous circulation on pressurized live wells

215

OWS3.216 **What is the primary distinction between coiled tubing and conventional pipe?**

Answer

- B** (A) They perform identical functions
- (B) Coiled tubing maintains continuous circulation while conventional pipe requires connections
- (C) Conventional pipe is always stronger
- (D) Coiled tubing cannot be used on live wells

216

OWS3.217 **Why do snubbing units remain essential control at balance point?**

Answer

- B** (A) For pressure reduction exclusively
- (B) Equilibrium is unstable and can quickly shift with minor changes in pressure or string weight
- (C) For monitoring purposes only
- (D) For increasing circulation rates

217

OWS3.218 **Why are dual strippers used in coiled tubing operations?**

Answer

- B** (A) To increase circulation rates
- (B) To ensure operational safety through redundancy if one stripper element fails
- (C) To reduce equipment operating costs
- (D) To eliminate pressure requirements

218

OWS3.219 **What is the primary advantage of braided wireline in well interventions?**

Answer

- C** (A) Eliminates pressure control requirements
- (B) Increases circulation rates exclusively
- (C) Provides enhanced flexibility and reduced diameter for certain well geometries
- (D) Cannot be sheared by rams

219

OWS3.220 **Which factors must be managed in snubbing?**

Answer

- D** (A) Only pressure
- (B) Pressure considerations during pipe movement
- (C) Proper Force calculations for safety
- (D) Both B and C

220

OWS3.221 **What critical consideration affects Bullhead pump rates and pressures?**

Answer

- C** (A) Casing pressure exclusively
- (B) Wellhead design only
- (C) Formation damage minimization while accounting for higher frictional pressure losses
- (D) Personnel comfort during operations

221

OWS3.222 **What must pressure inside the coiled tubing remain above during Bullhead Method?**

Answer

- C** (A) Formation pressure
- (B) Hydrostatic pressure of the wellbore
- (C) The coiled tubing collapse rating
- (D) The annular pressure

222

OWS3.223 **Which type of snubbing unit has the largest operational footprint capability?**

Answer

- D** (A) Portable snubbing units
- (B) Rig Assist Space Saver units
- (C) Rig Assist Snubbing Units only
- (D) Stand Alone Snubbing Units (Hydraulic Workover Units)

223

OWS3.224 **Why is an anti-buckling guide used during coiled tubing deployment?**

Answer

- B** (A) To increase tubing tensile strength
- (B) To provide additional lateral support and prevent buckling from compressive loads
- (C) To reduce pump friction pressures
- (D) To facilitate faster deployment rates

224

OWS3.225 **When performing maintenance on stripper elements, which action occurs first?**

Answer

- B** (A) Increase wellbore pressure
- (B) Close both Upper and Lower Safety Rams to provide two-barrier isolation
- (C) Activate emergency shutdown
- (D) Increase pump rates significantly

225

OWS3.226 **Which must be considered in well design?**

Answer

- D** (A) Only formation pressure
- (B) Formation pressure gradients and characteristics
- (C) Wellbore stability and geometry
- (D) Both B and C

226

OWS3.227 **Why are spherical preventers typically used as strippers in snubbing?**

Answer

- A** (A) They provide longer contact area which enhances durability and reduces wear during operations
- (B) They require less maintenance
- (C) They operate at lower pressures
- (D) They are less expensive

227

OWS3.228 **What indicator might suggest compromised well integrity?**

Answer

- C** (A) Increased wellhead temperature
- (B) Reduced circulation rates
- (C) Identification of sustained casing pressure (CP) exceeding 500 psi
- (D) Low surface pressure only

228